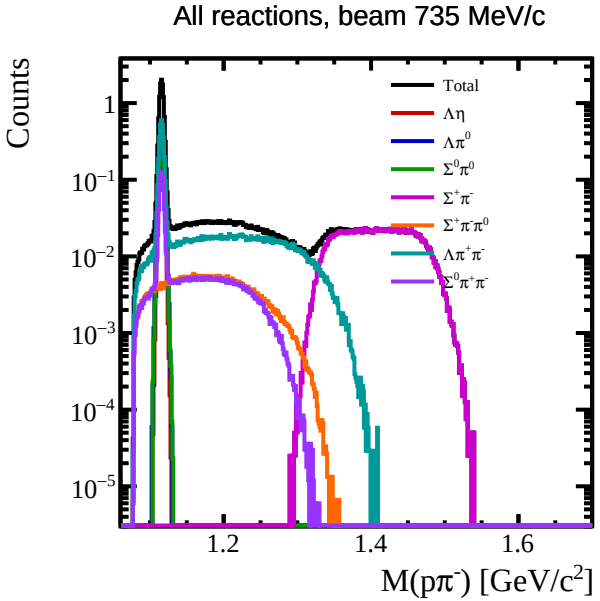
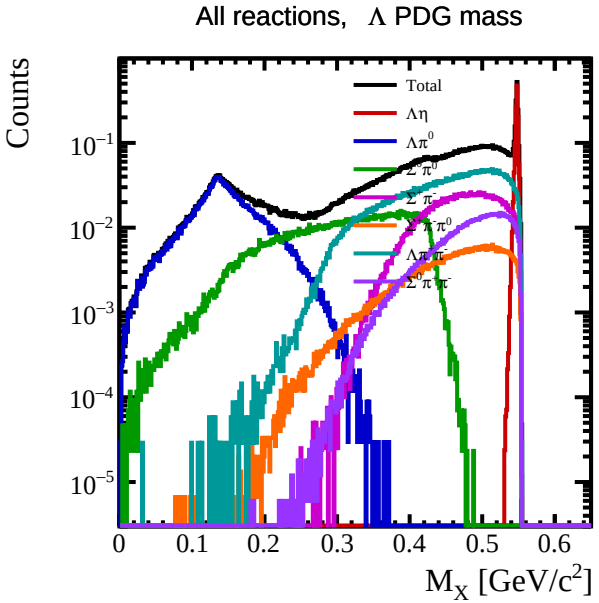
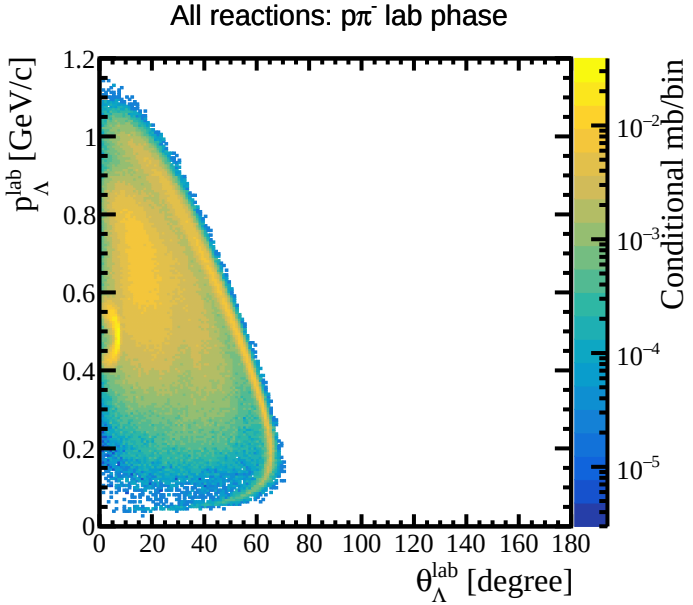
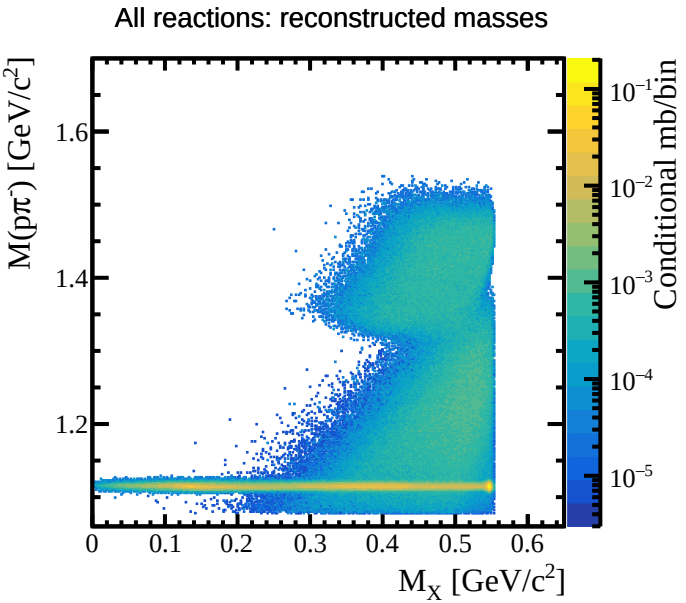
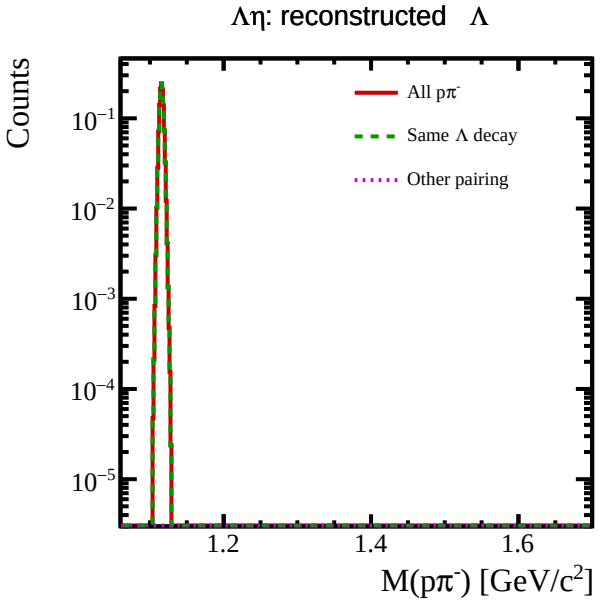
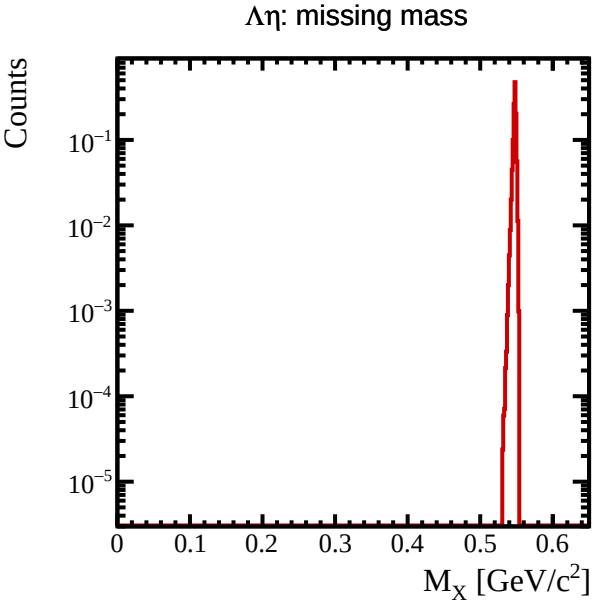

All reactions, beam 735 MeV/c
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π-
Σ+π-π0
Λπ+π-
Σ0π+π-
M(pπ-) [GeV/c2]

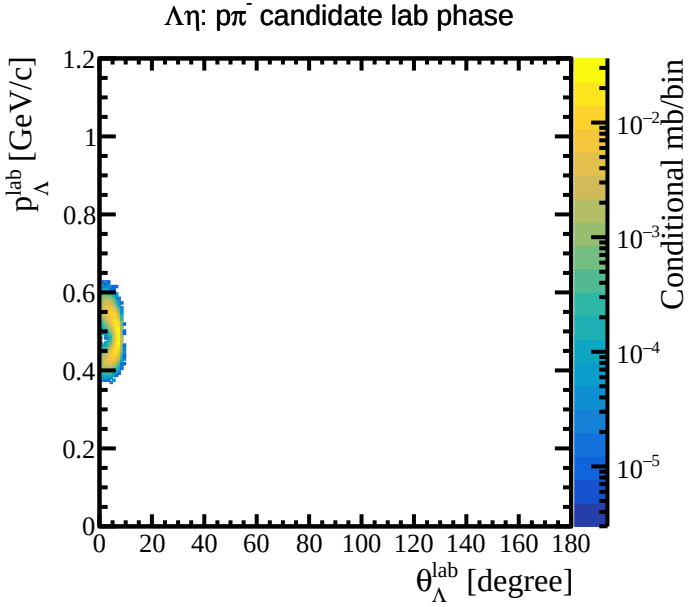

All reactions, Λ PDG mass
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π-
Σ+π-π0
Λπ+π-
Σ0π+π-
MX [GeV/c2]

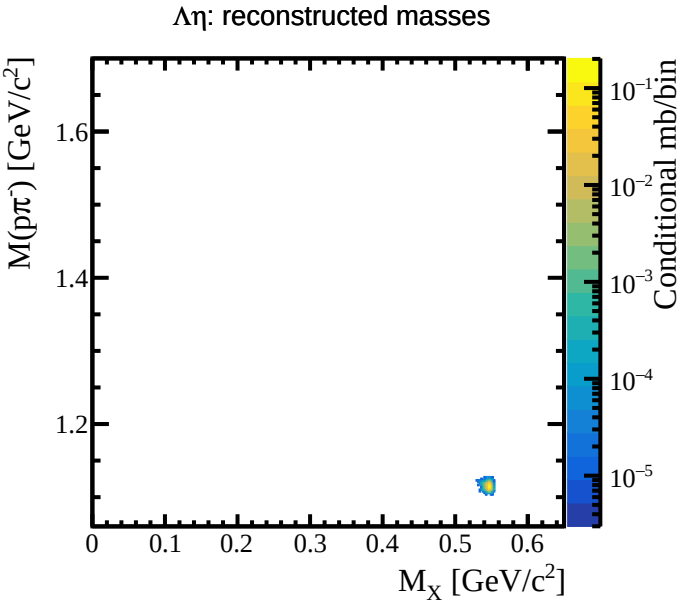

All reactions: pπ- lab phase
pΛlab [GeV/c]
θΛlab [degree]
Conditional mb/bin

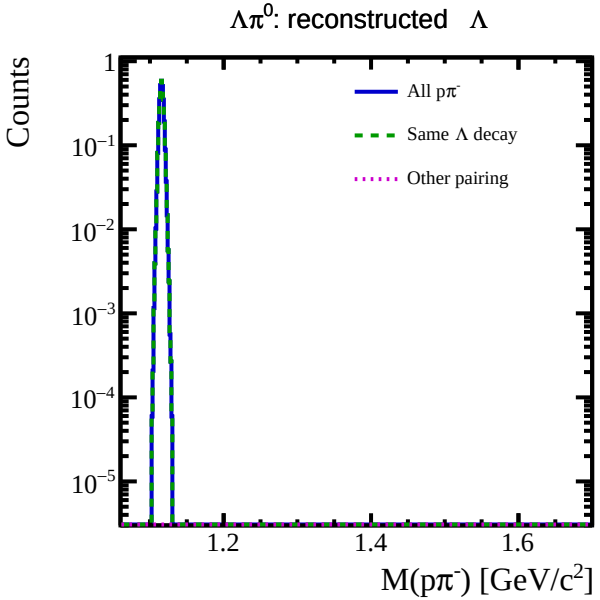

All reactions: reconstructed masses
M(pπ-) [GeV/c2]
MX [GeV/c2]
Conditional mb/bin

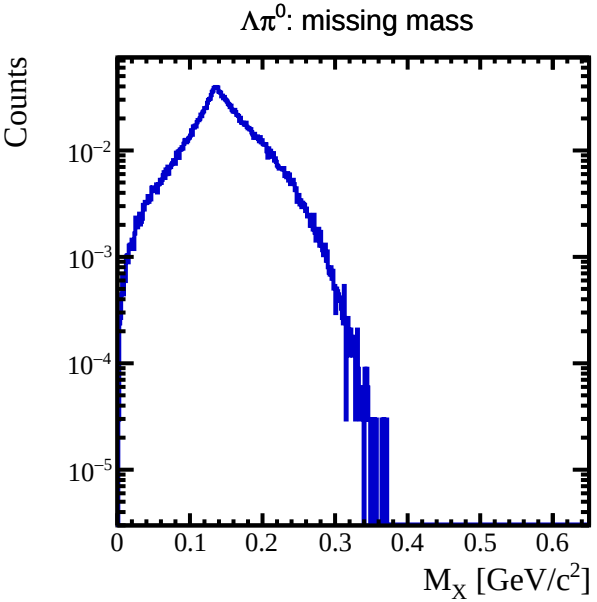

Λη: reconstructed Λ
Counts
All pπ⁻
Same Λ decay
Other pairing
M(pπ⁻) [GeV/c²]

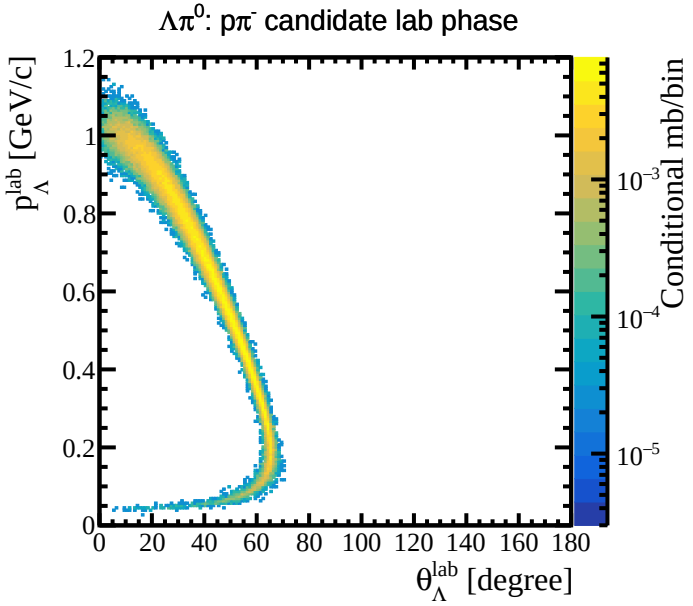

Λη: missing mass
Counts
M_X [GeV/c²]

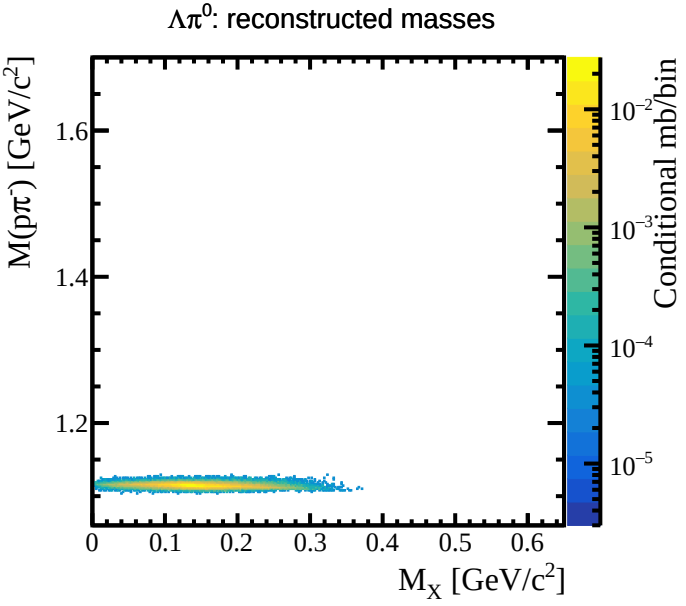

Λη: pπ⁻ candidate lab phase
p_Λ^lab [GeV/c]
θ_Λ^lab [degree]
Conditional mb/bin

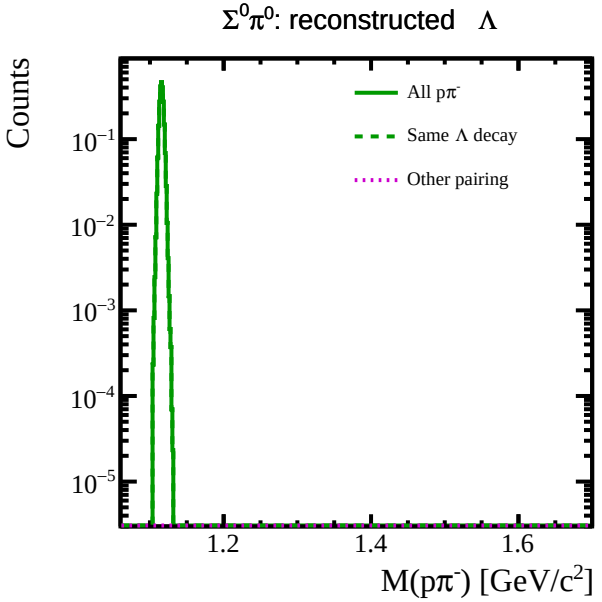

Λη: reconstructed masses
M(pπ⁻) [GeV/c²]
M_X [GeV/c²]
Conditional mb/bin

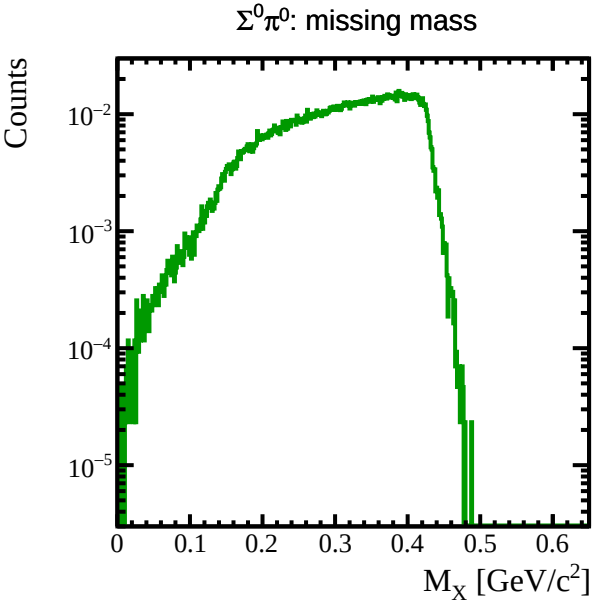

Λπ0: reconstructed Λ
Counts
All pπ-
Same Λ decay
Other pairing
M(pπ-) [GeV/c2]

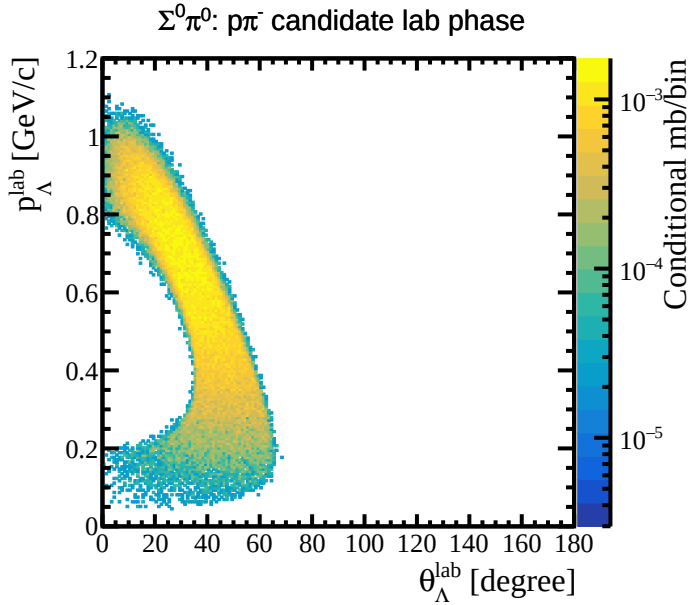

Λπ0: missing mass
Counts
MX [GeV/c2]

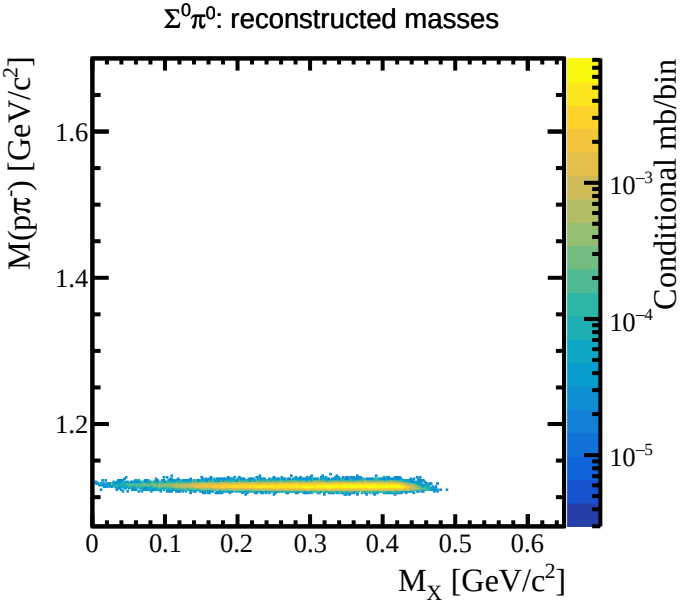

Λπ0: pπ- candidate lab phase
pΛlab [GeV/c]
θΛlab [degree]
Conditional mb/bin

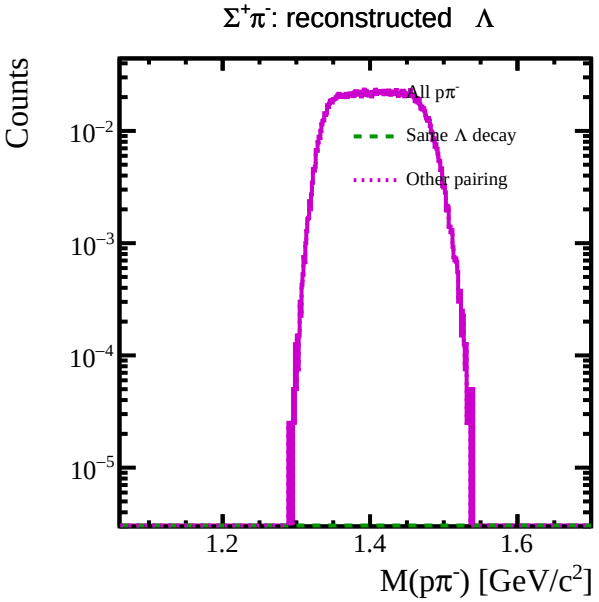

Λπ0: reconstructed masses
M(pπ-) [GeV/c2]
MX [GeV/c2]
Conditional mb/bin

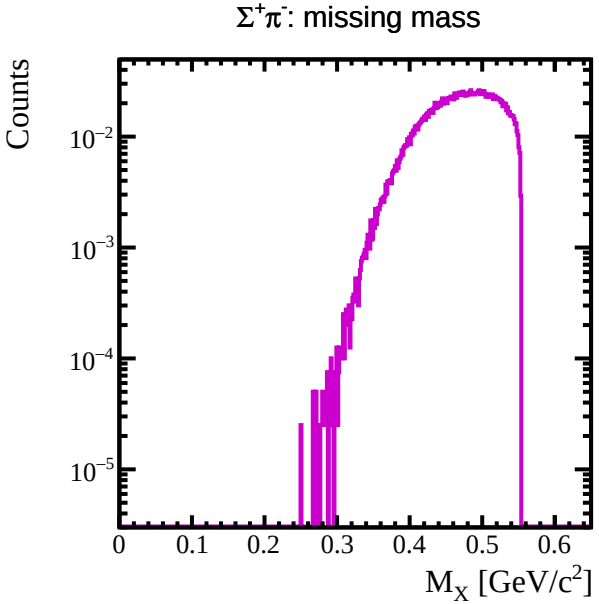

$\Sigma^0\pi^0$: reconstructed Λ
Counts
All $p\pi^-$
Same Λ decay
Other pairing
$M(p\pi^-)$ [GeV/c^2]

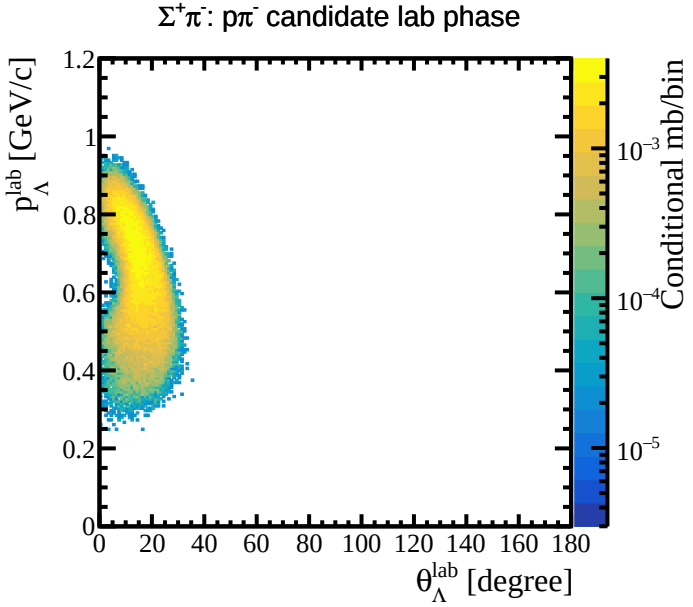

$\Sigma^0\pi^0$: missing mass
Counts
M_X [GeV/c^2]

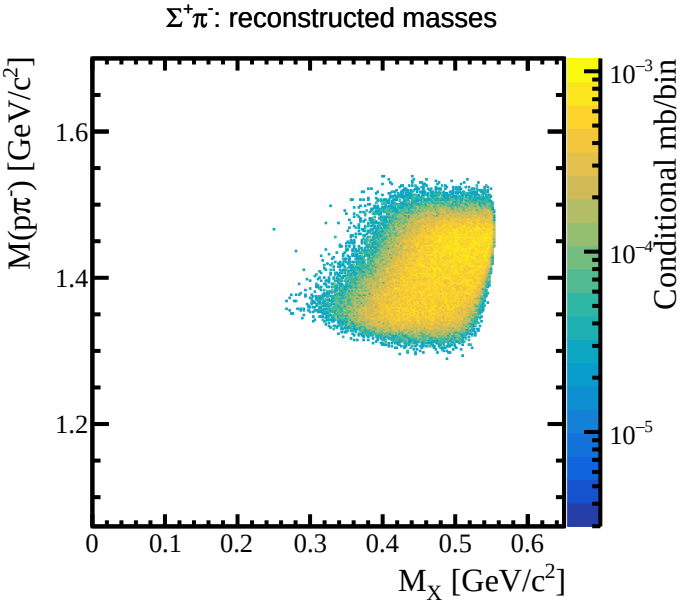

$\Sigma^0\pi^0$: $p\pi^-$ candidate lab phase
p_Λ^{lab} [GeV/c]
θ_Λ^{lab} [degree]
Conditional mb/bin

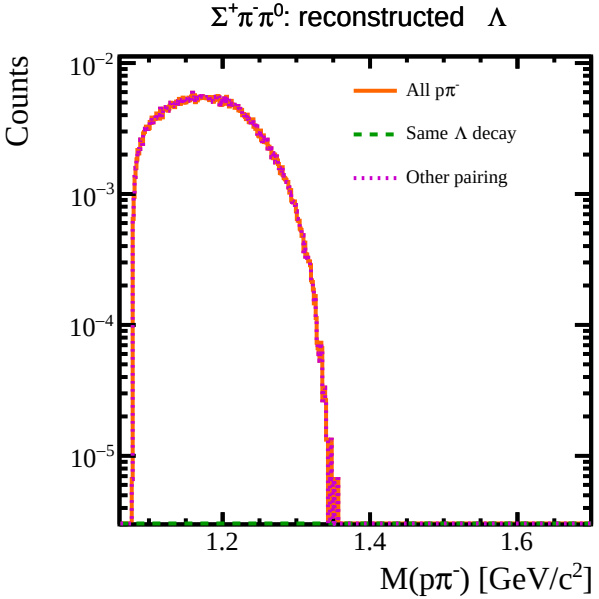

$\Sigma^0\pi^0$: reconstructed masses
$M(p\pi^-)$ [GeV/c^2]
M_X [GeV/c^2]
Conditional mb/bin

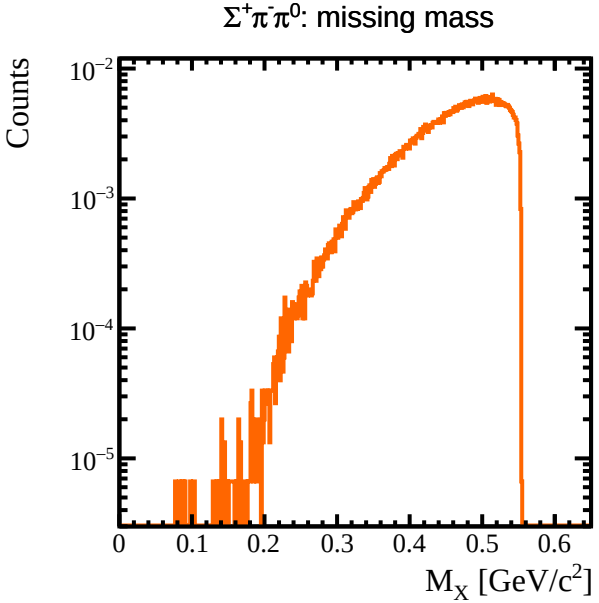

Σ+π-: reconstructed Λ
Counts
All pπ-
Same Λ decay
Other pairing
M(pπ-) [GeV/c2]

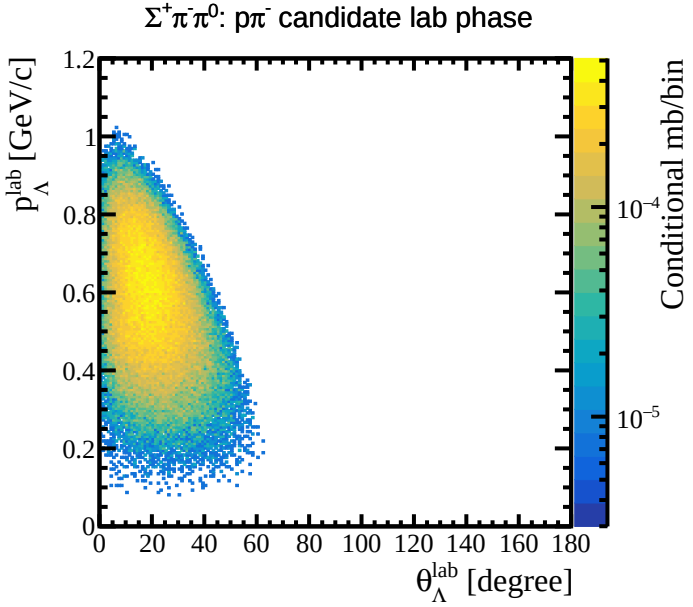

Σ+π-: missing mass
Counts
MX [GeV/c2]

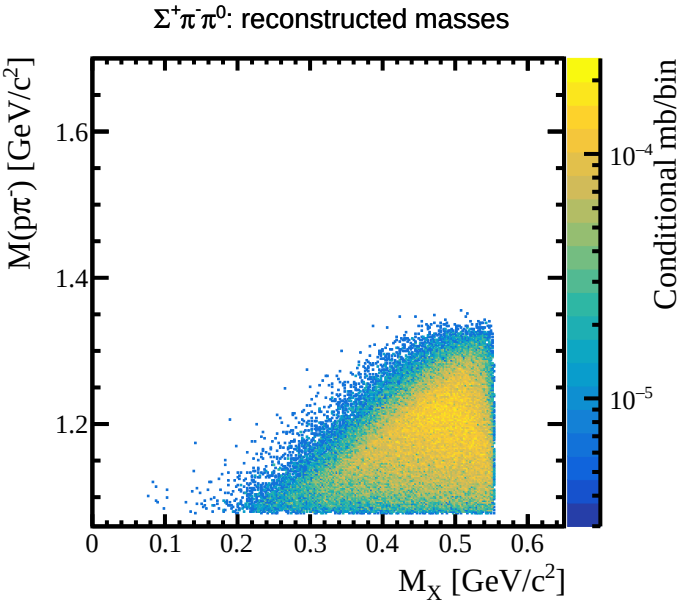

Σ+π-: pπ- candidate lab phase
pΛlab [GeV/c]
θΛlab [degree]
Conditional mb/bin

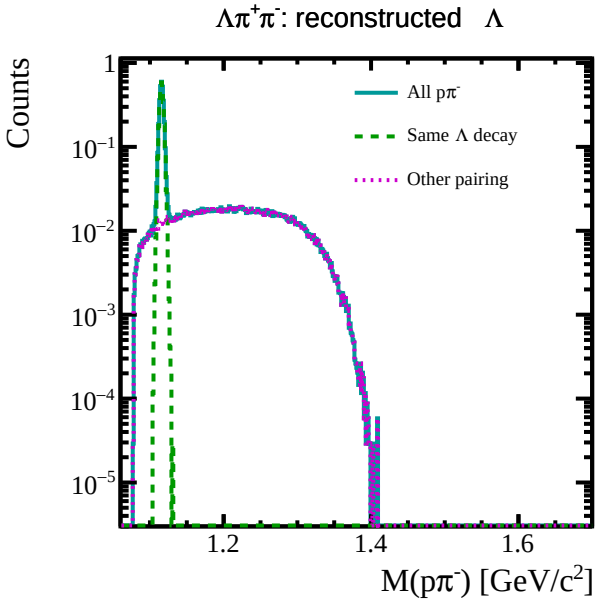

Σ+π-: reconstructed masses
M(pπ-) [GeV/c2]
MX [GeV/c2]
Conditional mb/bin

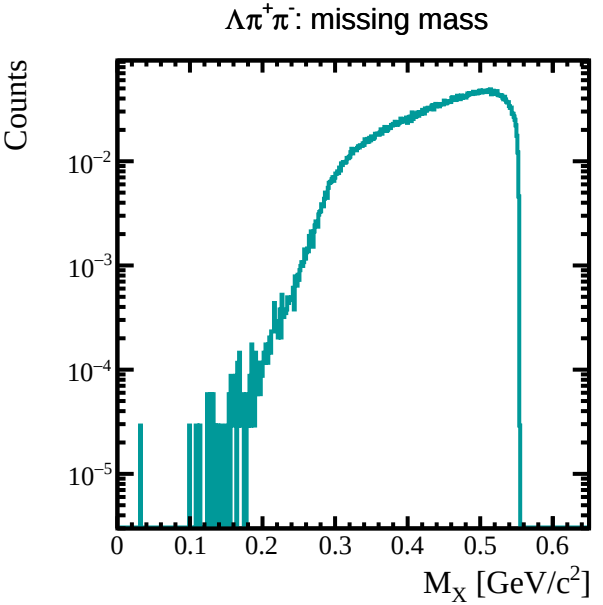

Σ+π-π0: reconstructed Λ
Counts
All pπ-
Same Λ decay
Other pairing
M(pπ-) [GeV/c2]

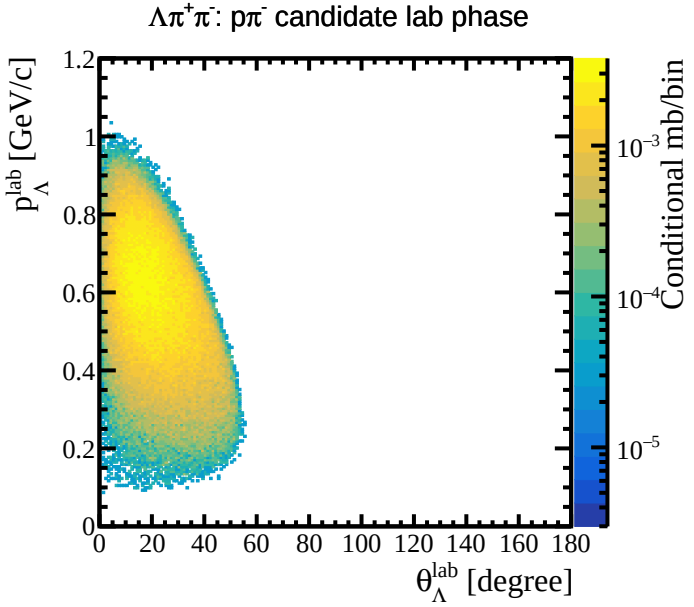

Σ+π-π0: missing mass
Counts
MX [GeV/c2]

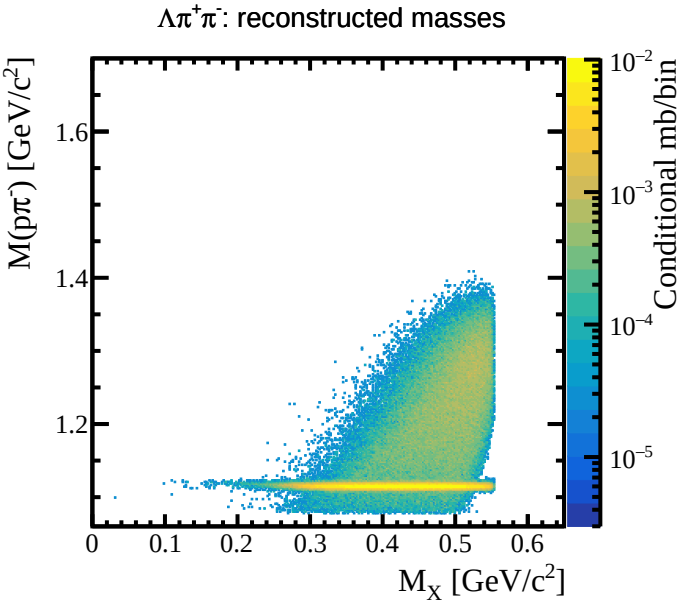

Σ+π-π0: pπ- candidate lab phase
pΛlab [GeV/c]
θΛlab [degree]
Conditional mb/bin

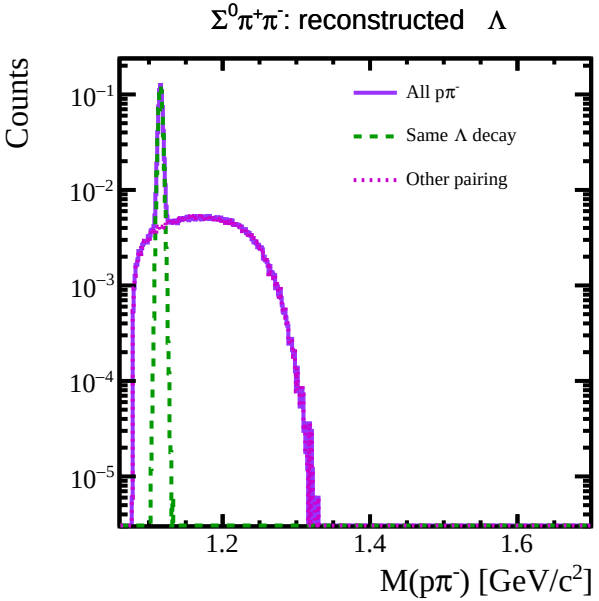

Σ+π-π0: reconstructed masses
M(pπ-) [GeV/c2]
MX [GeV/c2]
Conditional mb/bin

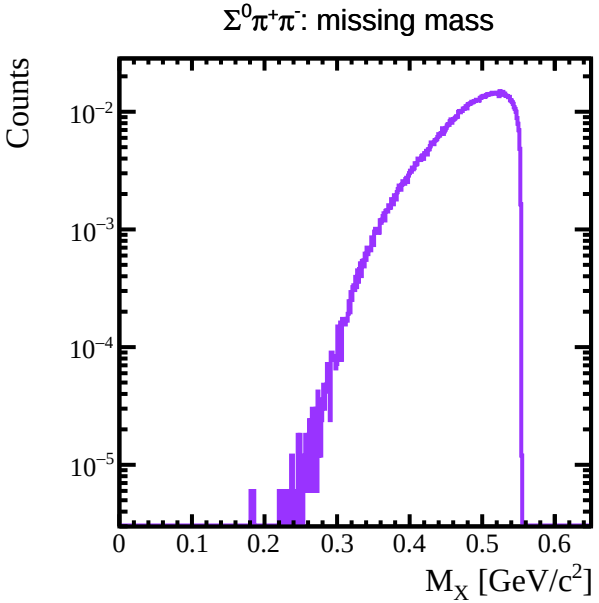

Λπ+π-: reconstructed Λ
Counts
All pπ-
Same Λ decay
Other pairing
M(pπ-) [GeV/c2]

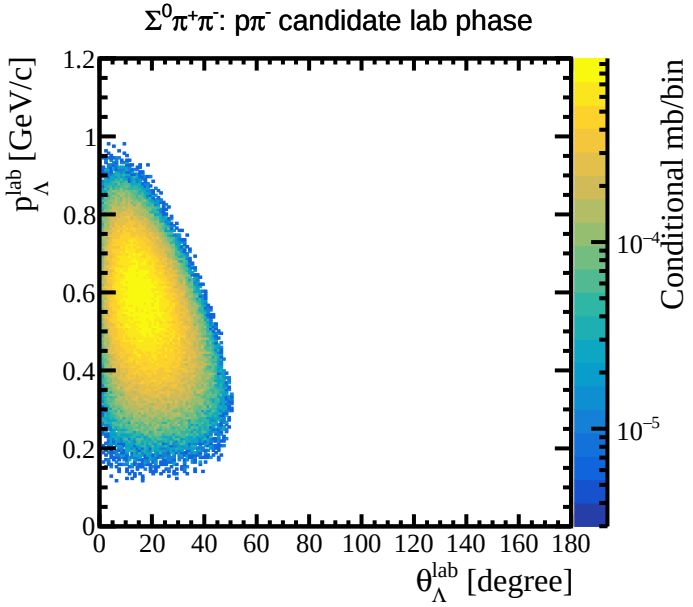

Λπ+π-: missing mass
Counts
MX [GeV/c2]

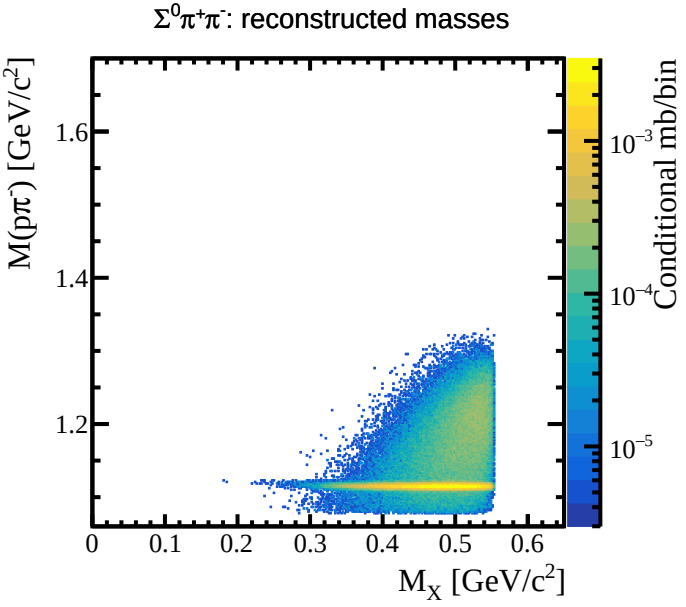

Λπ+π-: pπ- candidate lab phase
pΛlab [GeV/c]
θΛlab [degree]
Conditional mb/bin

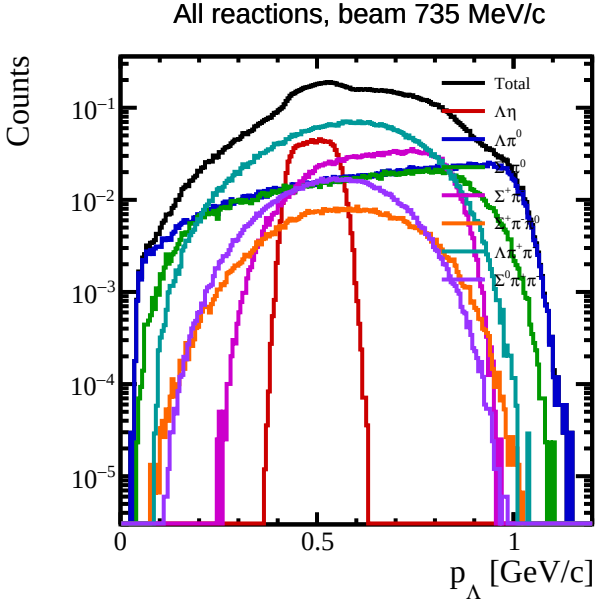

Λπ+π-: reconstructed masses
M(pπ-) [GeV/c2]
MX [GeV/c2]
Conditional mb/bin

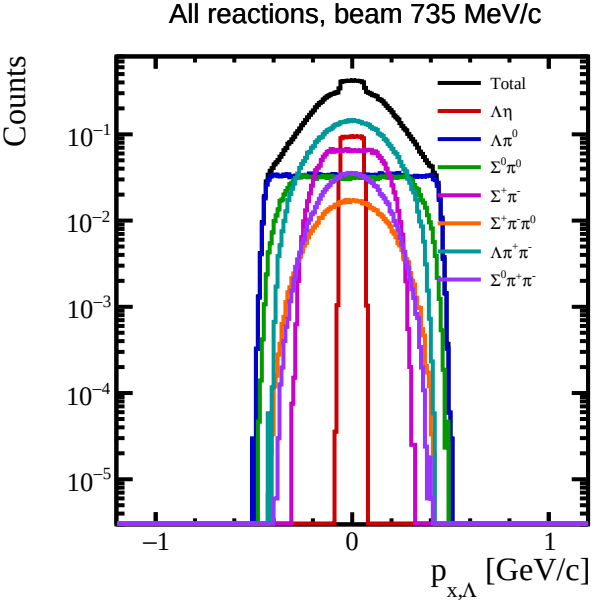

$\Sigma^0\pi^+\pi^-$: reconstructed Λ
Counts
All $p\pi^-$
Same Λ decay
Other pairing
$M(p\pi^-)$ [GeV/c^2]

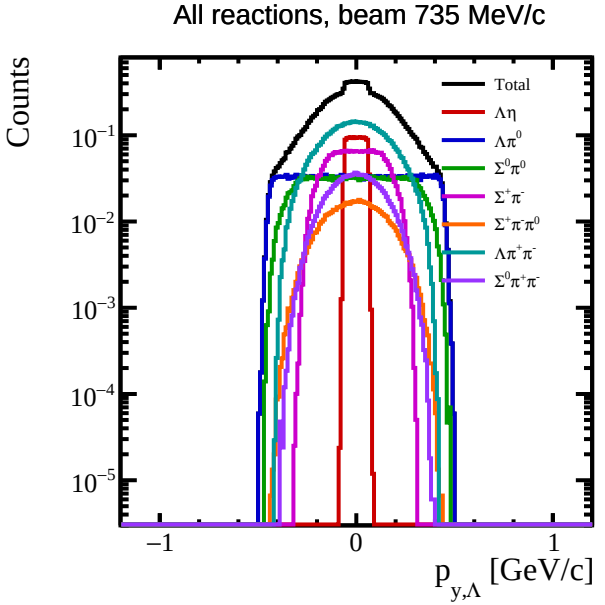

$\Sigma^0\pi^+\pi^-$: missing mass
Counts
M_X [GeV/c^2]

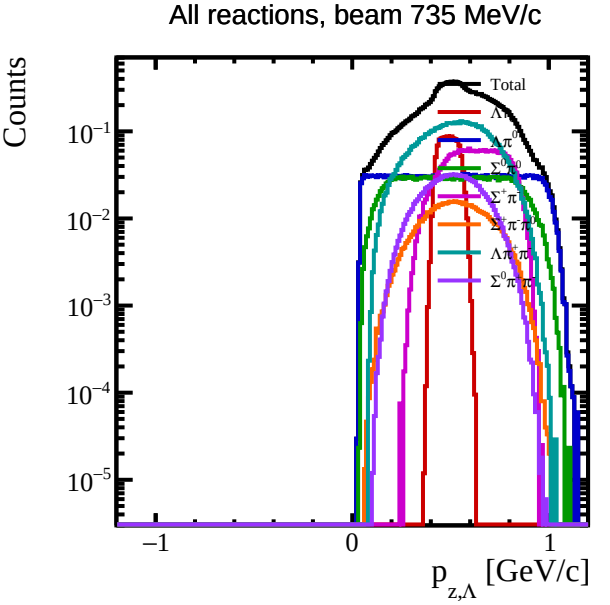

$\Sigma^0\pi^+\pi^-$: $p\pi^-$ candidate lab phase
p_Λ^{lab} [GeV/c]
θ_Λ^{lab} [degree]
Conditional mb/bin

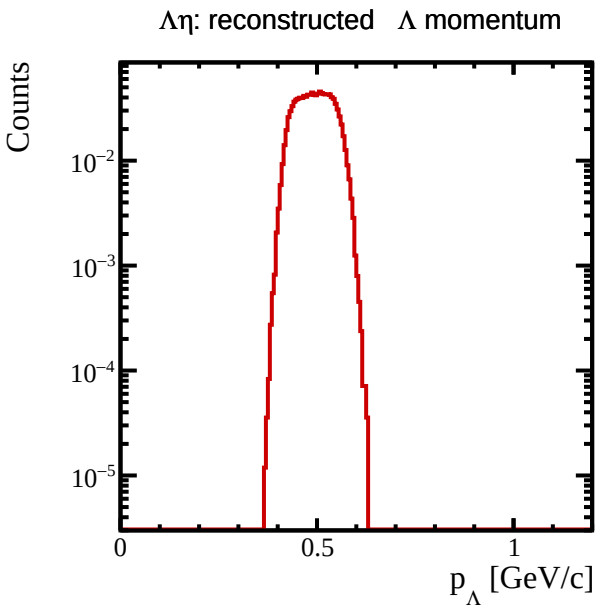

$\Sigma^0\pi^+\pi^-$: reconstructed masses
$M(p\pi^-)$ [GeV/c^2]
M_X [GeV/c^2]
Conditional mb/bin

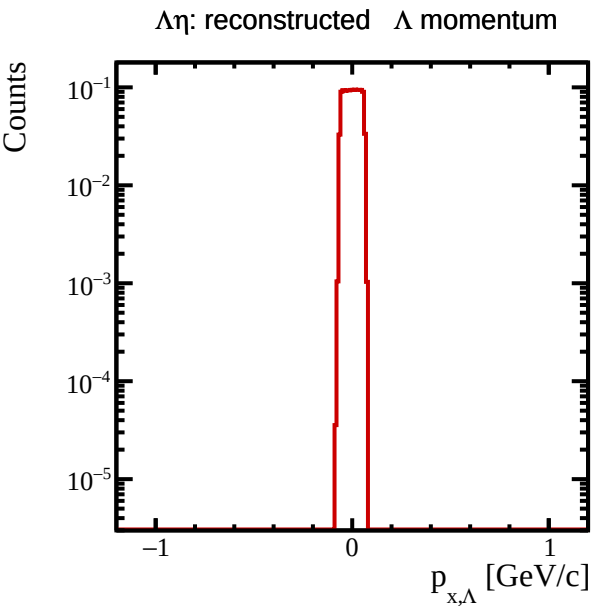

All reactions, beam 735 MeV/c
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π-
Σ+π-π0
Λπ+π-
Σ0π+π-
p Λ [GeV/c]

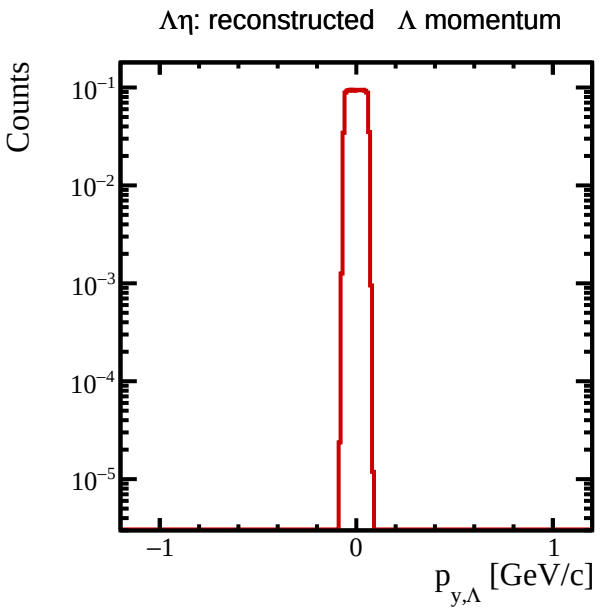

All reactions, beam 735 MeV/c
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π-
Σ+π-π0
Λπ+π-
Σ0π+π-
p x,Λ [GeV/c]

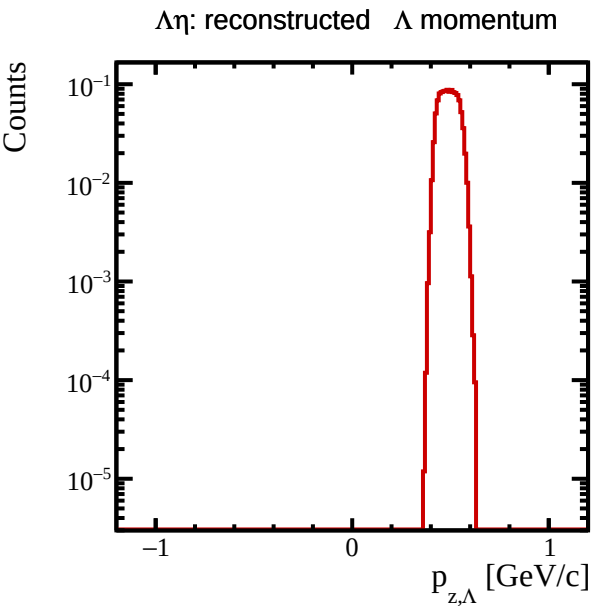

All reactions, beam 735 MeV/c
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π-
Σ+π-π0
Λπ+π-
Σ0π+π-
p y,Λ [GeV/c]

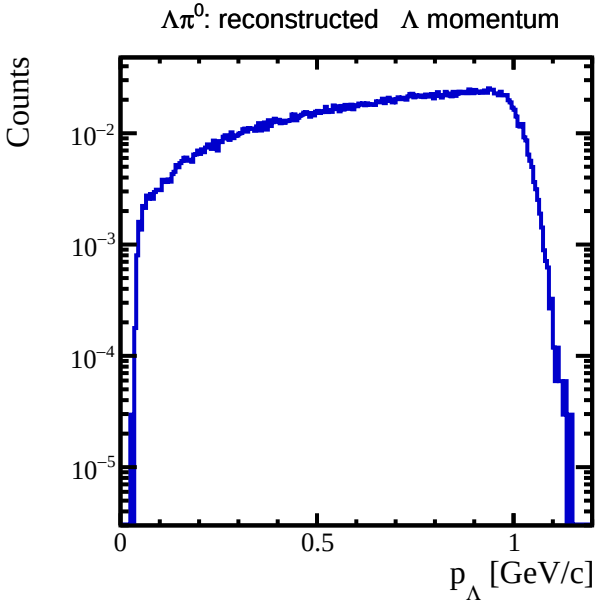

All reactions, beam 735 MeV/c
Counts
Total
p z,Λ [GeV/c]

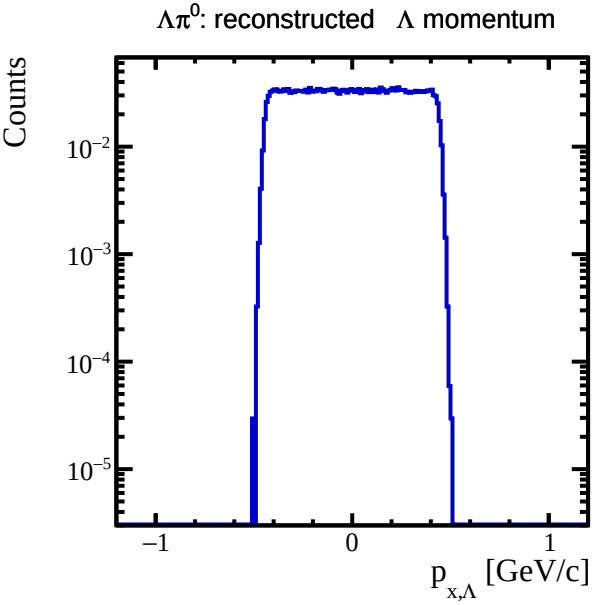

Λη: reconstructed Λ momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
0.5
1
p_{Λ} [GeV/c]

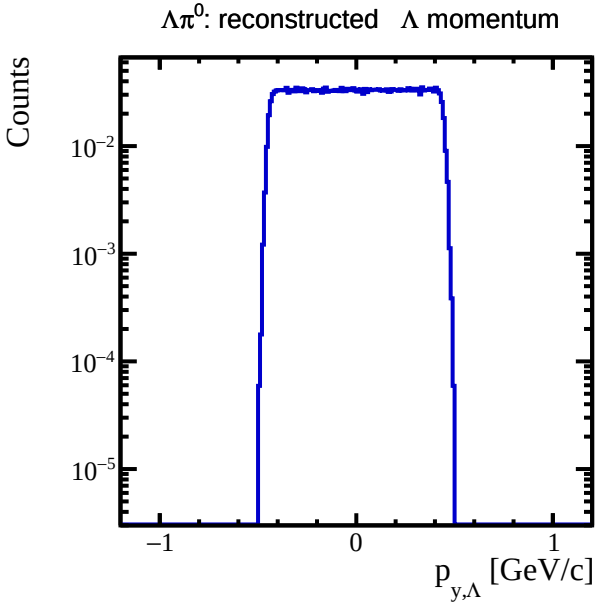

Λη: reconstructed Λ momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
$p_{x,\Lambda}$ [GeV/c]

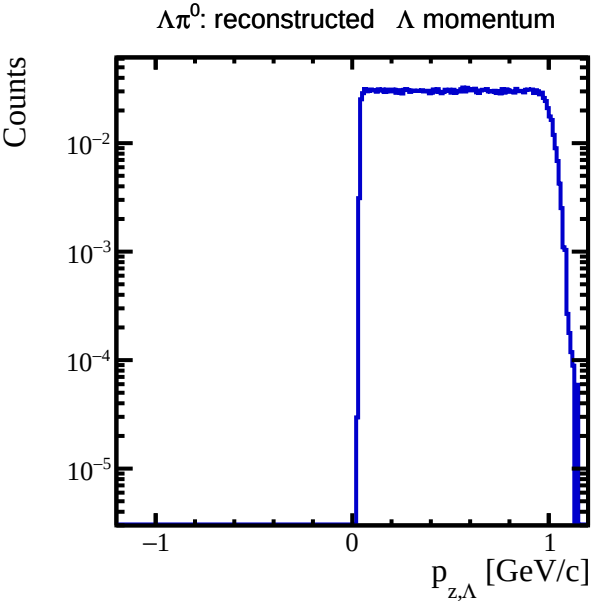

Λη: reconstructed Λ momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
$p_{y,\Lambda}$ [GeV/c]

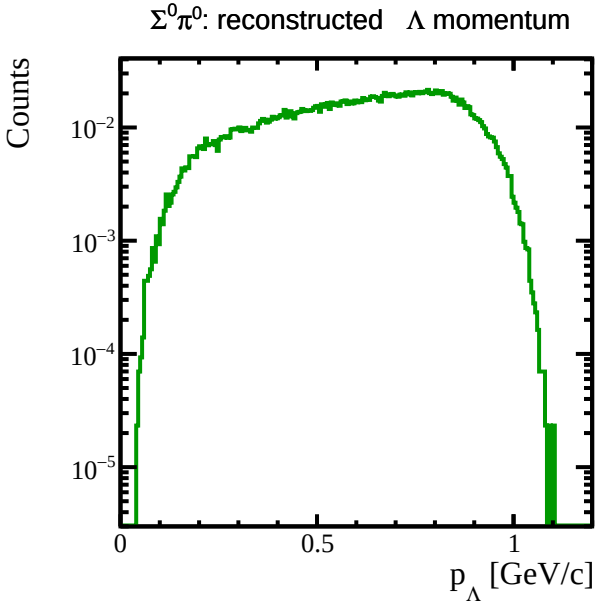

Λη: reconstructed Λ momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
$p_{z,\Lambda}$ [GeV/c]

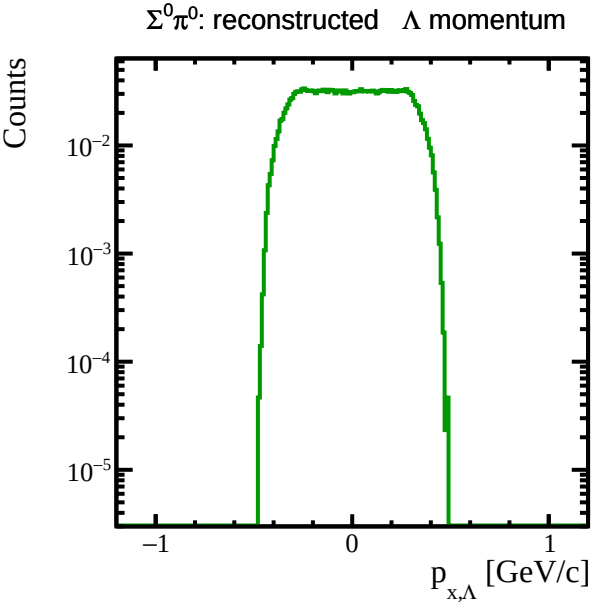

Λπ0: reconstructed Λ momentum
Counts
10−2
10−3
10−4
10−5
0
0.5
1
pΛ [GeV/c]

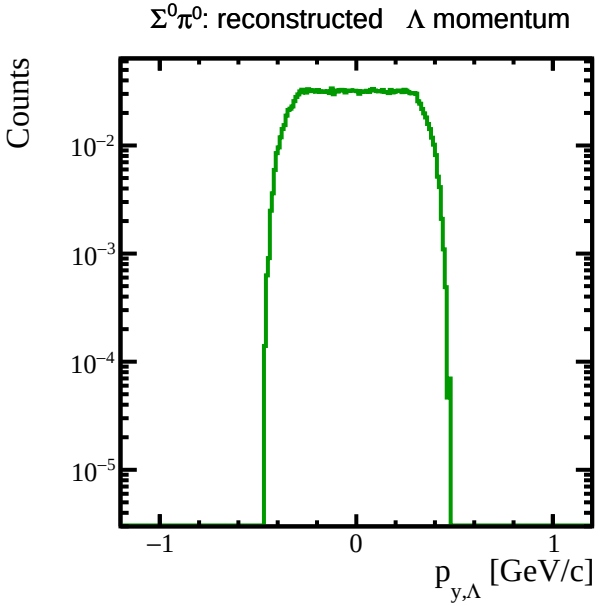

Λπ0: reconstructed Λ momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
px,Λ [GeV/c]

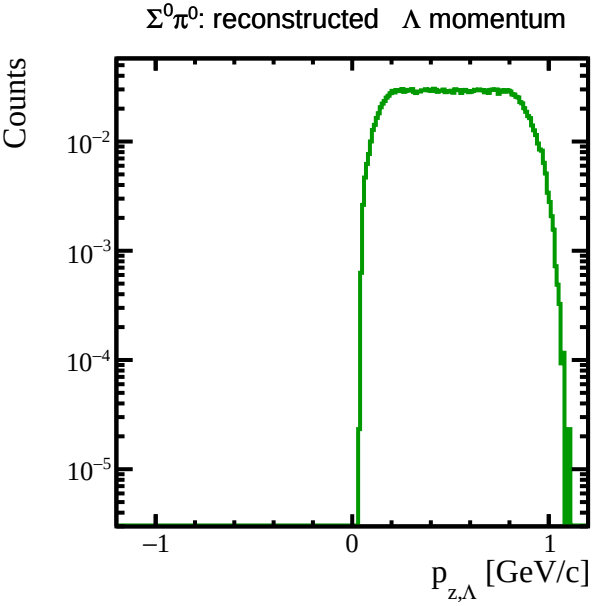

Λπ0: reconstructed Λ momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
py,Λ [GeV/c]

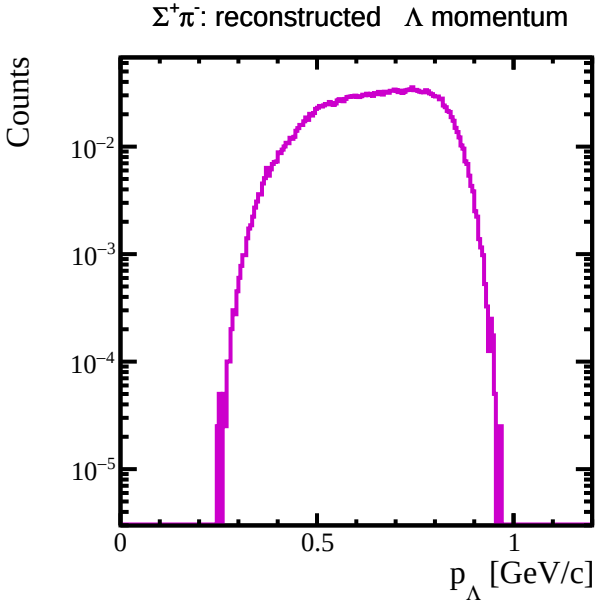

Λπ0: reconstructed Λ momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
pz,Λ [GeV/c]

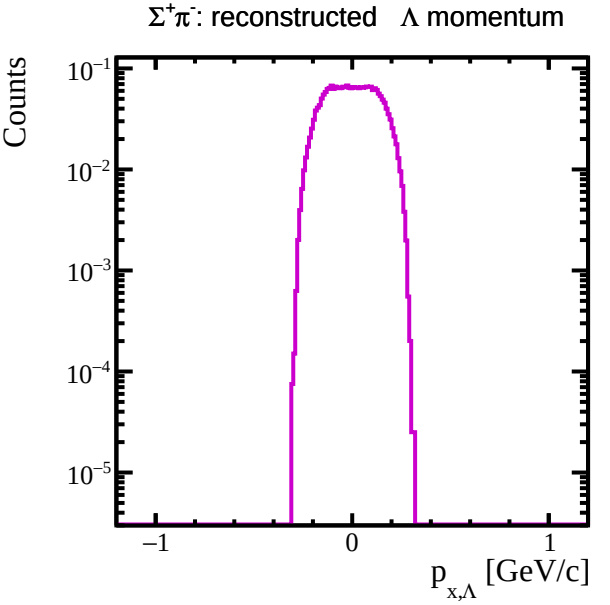

$\Sigma^0\pi^0$: reconstructed Λ momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
0.5
1
p_Λ [GeV/c]

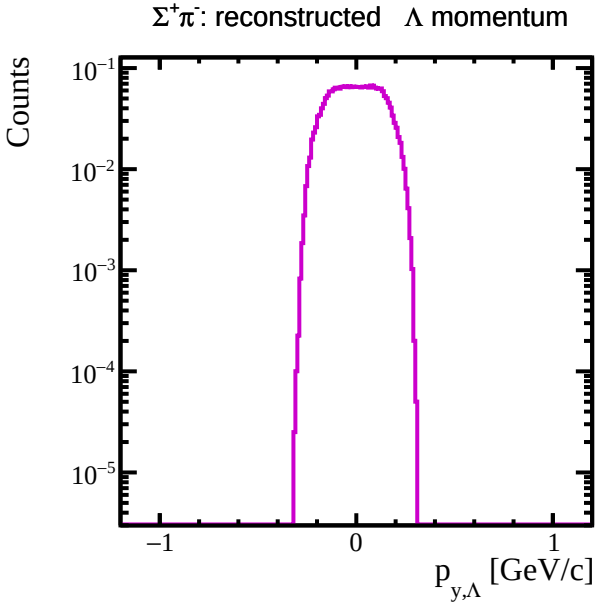

$\Sigma^0\pi^0$: reconstructed Λ momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
$p_{x,\Lambda}$ [GeV/c]

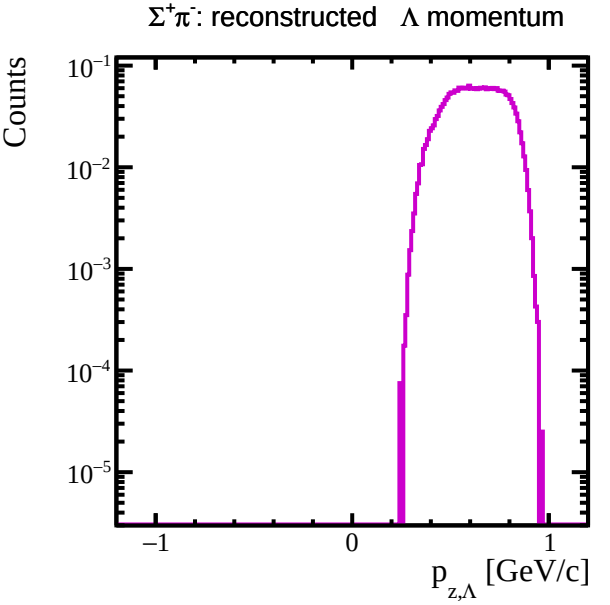

$\Sigma^0\pi^0$: reconstructed Λ momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
$p_{y,\Lambda}$ [GeV/c]

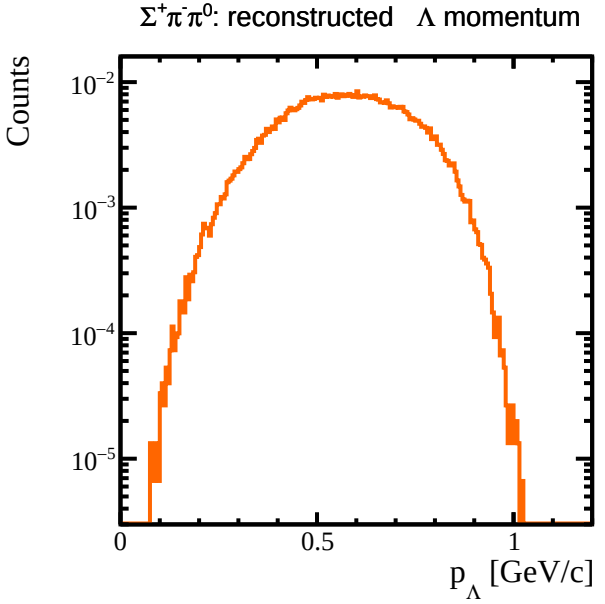

$\Sigma^0\pi^0$: reconstructed Λ momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
$p_{z,\Lambda}$ [GeV/c]

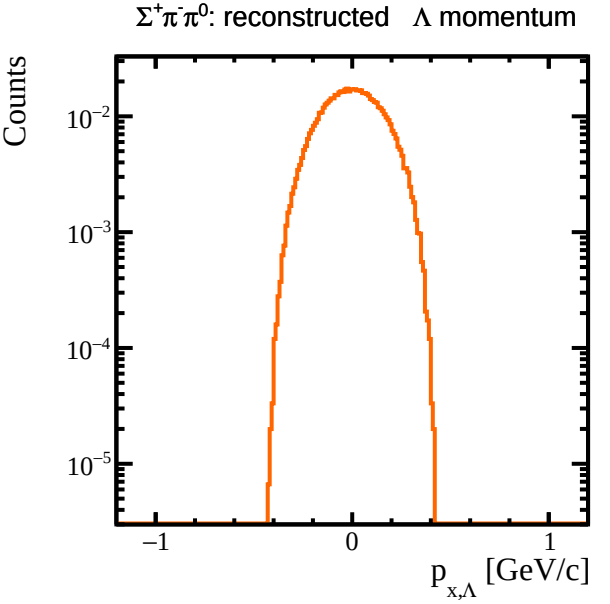

Σ+π-: reconstructed Λ momentum
Counts
10−2
10−3
10−4
10−5
0
0.5
1
p Λ [GeV/c]

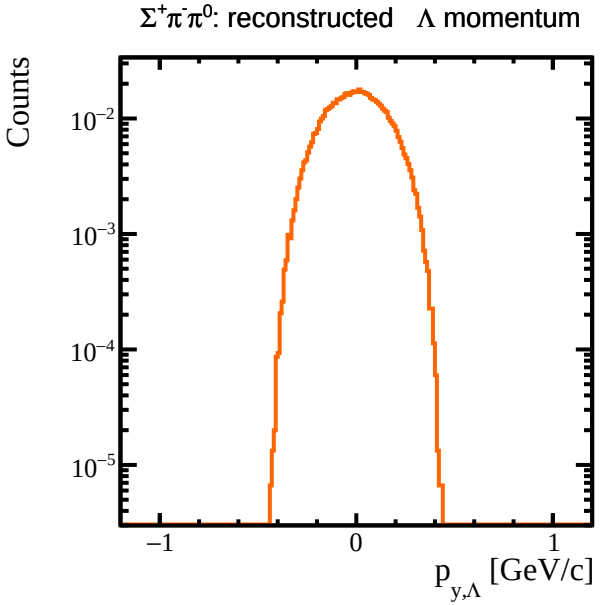

Σ+π-: reconstructed Λ momentum
Counts
10−1
10−2
10−3
10−4
10−5
−1
0
1
p x,Λ [GeV/c]

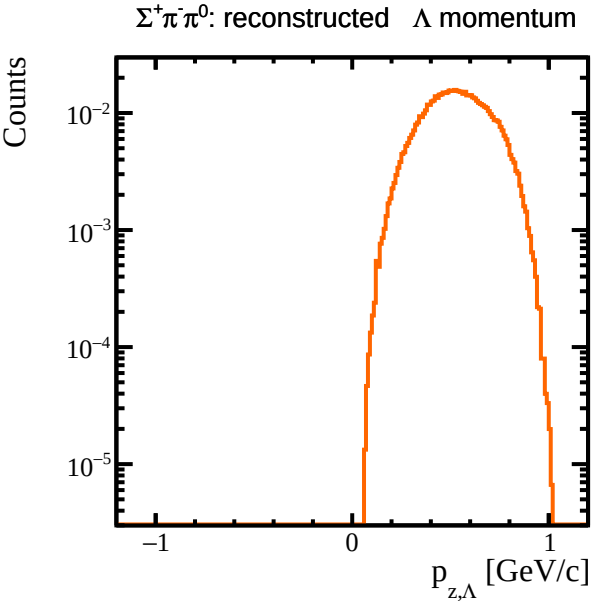

Σ+π-: reconstructed Λ momentum
Counts
10−1
10−2
10−3
10−4
10−5
−1
0
1
p y,Λ [GeV/c]

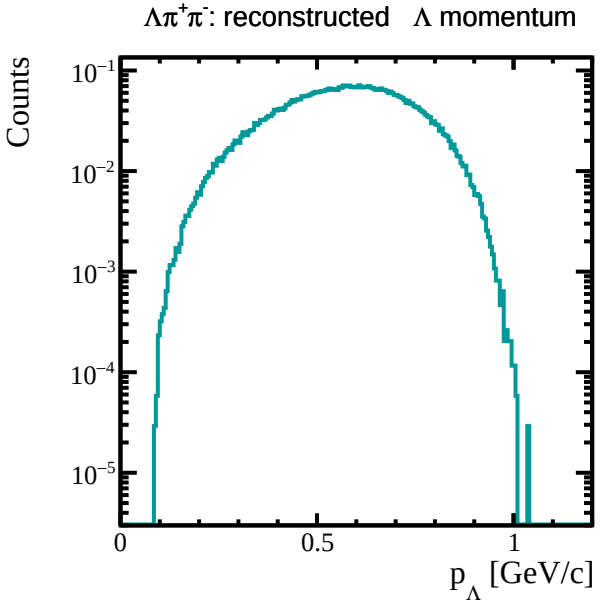

Σ+π-: reconstructed Λ momentum
Counts
10−1
10−2
10−3
10−4
10−5
−1
0
1
p z,Λ [GeV/c]

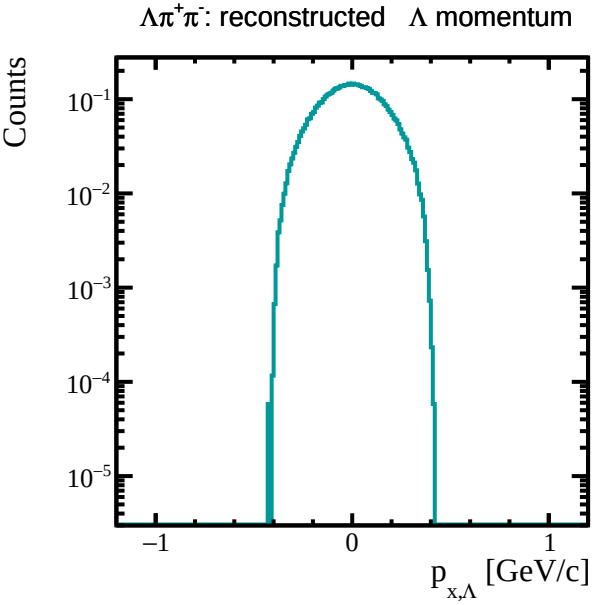

Σ+π-π0: reconstructed Λ momentum
Counts
10−2
10−3
10−4
10−5
0
0.5
1
pΛ [GeV/c]

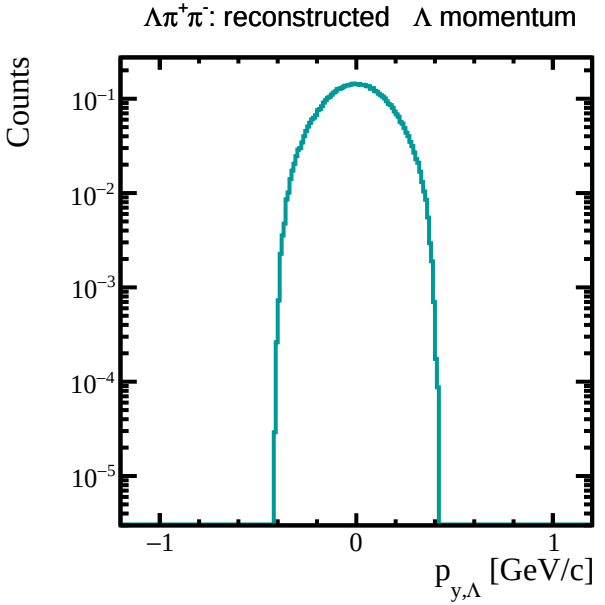

Σ+π-π0: reconstructed Λ momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
px,Λ [GeV/c]

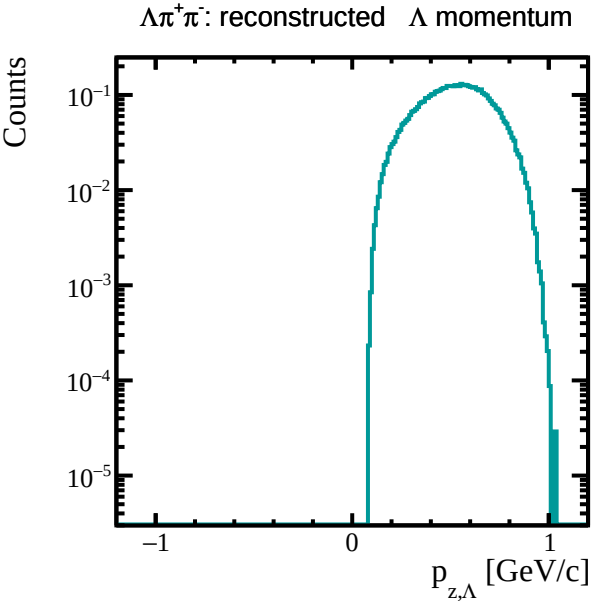

Σ+π-π0: reconstructed Λ momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
py,Λ [GeV/c]

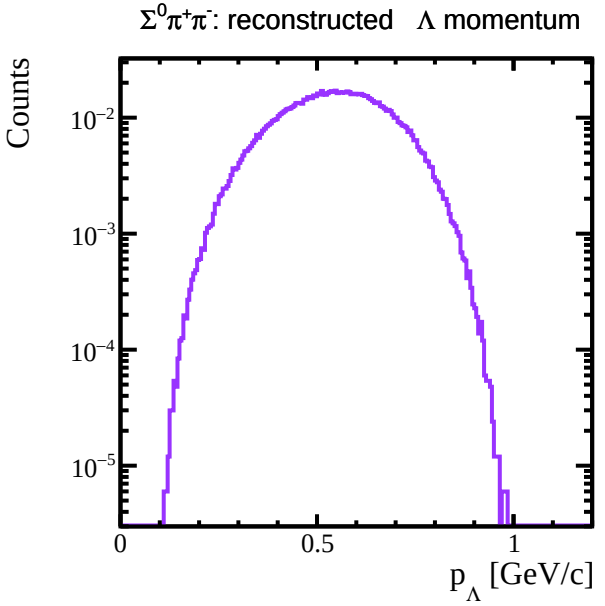

Σ+π-π0: reconstructed Λ momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
pz,Λ [GeV/c]

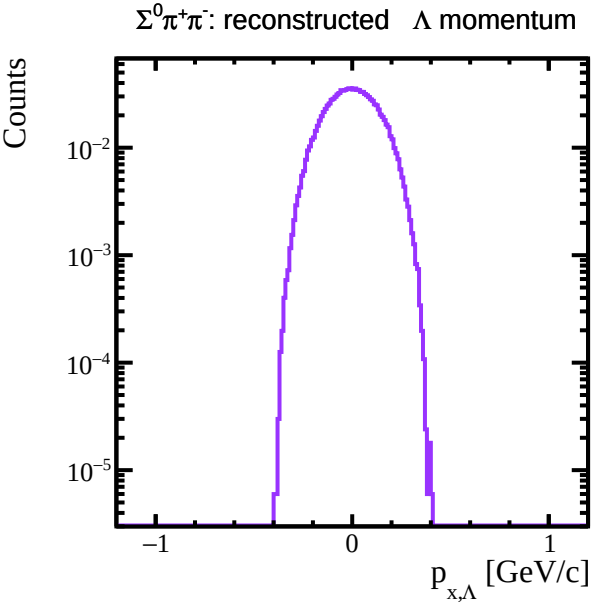

Λπ+π-: reconstructed Λ momentum
Counts
10−1
10−2
10−3
10−4
10−5
0
0.5
1
pΛ [GeV/c]

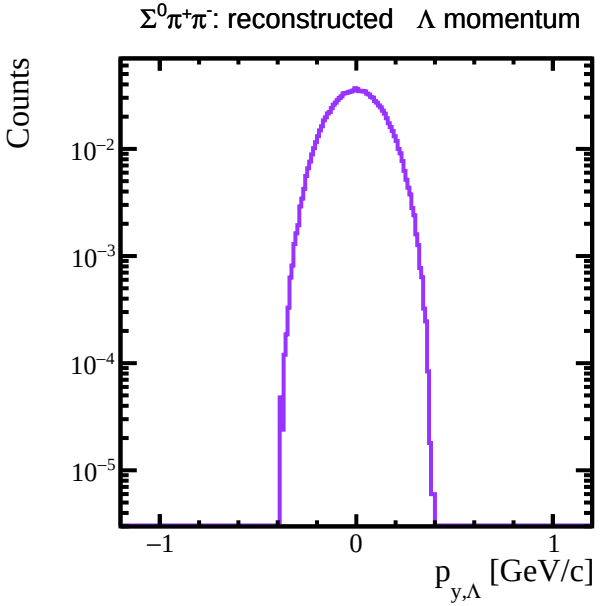

Λπ+π-: reconstructed Λ momentum
Counts
10−1
10−2
10−3
10−4
10−5
−1
0
1
px,Λ [GeV/c]

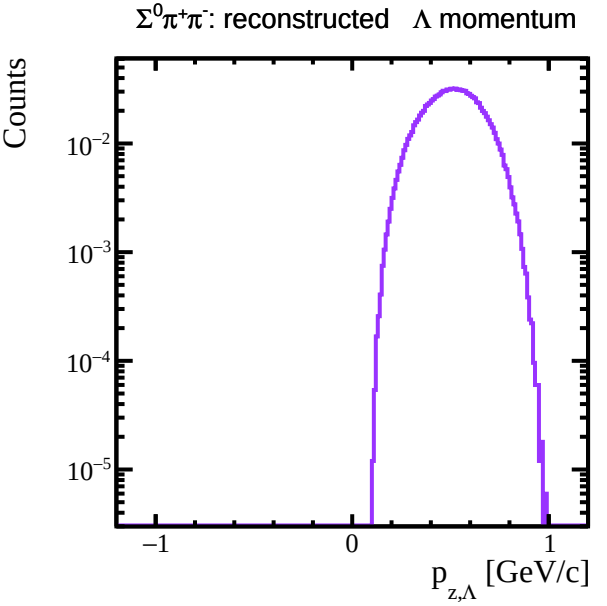

Λπ+π-: reconstructed Λ momentum
Counts
10−1
10−2
10−3
10−4
10−5
−1
0
1
py,Λ [GeV/c]

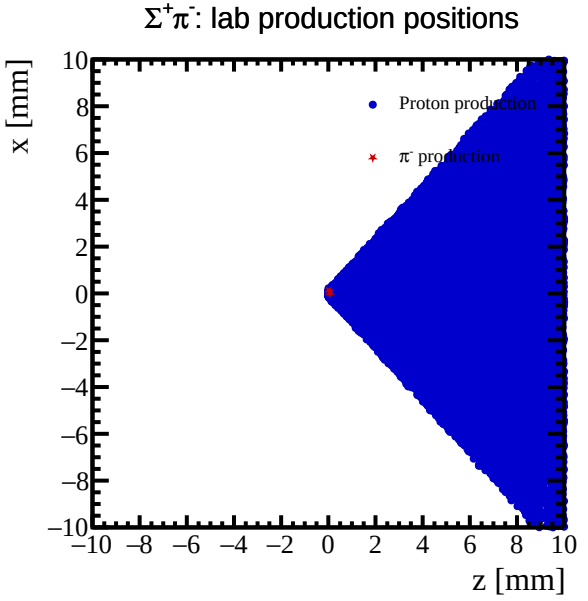

Λπ+π-: reconstructed Λ momentum
Counts
10−1
10−2
10−3
10−4
10−5
−1
0
1
pz,Λ [GeV/c]

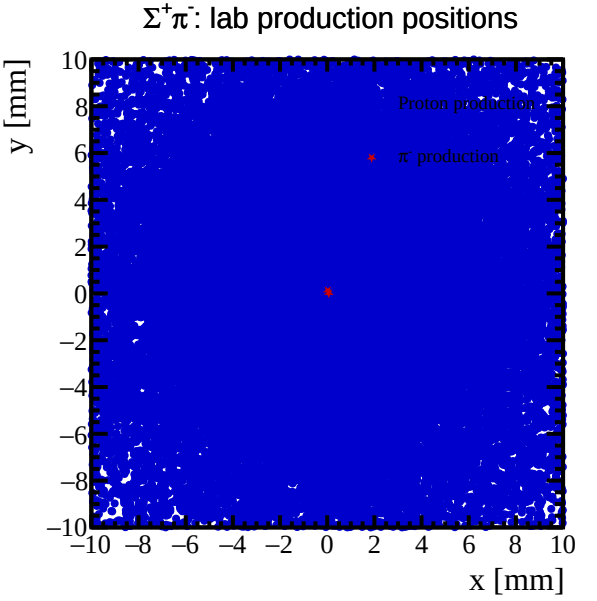

$\Sigma^0\pi^+\pi^-$: reconstructed Λ momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
0.5
1
p_Λ [GeV/c]

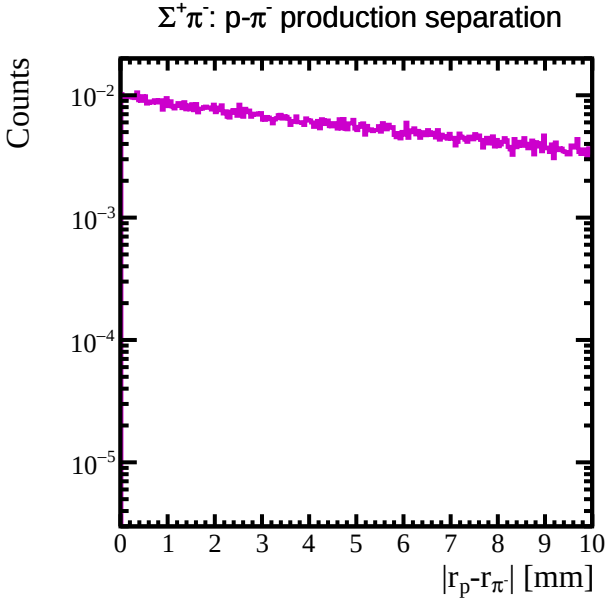

$\Sigma^0\pi^+\pi^-$: reconstructed Λ momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
$p_{x,\Lambda}$ [GeV/c]

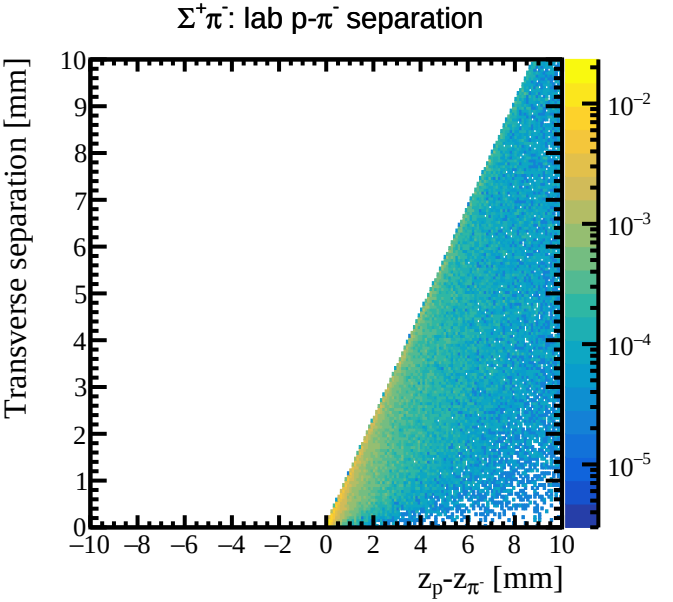

$\Sigma^0\pi^+\pi^-$: reconstructed Λ momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
$p_{y,\Lambda}$ [GeV/c]

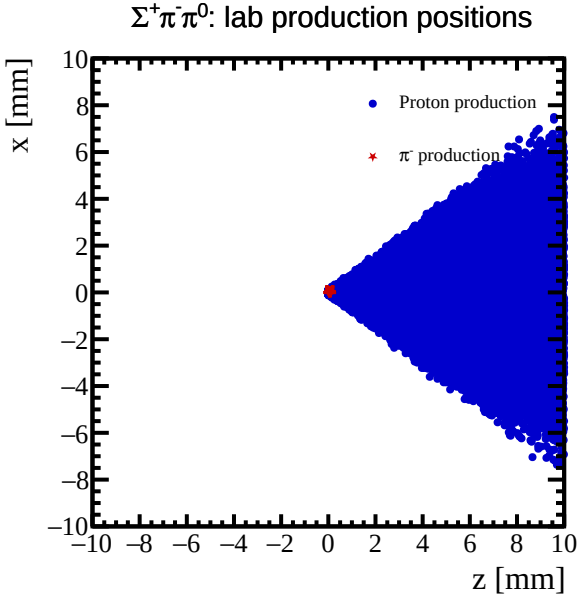

$\Sigma^0\pi^+\pi^-$: reconstructed Λ momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
$p_{z,\Lambda}$ [GeV/c]

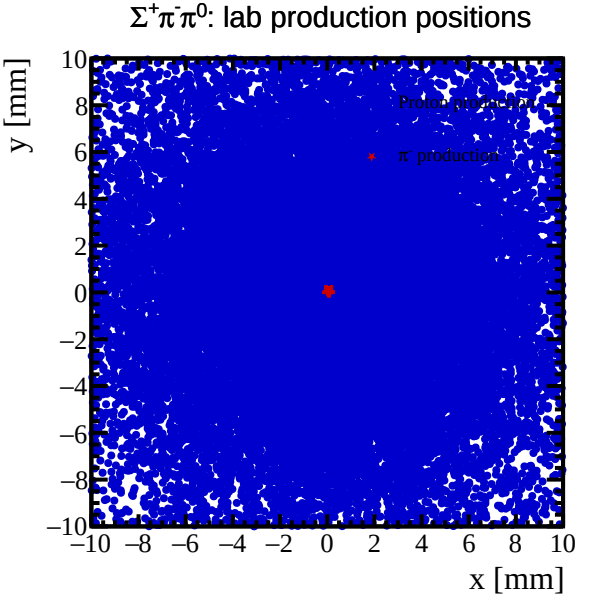

Σ+π-: lab production positions
Proton production
π- production
x [mm]
z [mm]

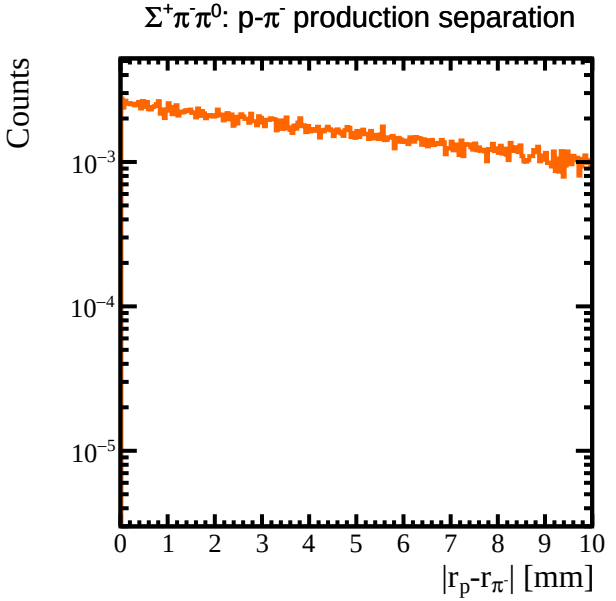

Σ+π-: lab production positions
Proton production
π- production
y [mm]
x [mm]

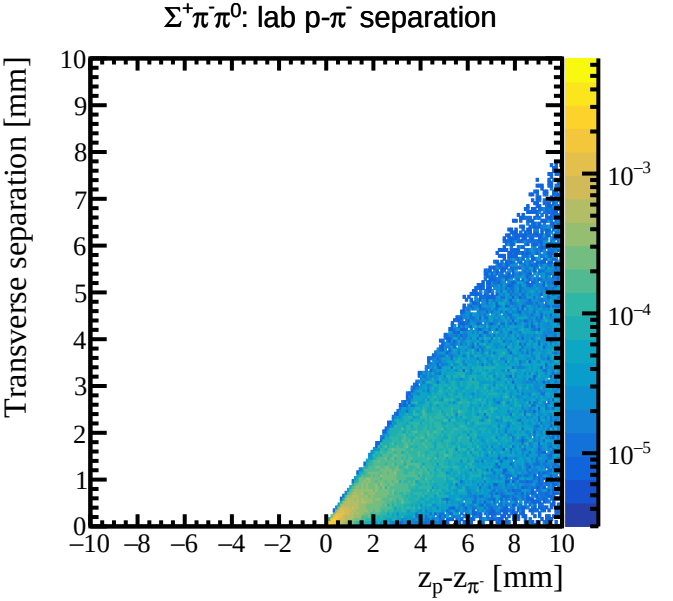

Σ+π-: p-π- production separation
Counts
|rp-rπ-| [mm]

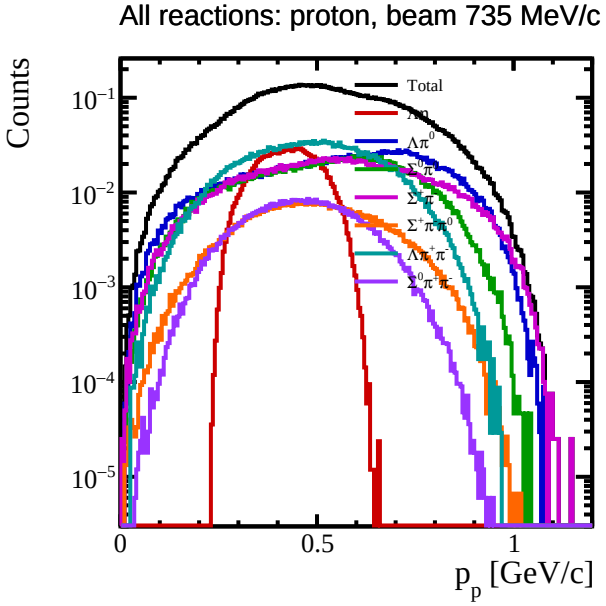

Σ+π-: lab p-π- separation
Transverse separation [mm]
zp-zπ- [mm]

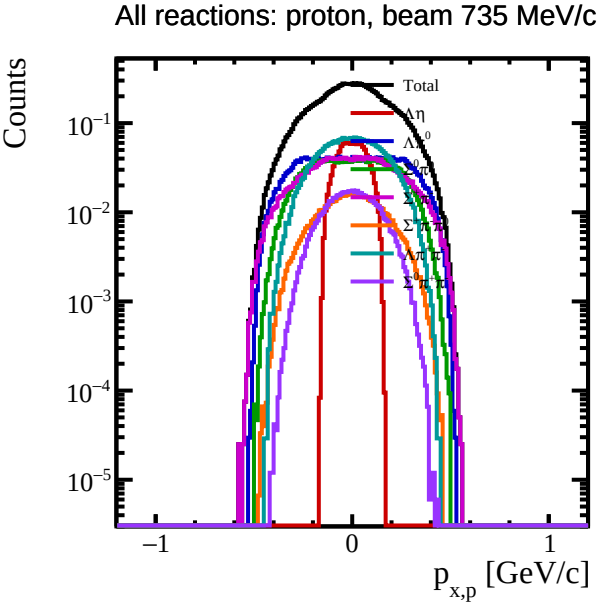

Σ+π-π0: lab production positions
Proton production
π- production
x [mm]
z [mm]

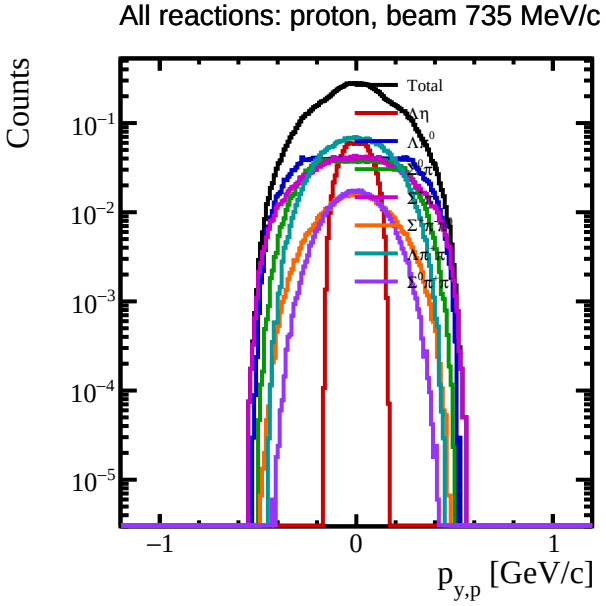

Σ+π-π0: lab production positions
Proton production
π- production
y [mm]
x [mm]

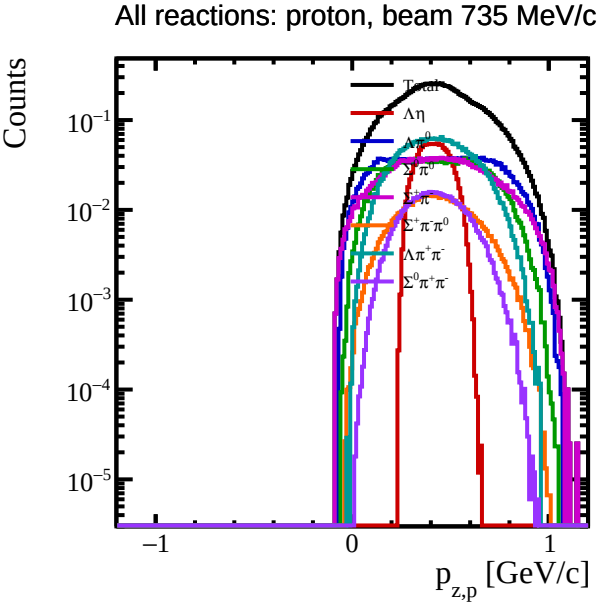

Σ+π-π0: p-π- production separation
Counts
|rp-rπ-| [mm]

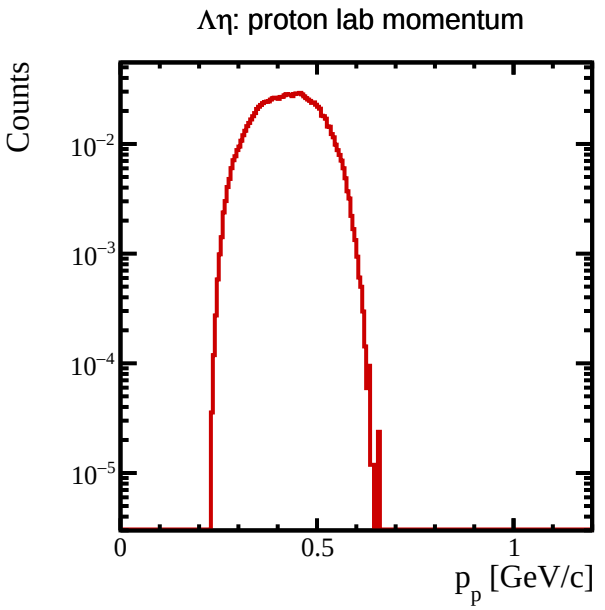

Σ+π-π0: lab p-π- separation
Transverse separation [mm]
zp-zπ- [mm]

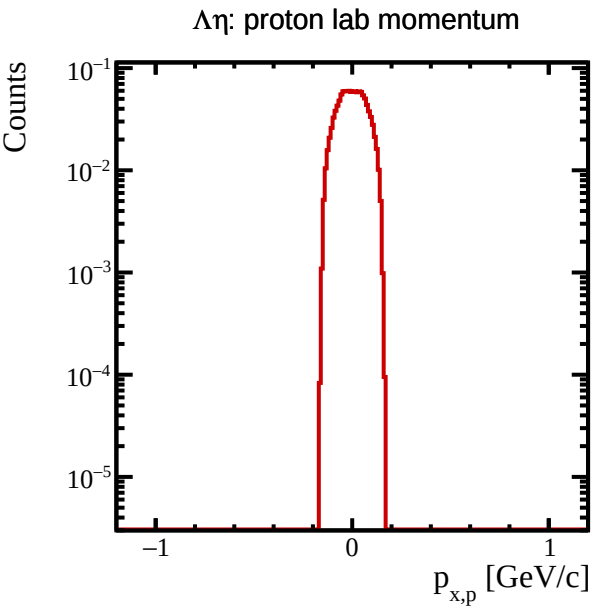

All reactions: proton, beam 735 MeV/c
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π−
Σ+π−π0
Λπ+π−
Σ0π+π−
p_p [GeV/c]

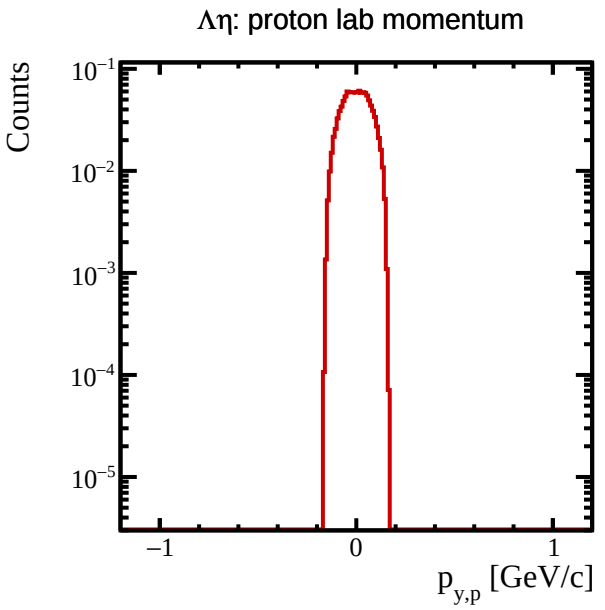

All reactions: proton, beam 735 MeV/c
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π−
Σ+π−π0
Λπ+π−
Σ0π+π−
p_x,p [GeV/c]

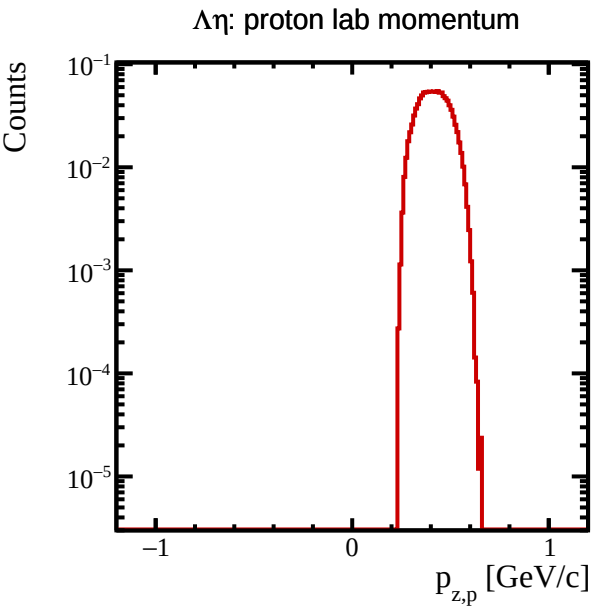

All reactions: proton, beam 735 MeV/c
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π−
Σ+π−π0
Λπ+π−
Σ0π+π−
p_y,p [GeV/c]

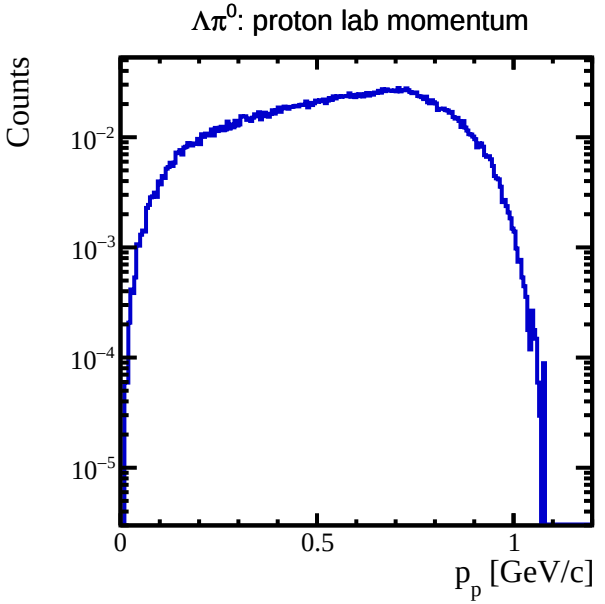

All reactions: proton, beam 735 MeV/c
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π−
Σ+π−π0
Λπ+π−
Σ0π+π−
p_z,p [GeV/c]

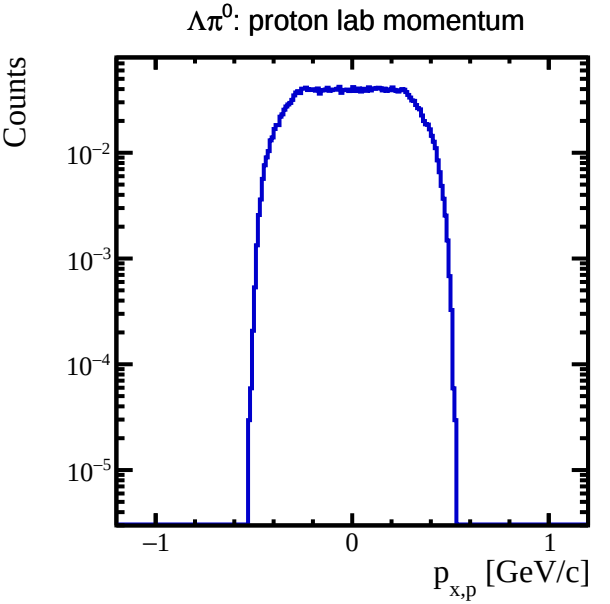

Λη: proton lab momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
0.5
1
p_p [GeV/c]

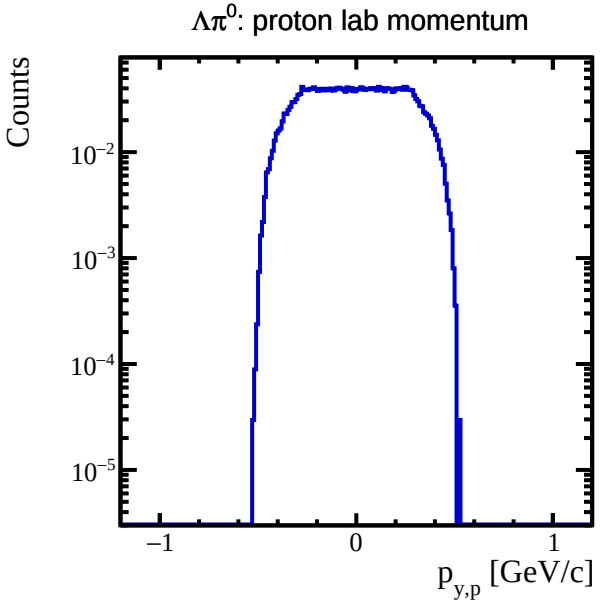

Λη: proton lab momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
$p_{x,p}$ [GeV/c]

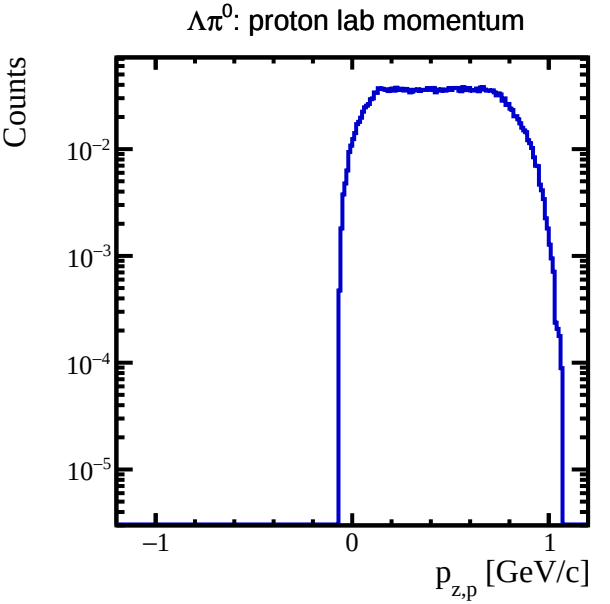

Λη: proton lab momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
$p_{y,p}$ [GeV/c]

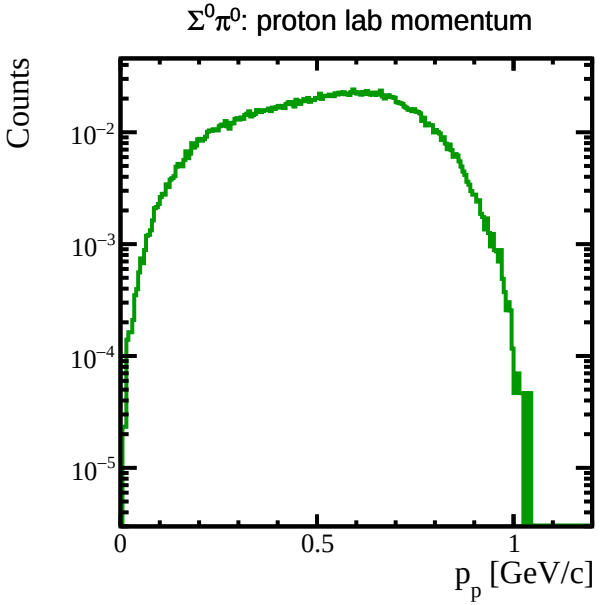

Λη: proton lab momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
$p_{z,p}$ [GeV/c]

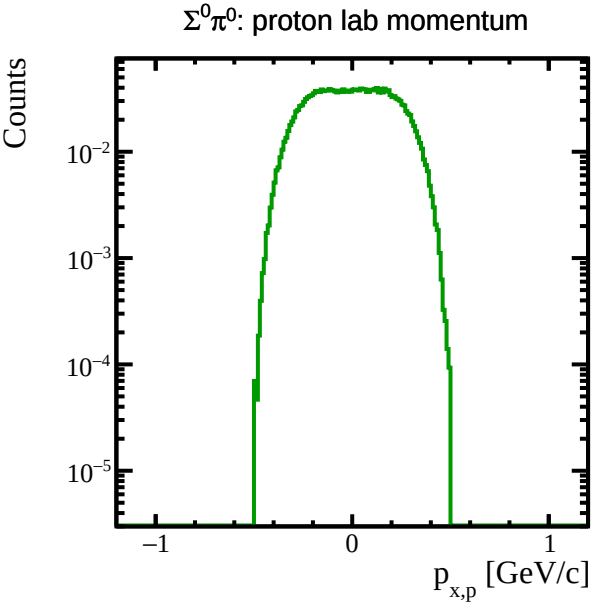

Λπ0: proton lab momentum
Counts
10−2
10−3
10−4
10−5
0
0.5
1
p_p [GeV/c]

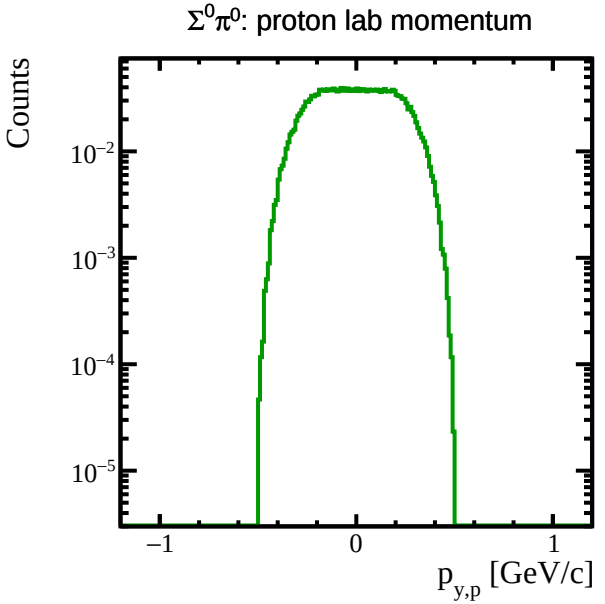

Λπ0: proton lab momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
p_x,p [GeV/c]

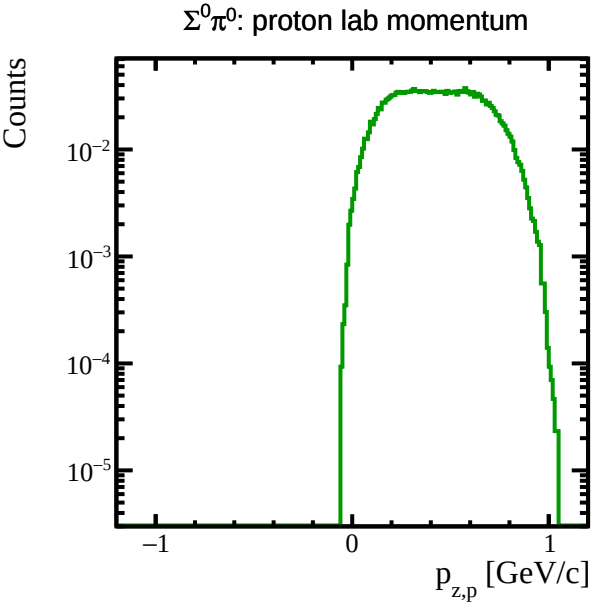

Λπ0: proton lab momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
p_y,p [GeV/c]

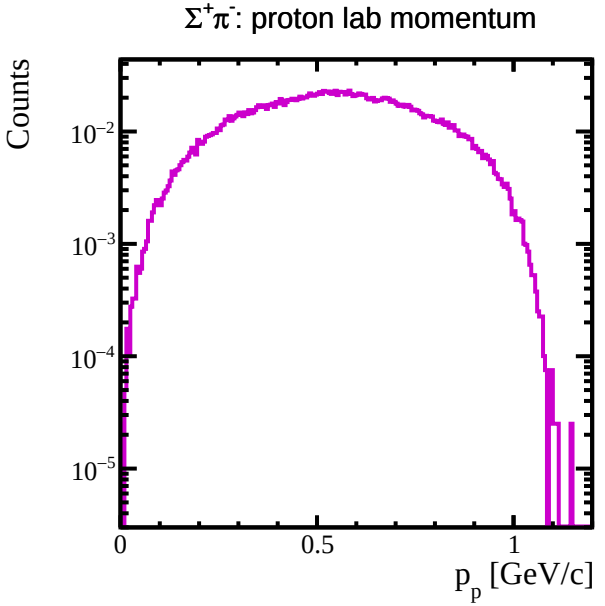

Λπ0: proton lab momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
p_z,p [GeV/c]

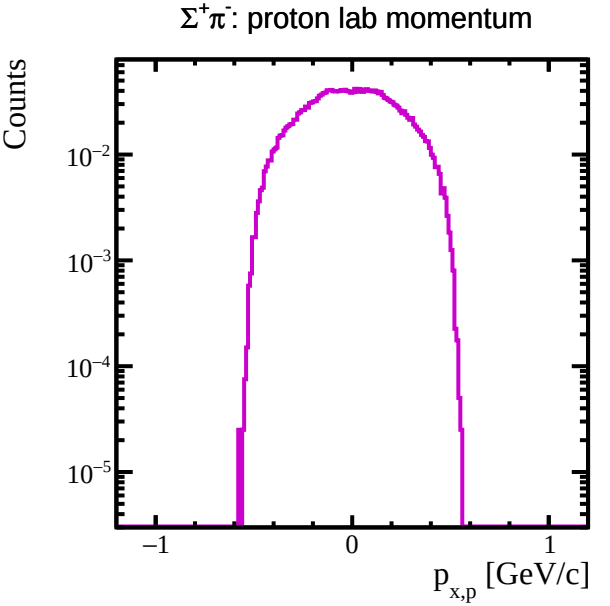

Σ⁰π⁰: proton lab momentum
Counts
10⁻²
10⁻³
10⁻⁴
10⁻⁵
0
0.5
1
p_p [GeV/c]

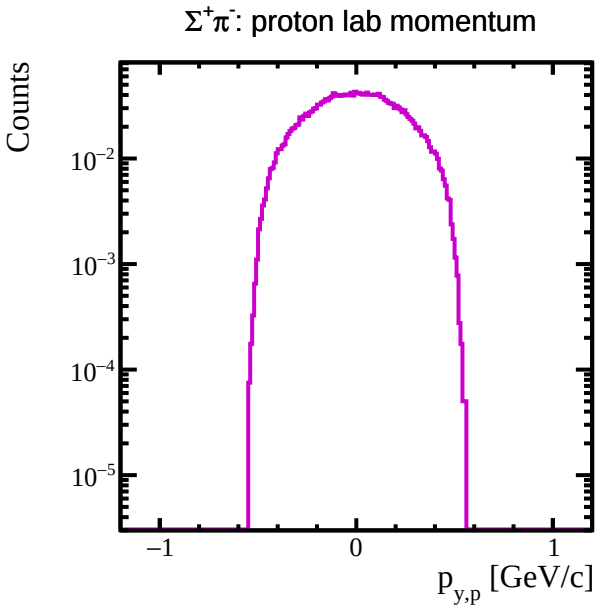

Σ⁰π⁰: proton lab momentum
Counts
10⁻²
10⁻³
10⁻⁴
10⁻⁵
−1
0
1
p_x,p [GeV/c]

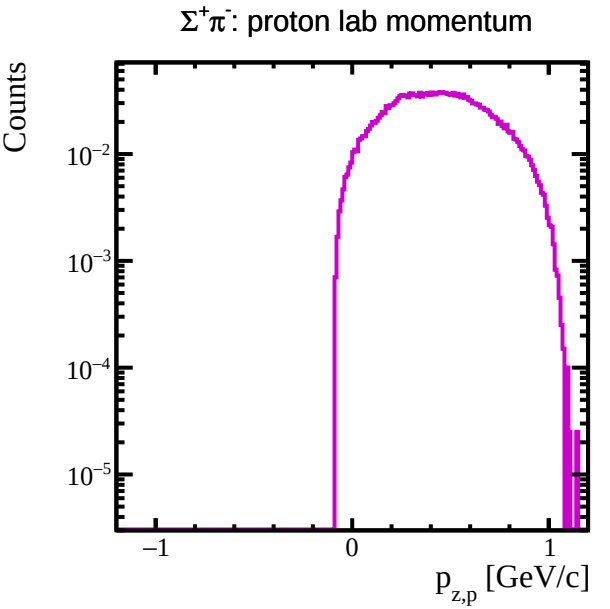

Σ⁰π⁰: proton lab momentum
Counts
10⁻²
10⁻³
10⁻⁴
10⁻⁵
−1
0
1
p_y,p [GeV/c]

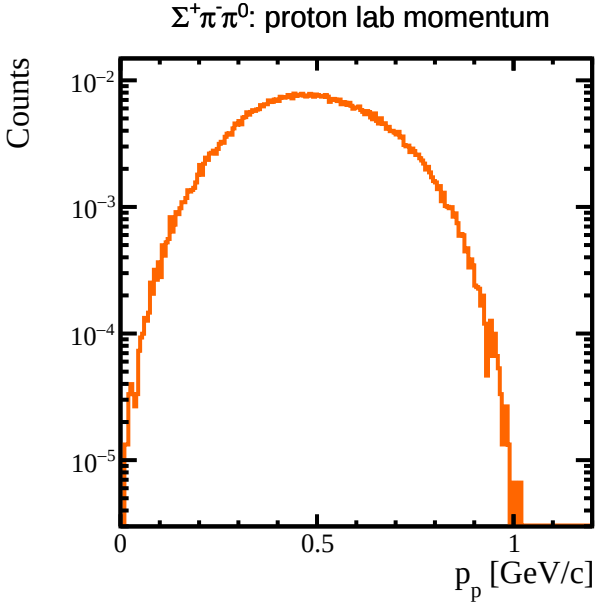

Σ⁰π⁰: proton lab momentum
Counts
10⁻²
10⁻³
10⁻⁴
10⁻⁵
−1
0
1
p_z,p [GeV/c]

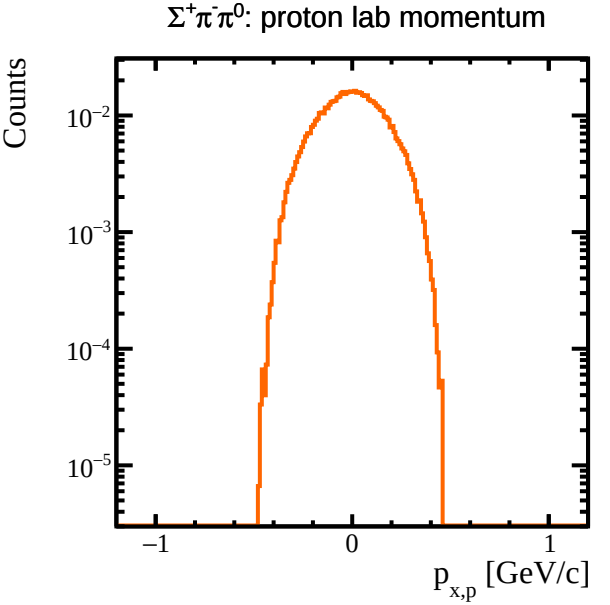

Σ+π-: proton lab momentum
Counts
10−2
10−3
10−4
10−5
0
0.5
1
p_p [GeV/c]

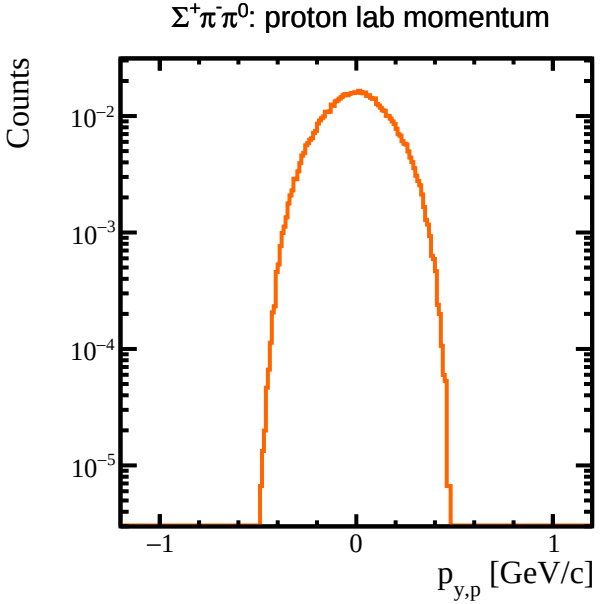

Σ+π-: proton lab momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
p_x,p [GeV/c]

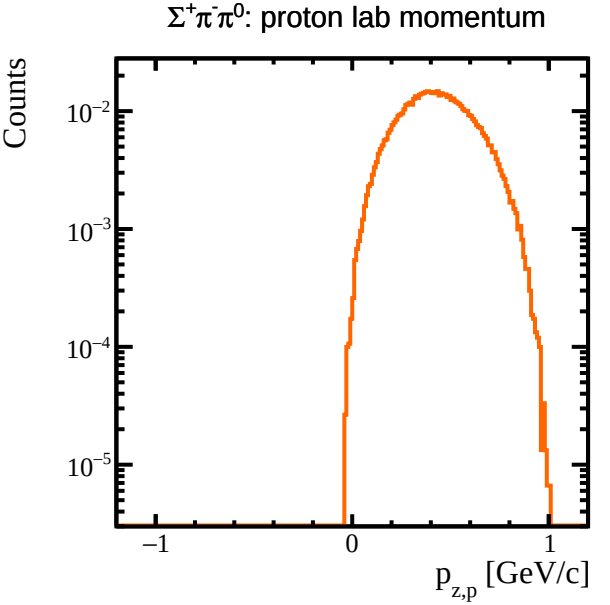

Σ+π-: proton lab momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
p_y,p [GeV/c]

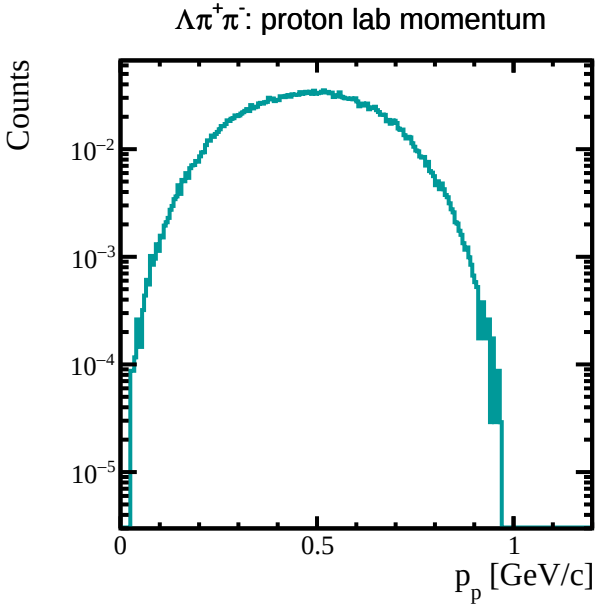

Σ+π-: proton lab momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
p_z,p [GeV/c]

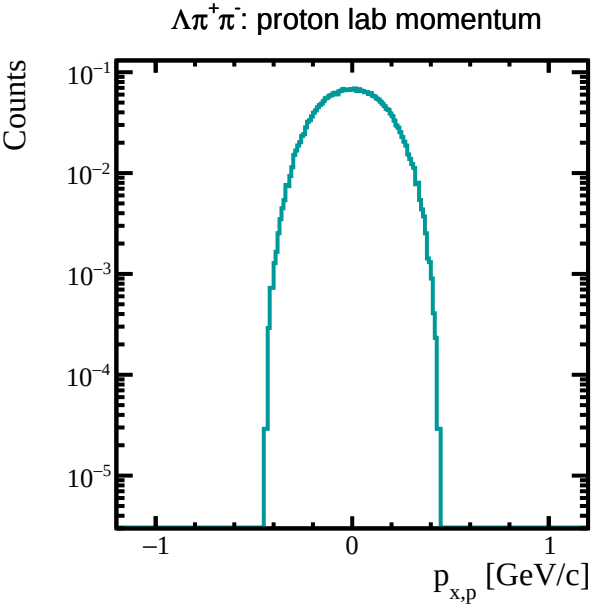

Σ+π-π0: proton lab momentum
Counts
10−2
10−3
10−4
10−5
0
0.5
1
p_p [GeV/c]

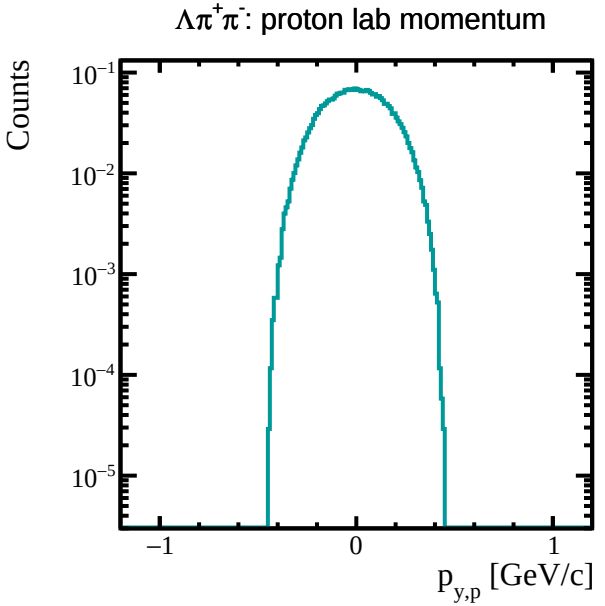

Σ+π-π0: proton lab momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
p_x,p [GeV/c]

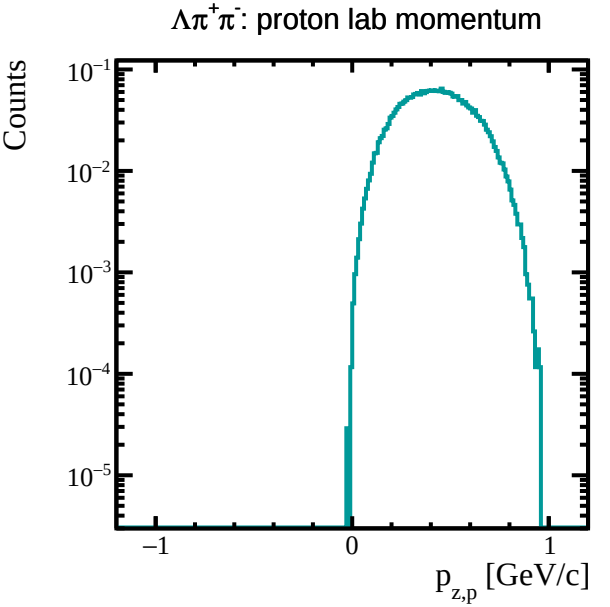

Σ+π-π0: proton lab momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
p_y,p [GeV/c]

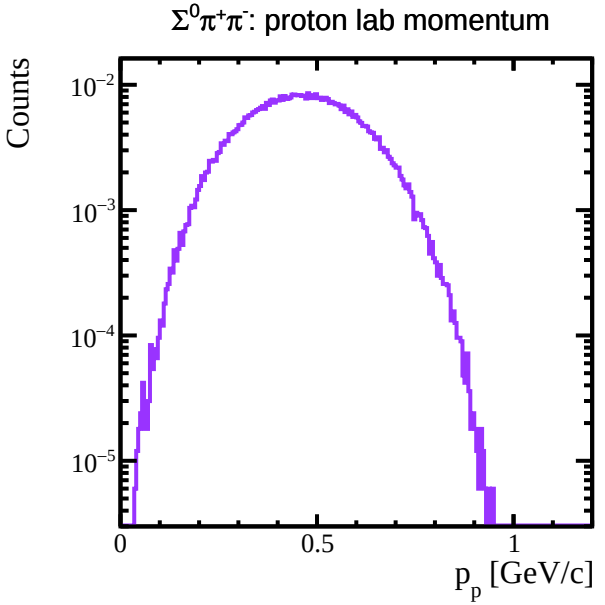

Σ+π-π0: proton lab momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
p_z,p [GeV/c]

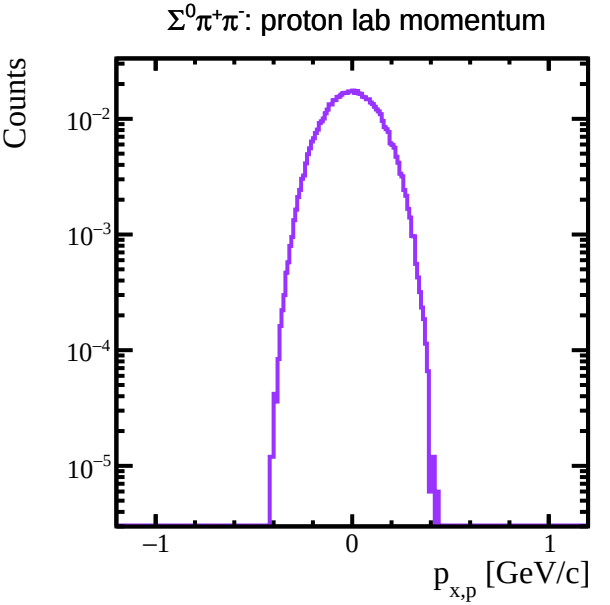

Λπ+π-: proton lab momentum
Counts
10−2
10−3
10−4
10−5
0
0.5
1
p_p [GeV/c]

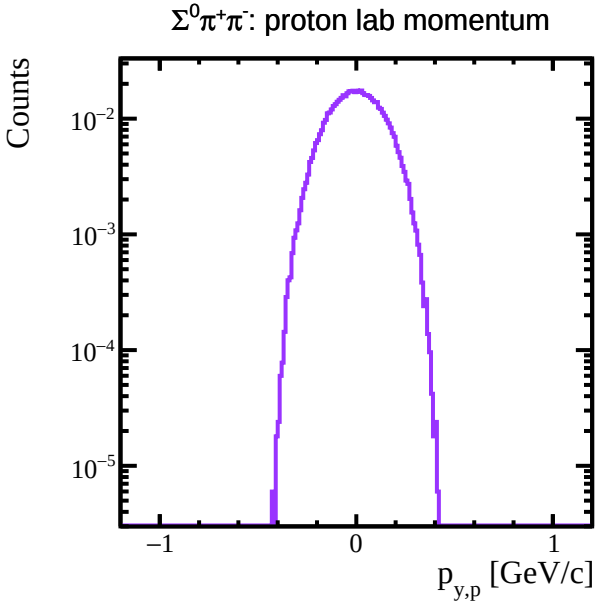

Λπ+π-: proton lab momentum
Counts
10−1
10−2
10−3
10−4
10−5
−1
0
1
p_x,p [GeV/c]

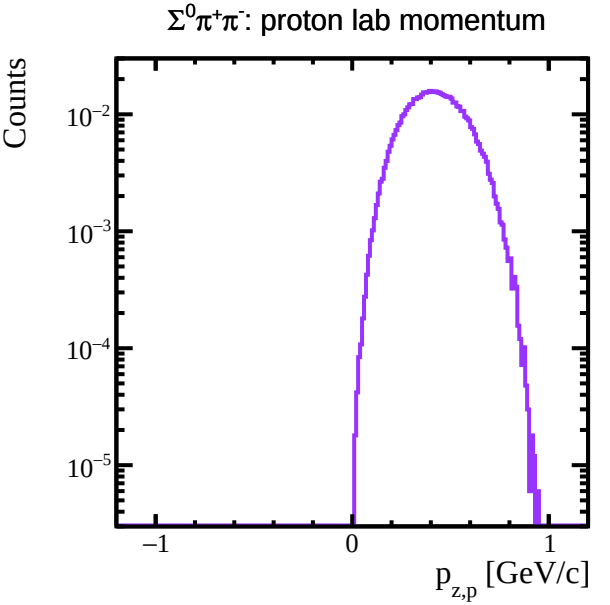

Λπ+π-: proton lab momentum
Counts
10−1
10−2
10−3
10−4
10−5
−1
0
1
p_y,p [GeV/c]

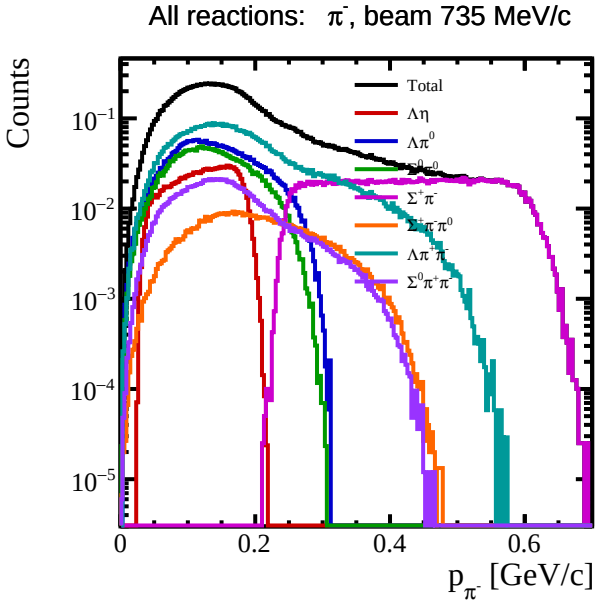

Λπ+π-: proton lab momentum
Counts
10−1
10−2
10−3
10−4
10−5
−1
0
1
p_z,p [GeV/c]

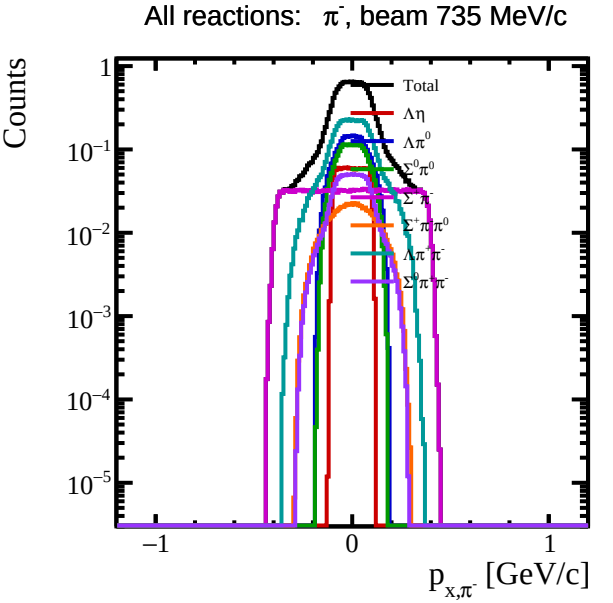

$\Sigma^0\pi^+\pi^-$: proton lab momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
0.5
1
p_p [GeV/c]

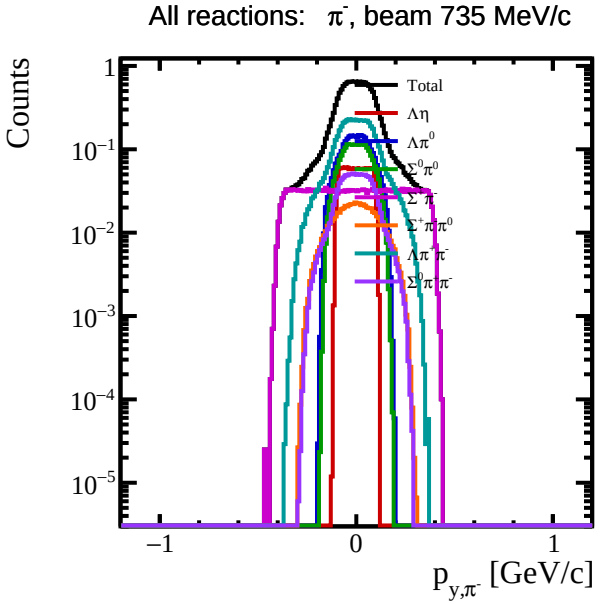

$\Sigma^0\pi^+\pi^-$: proton lab momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
$p_{x,p}$ [GeV/c]

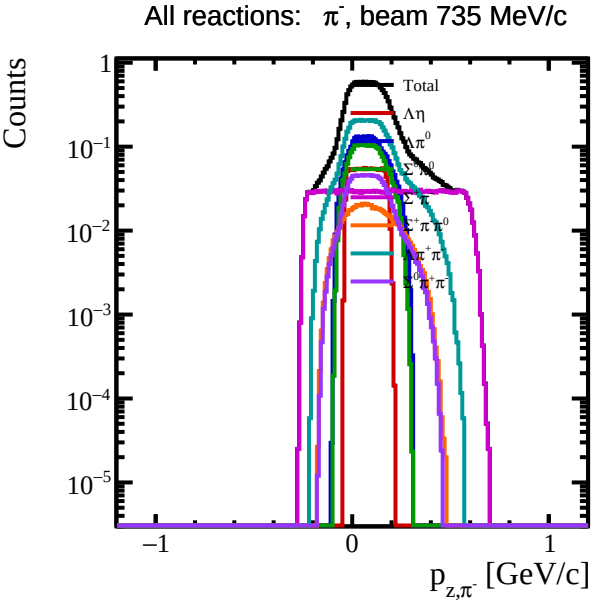

$\Sigma^0\pi^+\pi^-$: proton lab momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
$p_{y,p}$ [GeV/c]

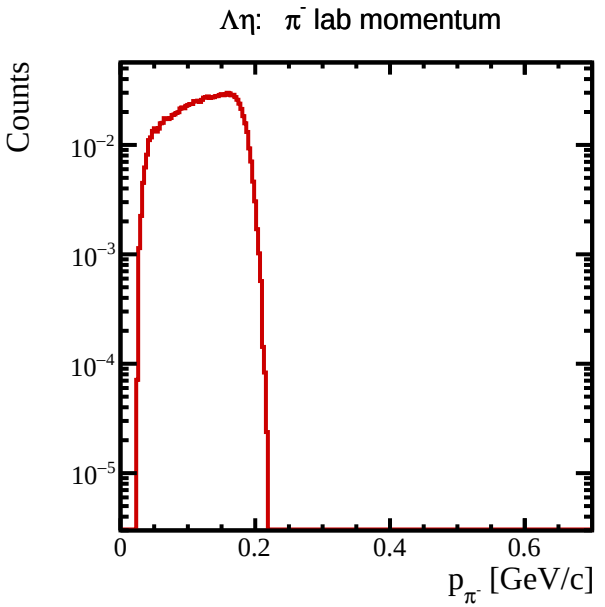

$\Sigma^0\pi^+\pi^-$: proton lab momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
$p_{z,p}$ [GeV/c]

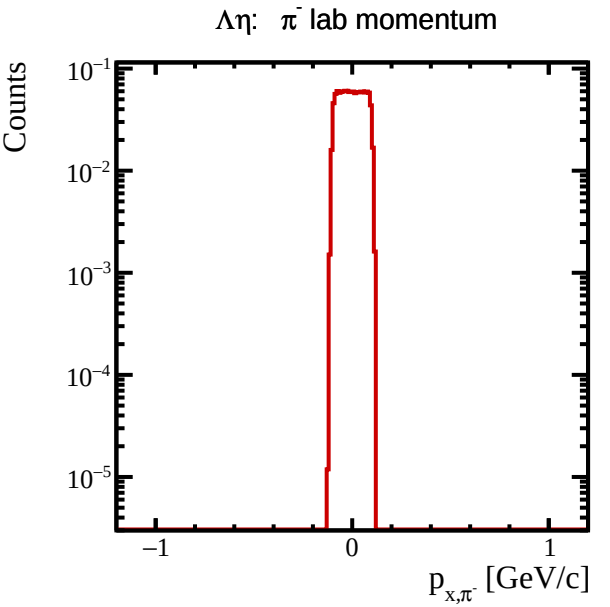

All reactions: π⁻, beam 735 MeV/c
Counts
Total
Λη
Λπ⁰
Σ⁰π⁰
Σ⁺π⁻
Σ⁺π⁻π⁰
Λπ⁺π⁻
Σ⁰π⁺π⁻
p_π⁻ [GeV/c]

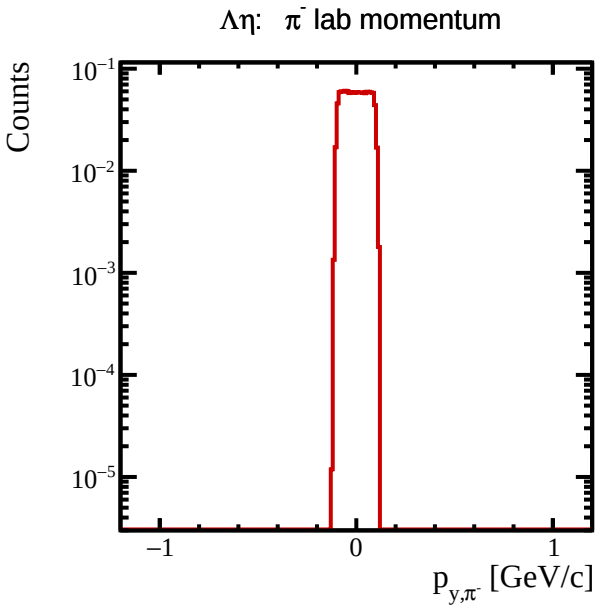

All reactions: π⁻, beam 735 MeV/c
Counts
Total
Λη
Λπ⁰
Σ⁰π⁰
Σ⁺π⁻
Σ⁺π⁻π⁰
Λπ⁺π⁻
Σ⁰π⁺π⁻
p_x,π⁻ [GeV/c]

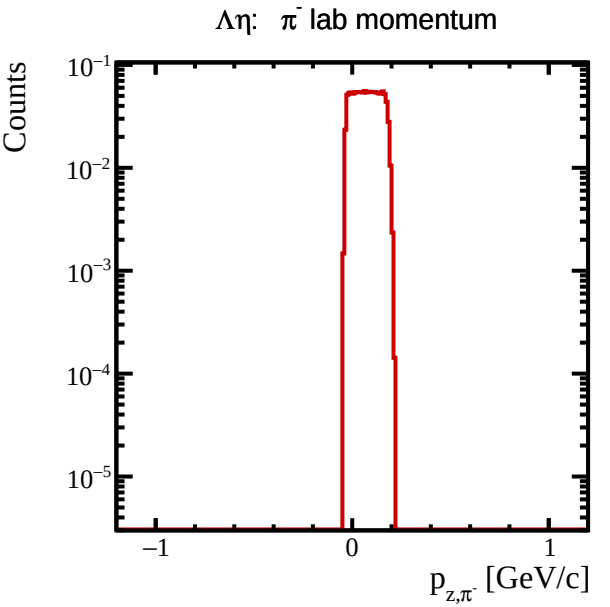

All reactions: π⁻, beam 735 MeV/c
Counts
Total
Λη
Λπ⁰
Σ⁰π⁰
Σ⁺π⁻
Σ⁺π⁻π⁰
Λπ⁺π⁻
Σ⁰π⁺π⁻
p_y,π⁻ [GeV/c]

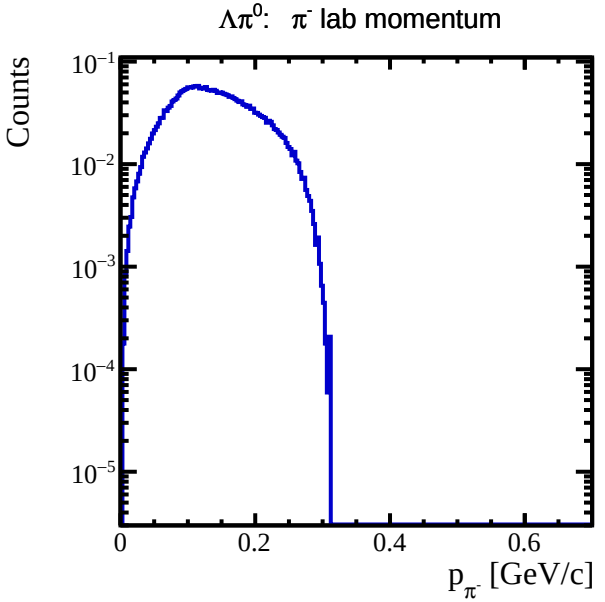

All reactions: π⁻, beam 735 MeV/c
Counts
Total
Λη
Λπ⁰
Σ⁰π⁰
Σ⁺π⁻
Σ⁺π⁻π⁰
Λπ⁺π⁻
Σ⁰π⁺π⁻
p_z,π⁻ [GeV/c]

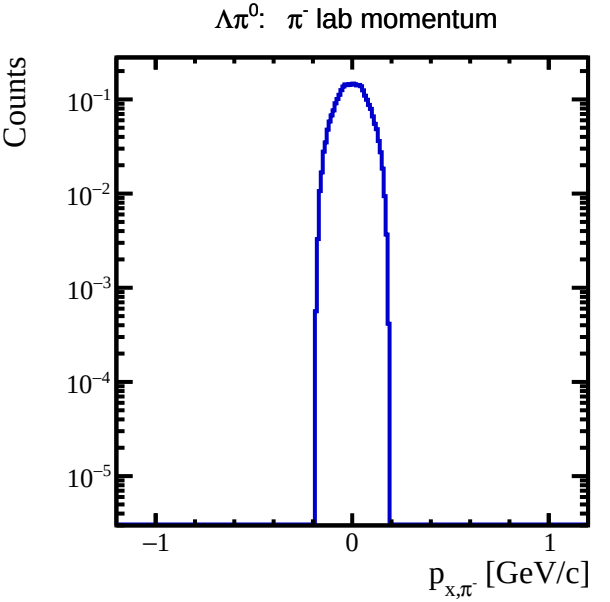

Λη: π⁻ lab momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
0.2
0.4
0.6
p_{π^-} [GeV/c]

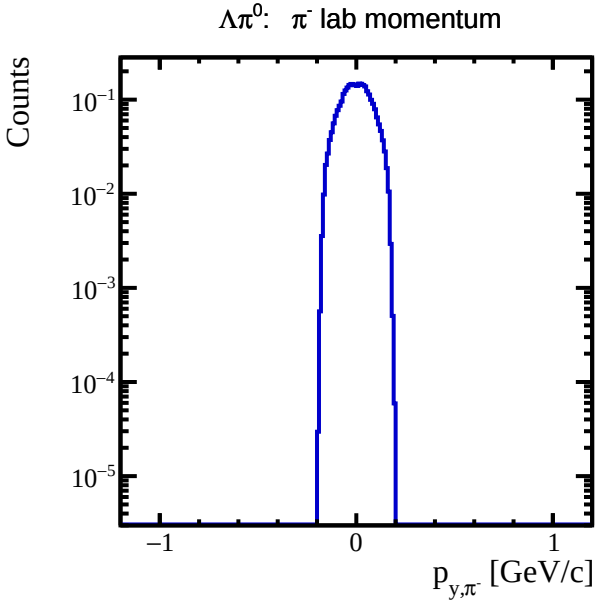

Λη: π⁻ lab momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
p_{x,π^-} [GeV/c]

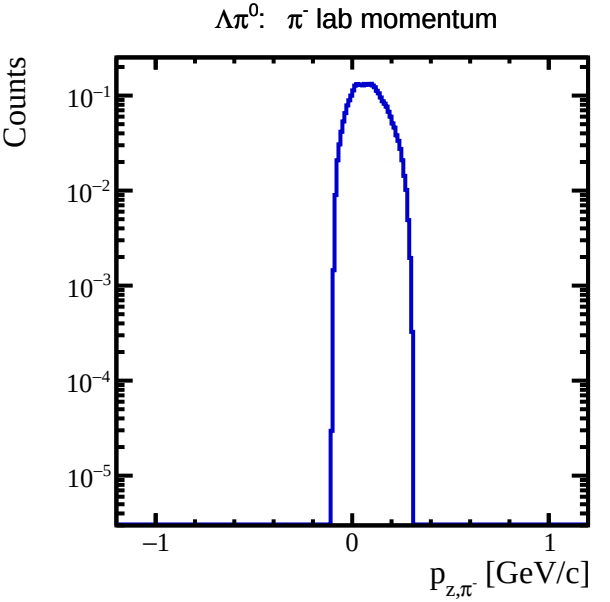

Λη: π⁻ lab momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
p_{y,π^-} [GeV/c]

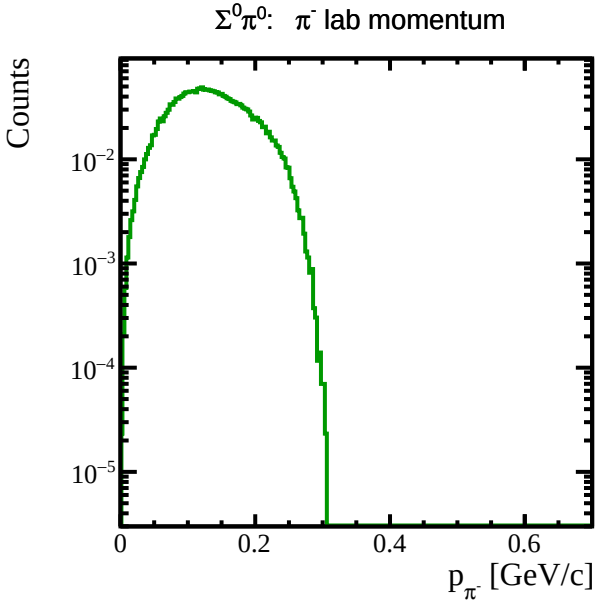

Λη: π⁻ lab momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
p_{z,π^-} [GeV/c]

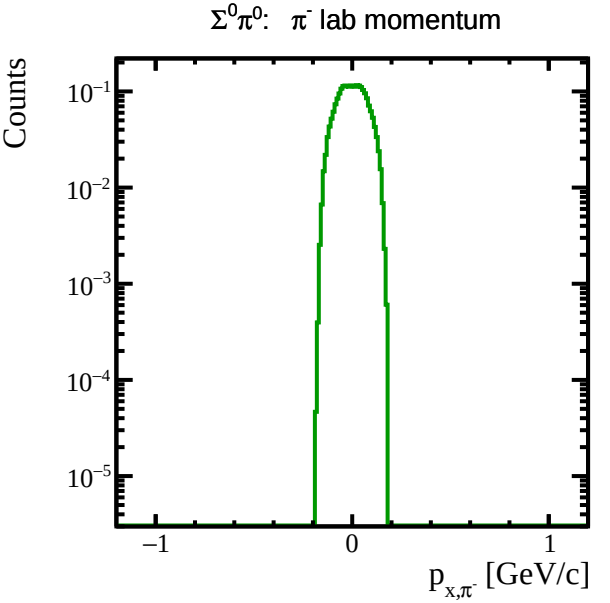

$\Lambda\pi^0$: π^- lab momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
0.2
0.4
0.6
p_{π^-} [GeV/c]

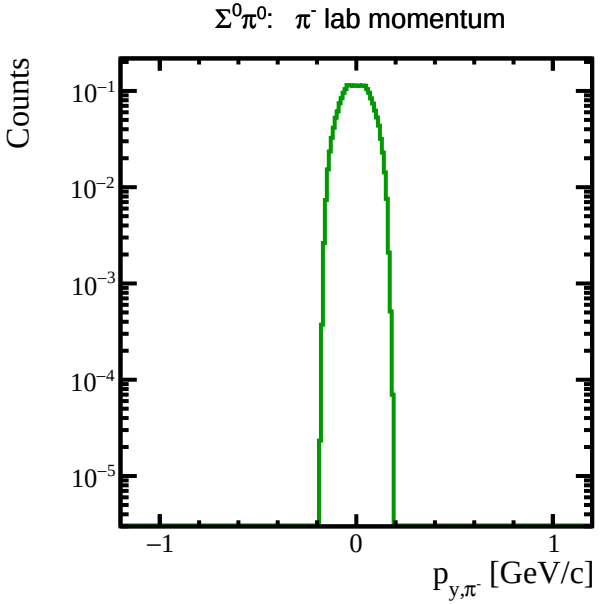

$\Lambda\pi^0$: π^- lab momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
p_{x,π^-} [GeV/c]

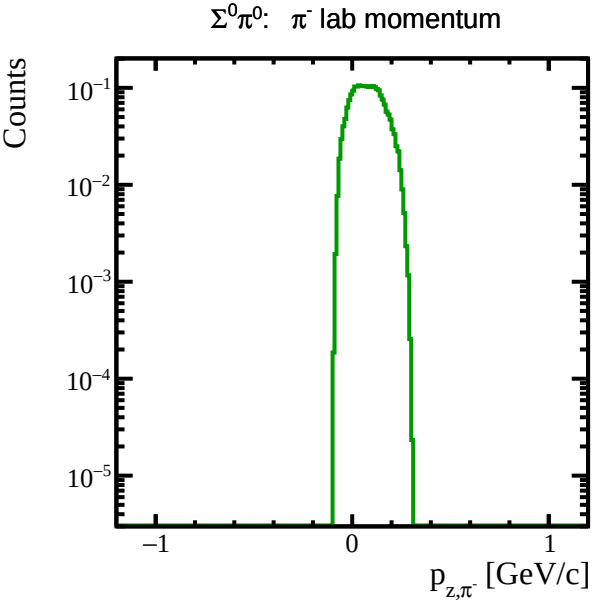

$\Lambda\pi^0$: π^- lab momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
p_{y,π^-} [GeV/c]

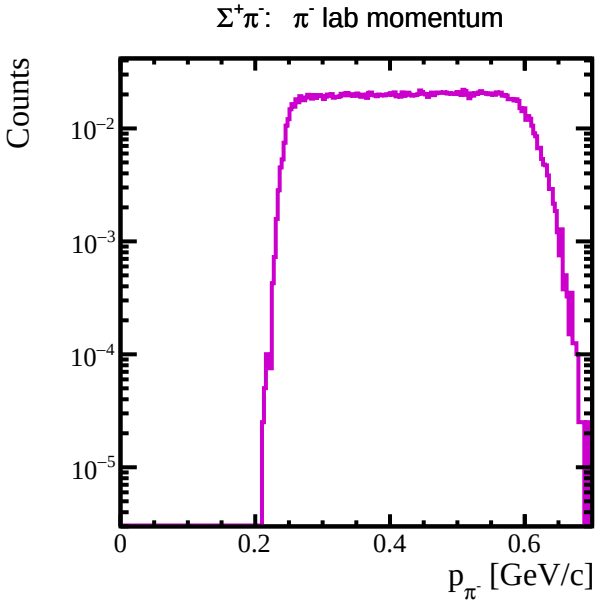

$\Lambda\pi^0$: π^- lab momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
p_{z,π^-} [GeV/c]

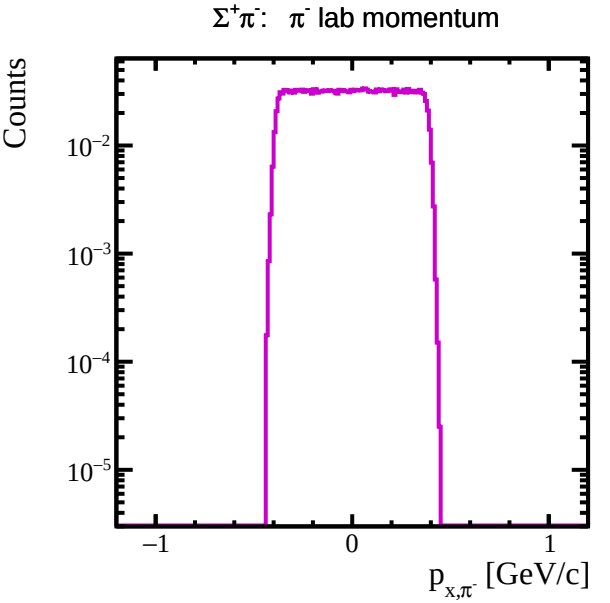

$\Sigma^0\pi^0$: π^- lab momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
0.2
0.4
0.6
p_{π^-} [GeV/c]

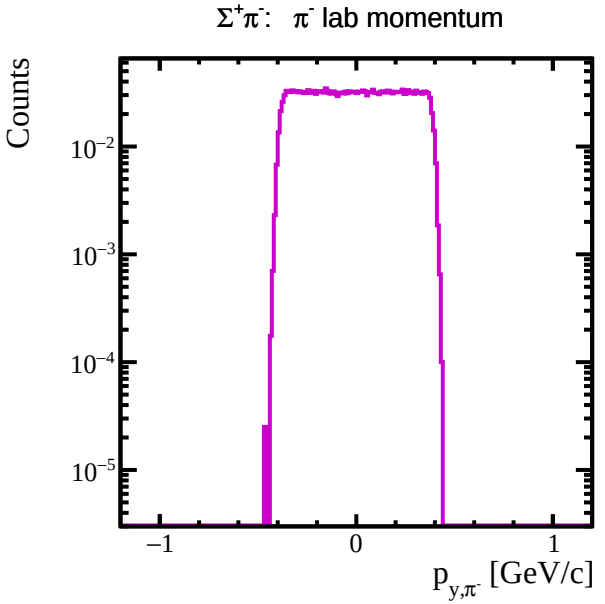

$\Sigma^0\pi^0$: π^- lab momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
p_{x,π^-} [GeV/c]

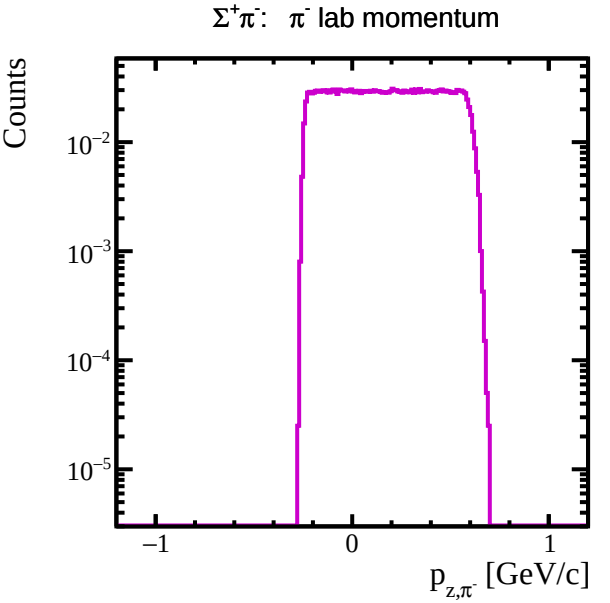

$\Sigma^0\pi^0$: π^- lab momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
p_{y,π^-} [GeV/c]

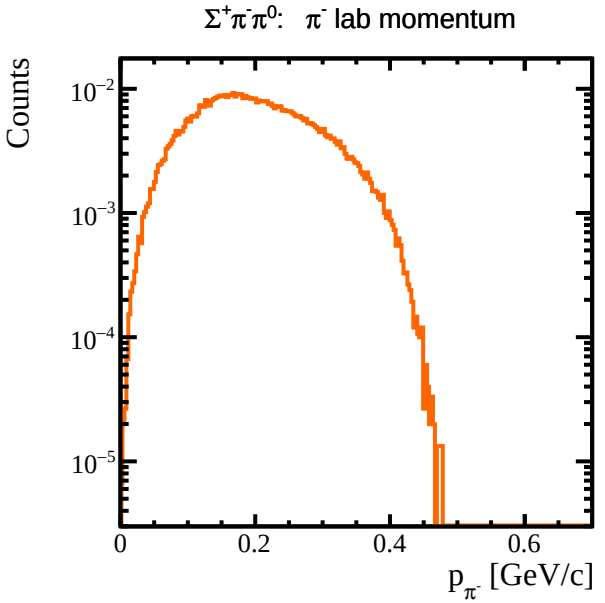

$\Sigma^0\pi^0$: π^- lab momentum
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
p_{z,π^-} [GeV/c]

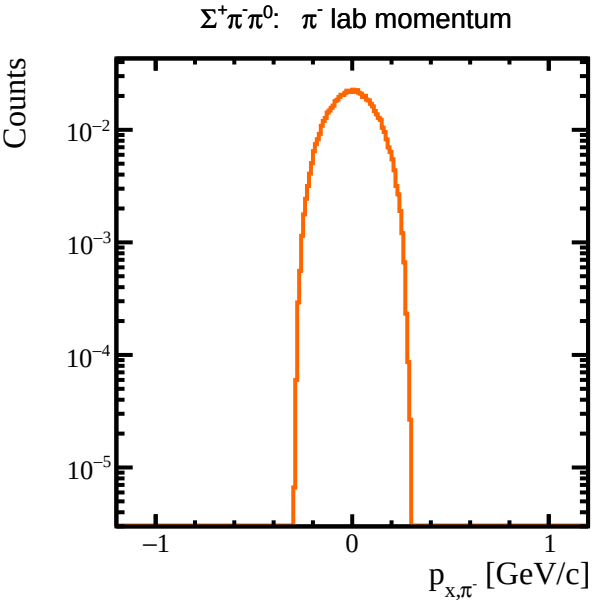

Σ+π-: π- lab momentum
Counts
10−2
10−3
10−4
10−5
0
0.2
0.4
0.6
pπ- [GeV/c]

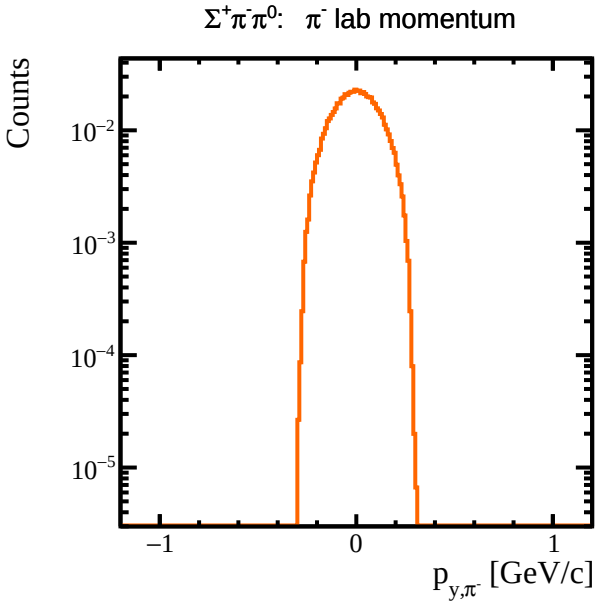

Σ+π-: π- lab momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
px,π- [GeV/c]

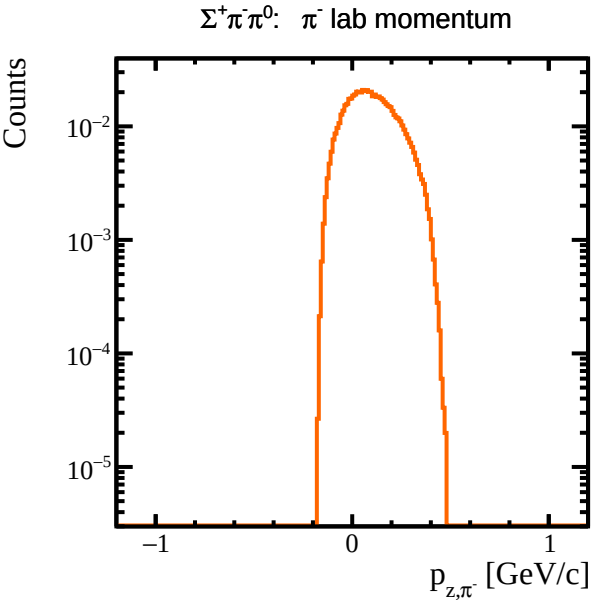

Σ+π-: π- lab momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
py,π- [GeV/c]

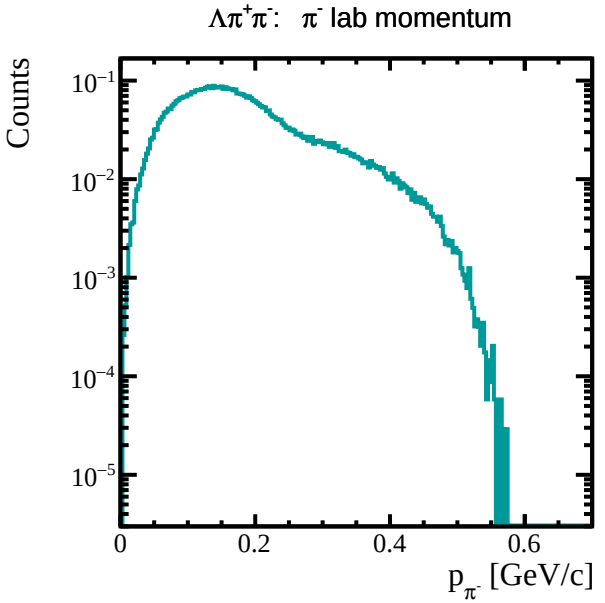

Σ+π-: π- lab momentum
Counts
10−2
10−3
10−4
10−5
−1
0
1
pz,π- [GeV/c]

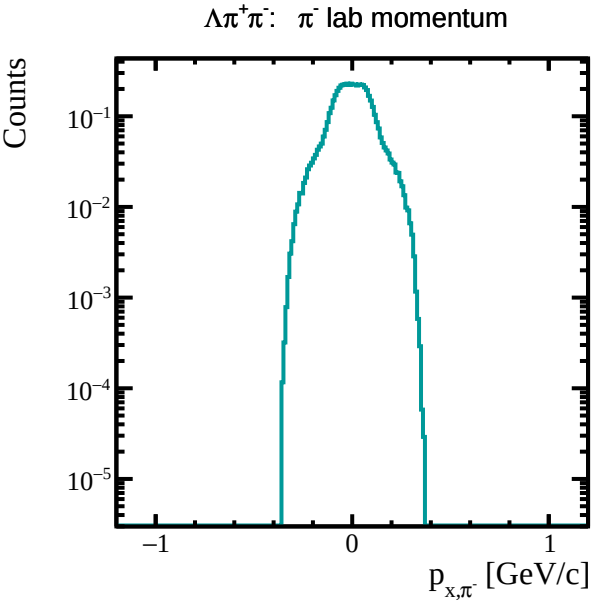

$\Sigma^+\pi^-\pi^0$: π^- lab momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
0.2
0.4
0.6
p_{π^-} [GeV/c]

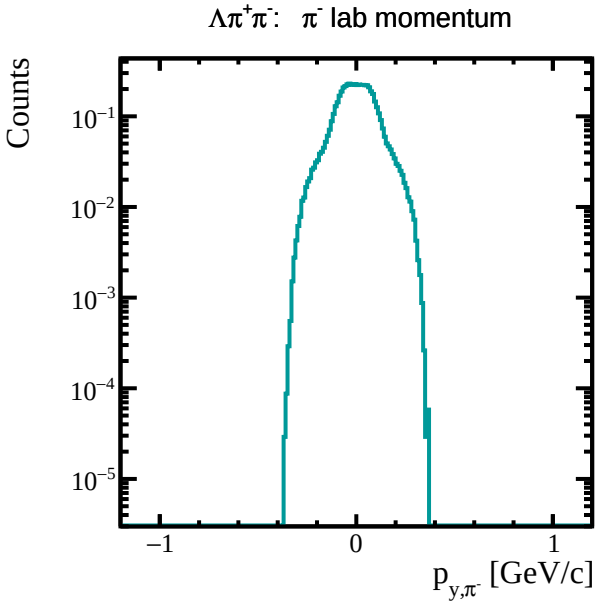

$\Sigma^+\pi^-\pi^0$: π^- lab momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
p_{x,π^-} [GeV/c]

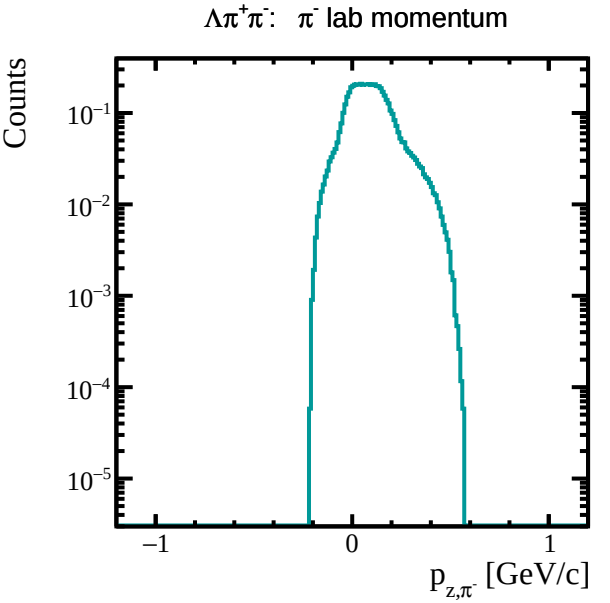

$\Sigma^+\pi^-\pi^0$: π^- lab momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
p_{y,π^-} [GeV/c]

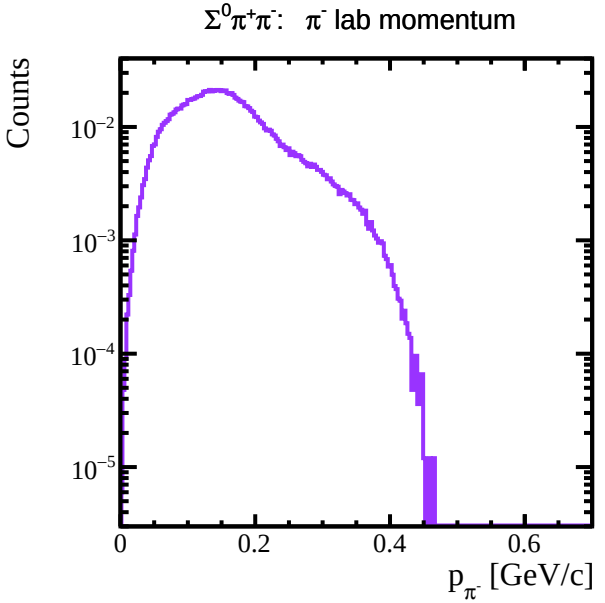

$\Sigma^+\pi^-\pi^0$: π^- lab momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
p_{z,π^-} [GeV/c]

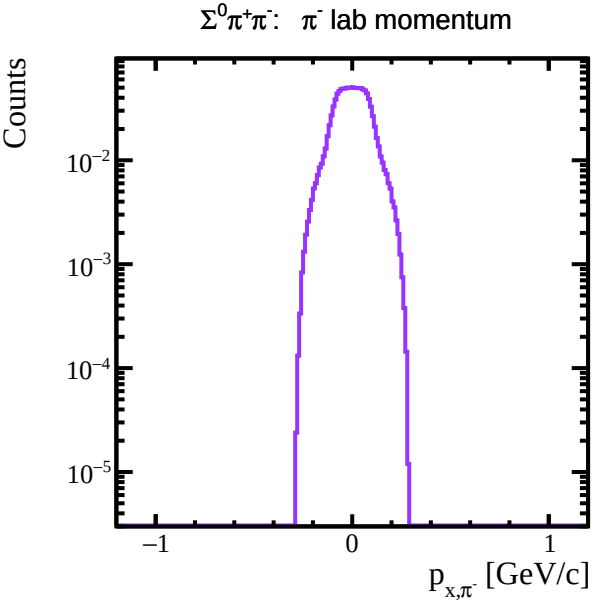

Λπ+π-: π- lab momentum
Counts
10−1
10−2
10−3
10−4
10−5
0
0.2
0.4
0.6
pπ- [GeV/c]

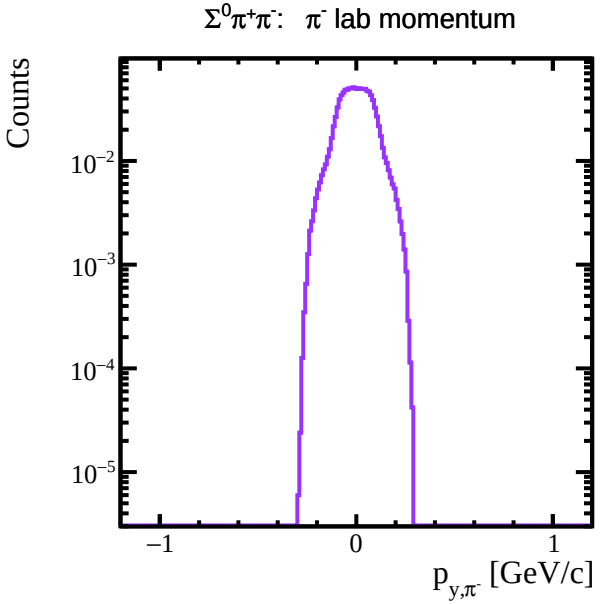

Λπ+π-: π- lab momentum
Counts
10−1
10−2
10−3
10−4
10−5
−1
0
1
px,π- [GeV/c]

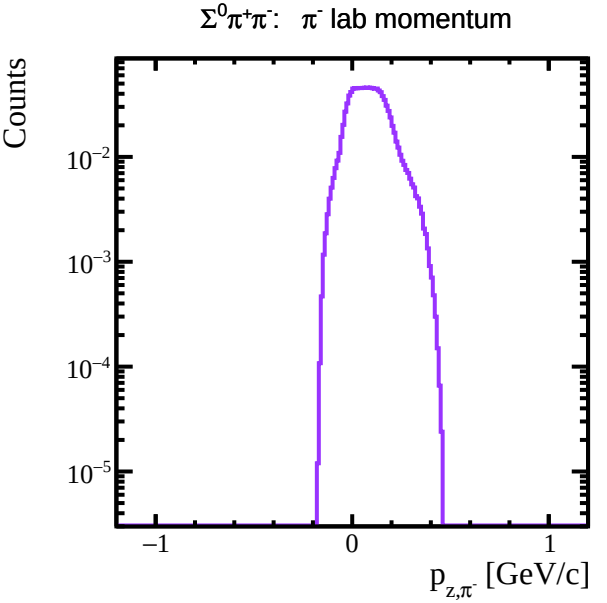

Λπ+π-: π- lab momentum
Counts
10−1
10−2
10−3
10−4
10−5
−1
0
1
py,π- [GeV/c]

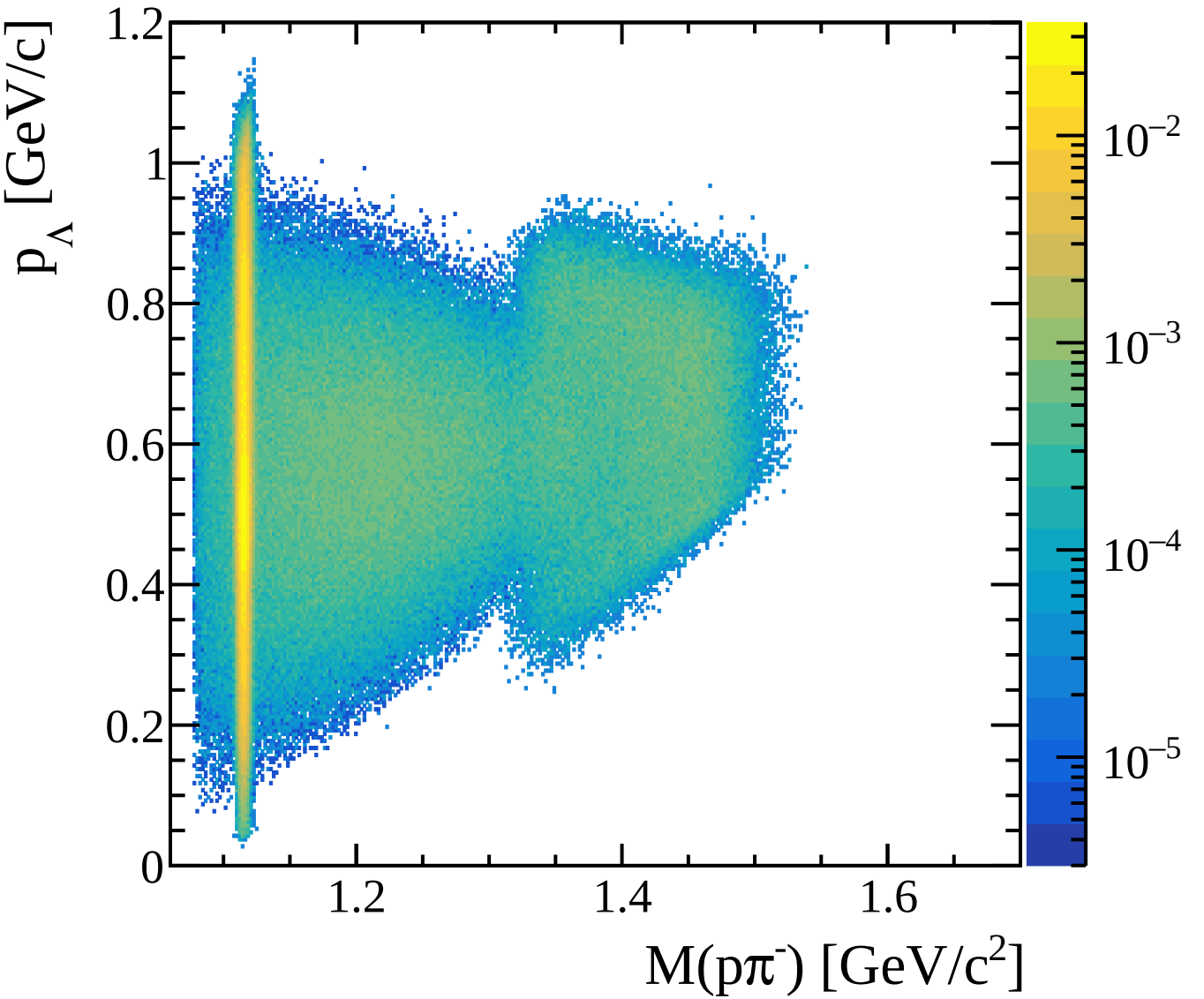

Λπ+π-: π- lab momentum
Counts
10−1
10−2
10−3
10−4
10−5
−1
0
1
pz,π- [GeV/c]

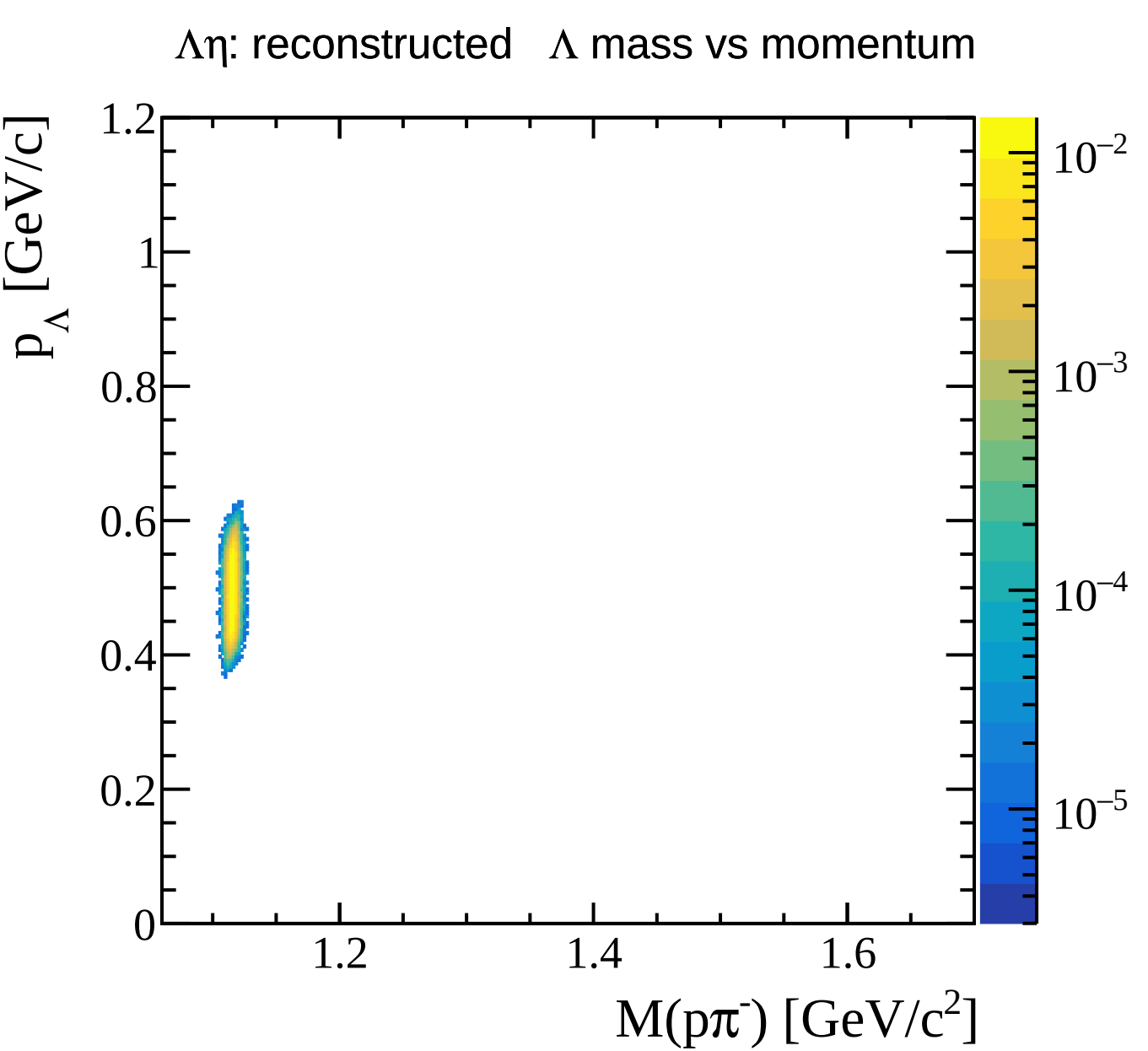

$\Sigma^0\pi^+\pi^-$: π^- lab momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
0.2
0.4
0.6
p_{π^-} [GeV/c]

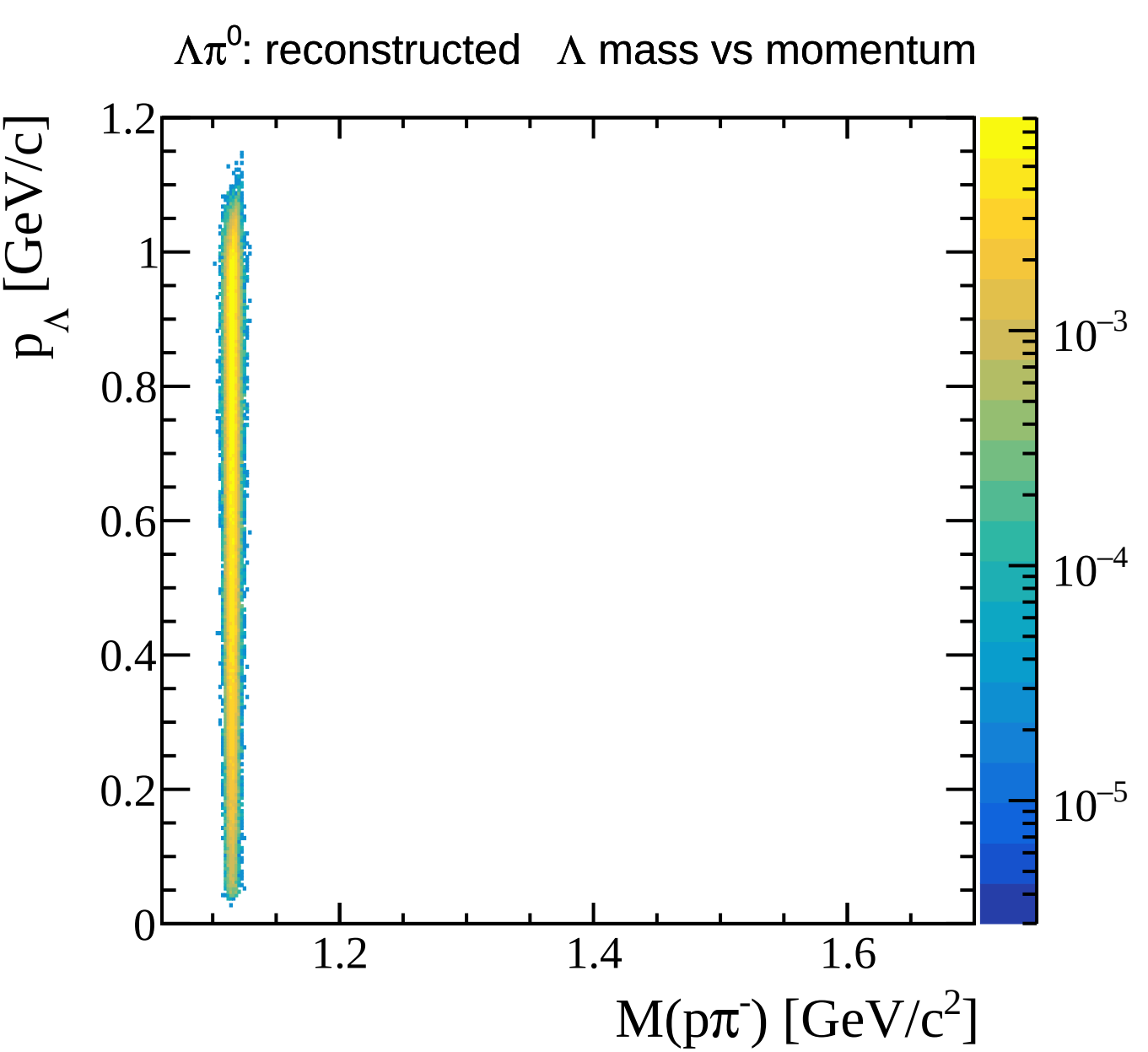

$\Sigma^0\pi^+\pi^-$: π^- lab momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
p_{x,π^-} [GeV/c]

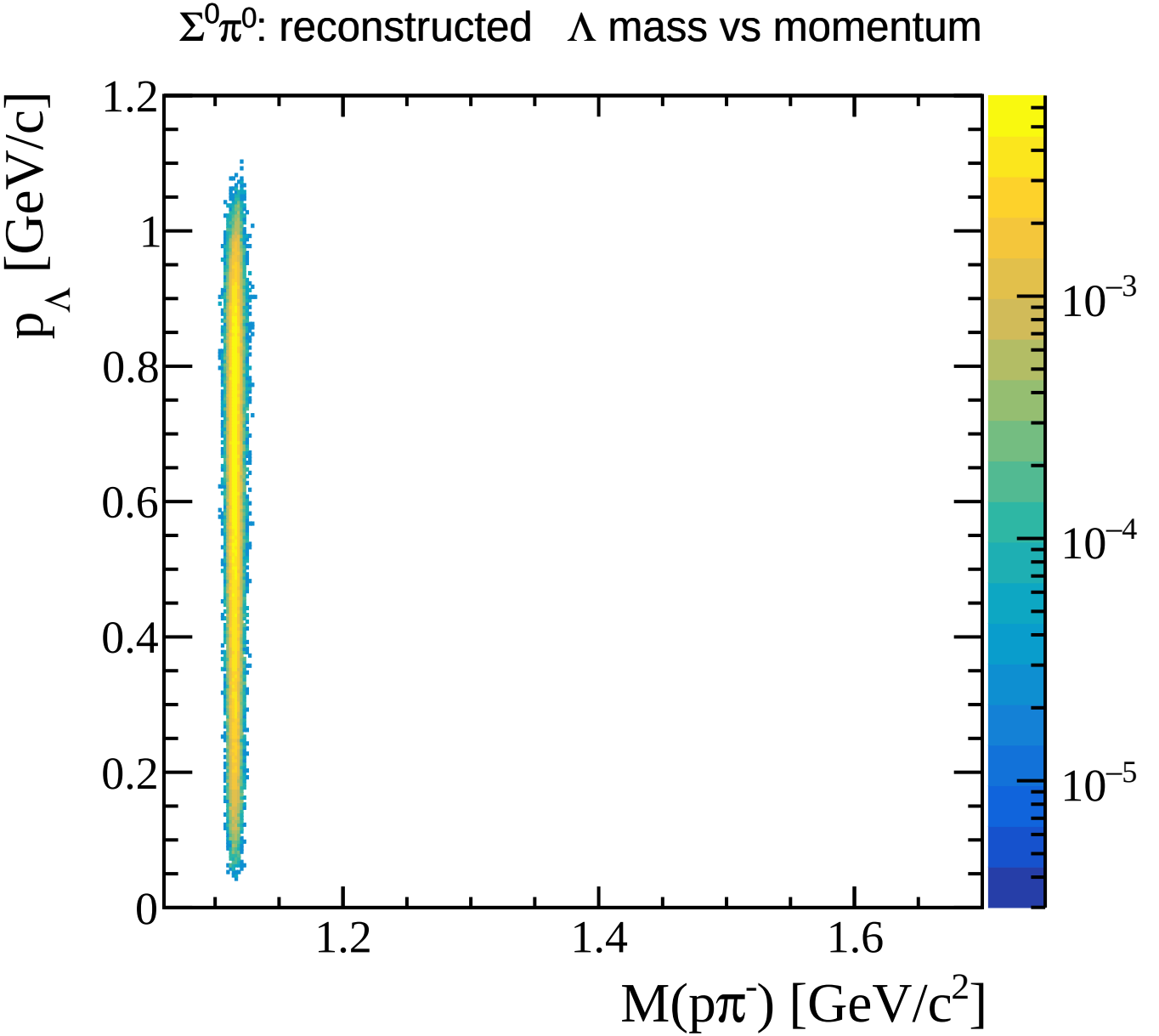

$\Sigma^0\pi^+\pi^-$: π^- lab momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
p_{y,π^-} [GeV/c]

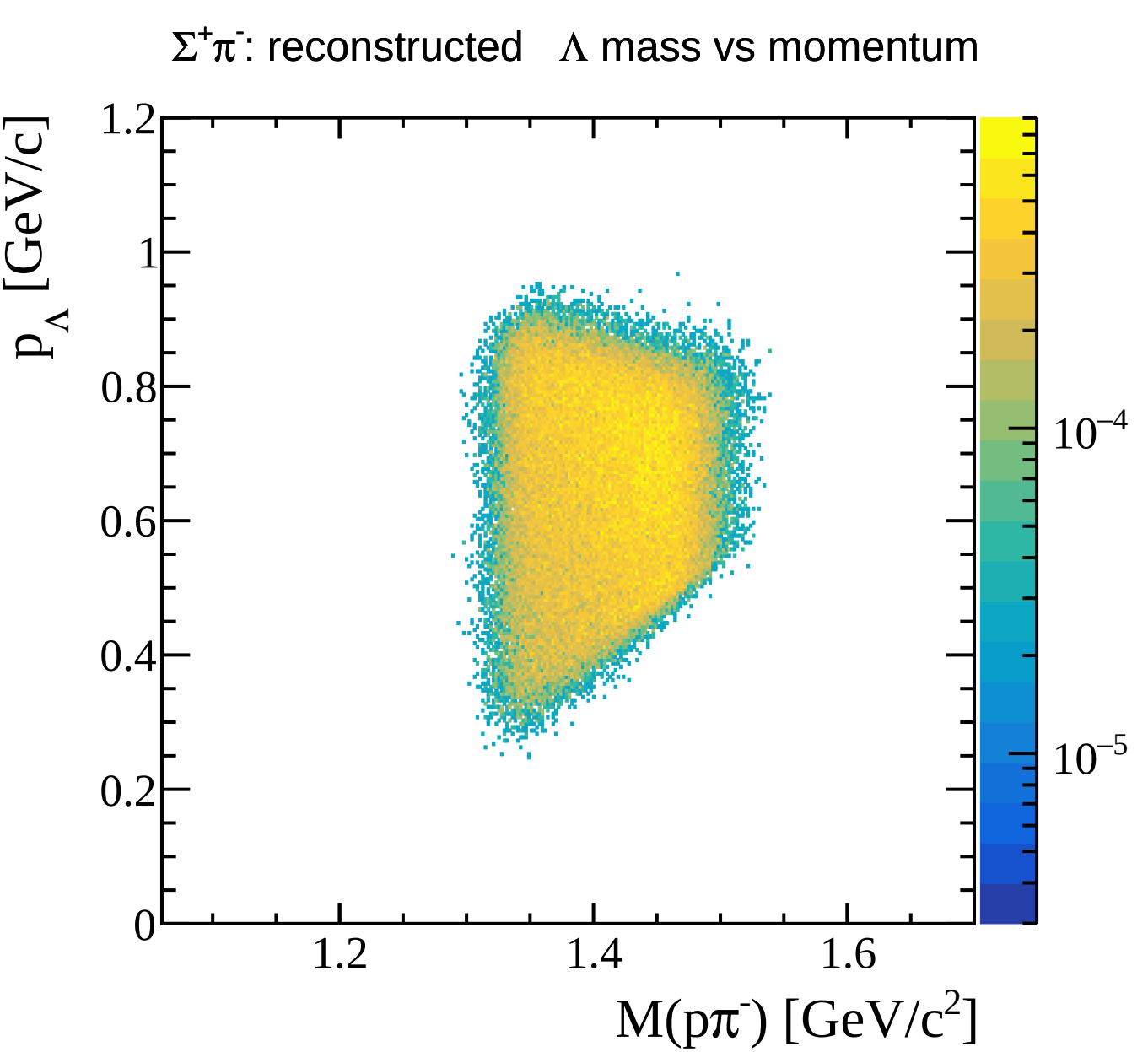

$\Sigma^0\pi^+\pi^-$: π^- lab momentum
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
−1
0
1
p_{z,π^-} [GeV/c]

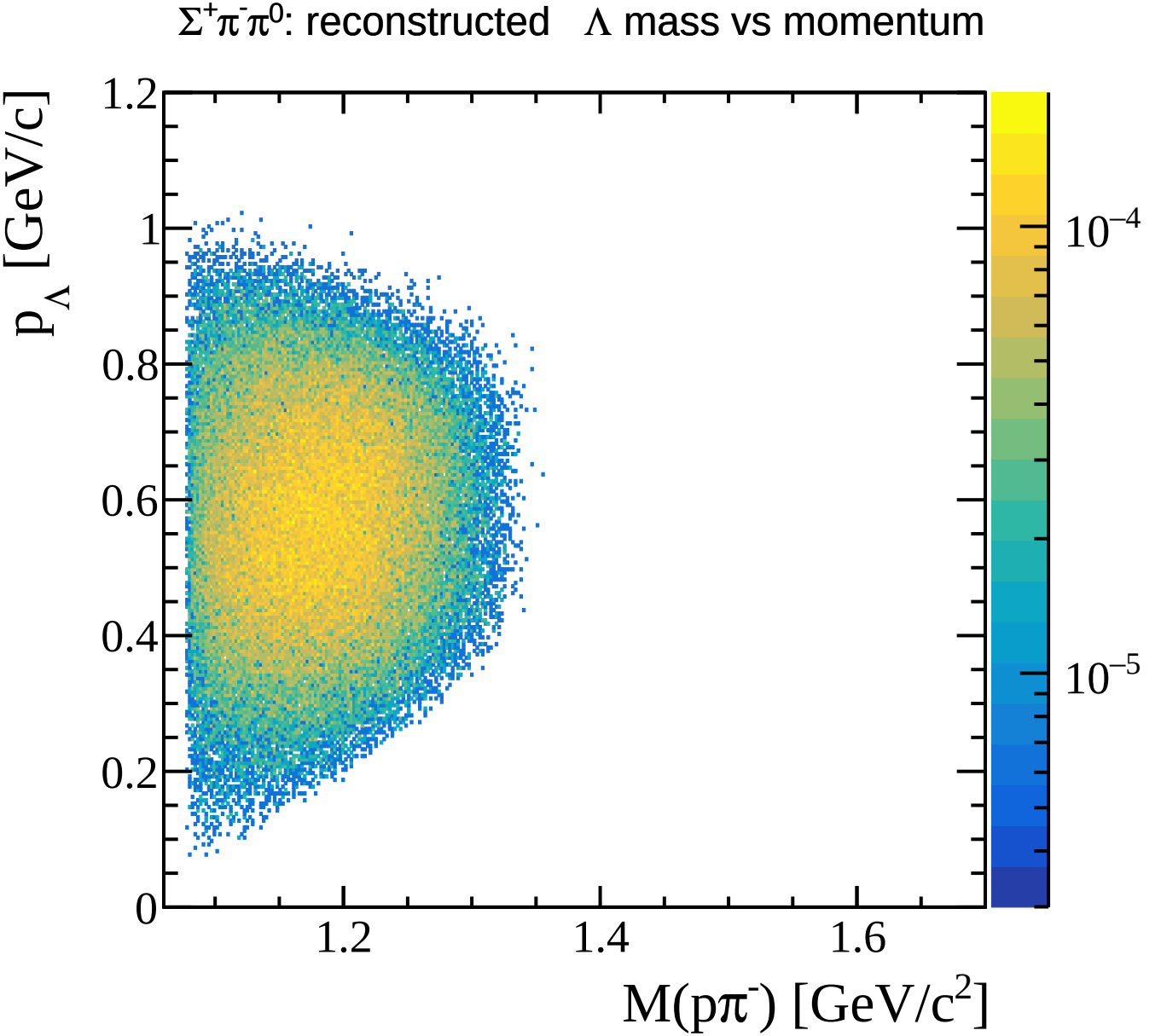

All reactions: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
M(pπ⁻) [GeV/c²]
10⁻²
10⁻³
10⁻⁴
10⁻⁵

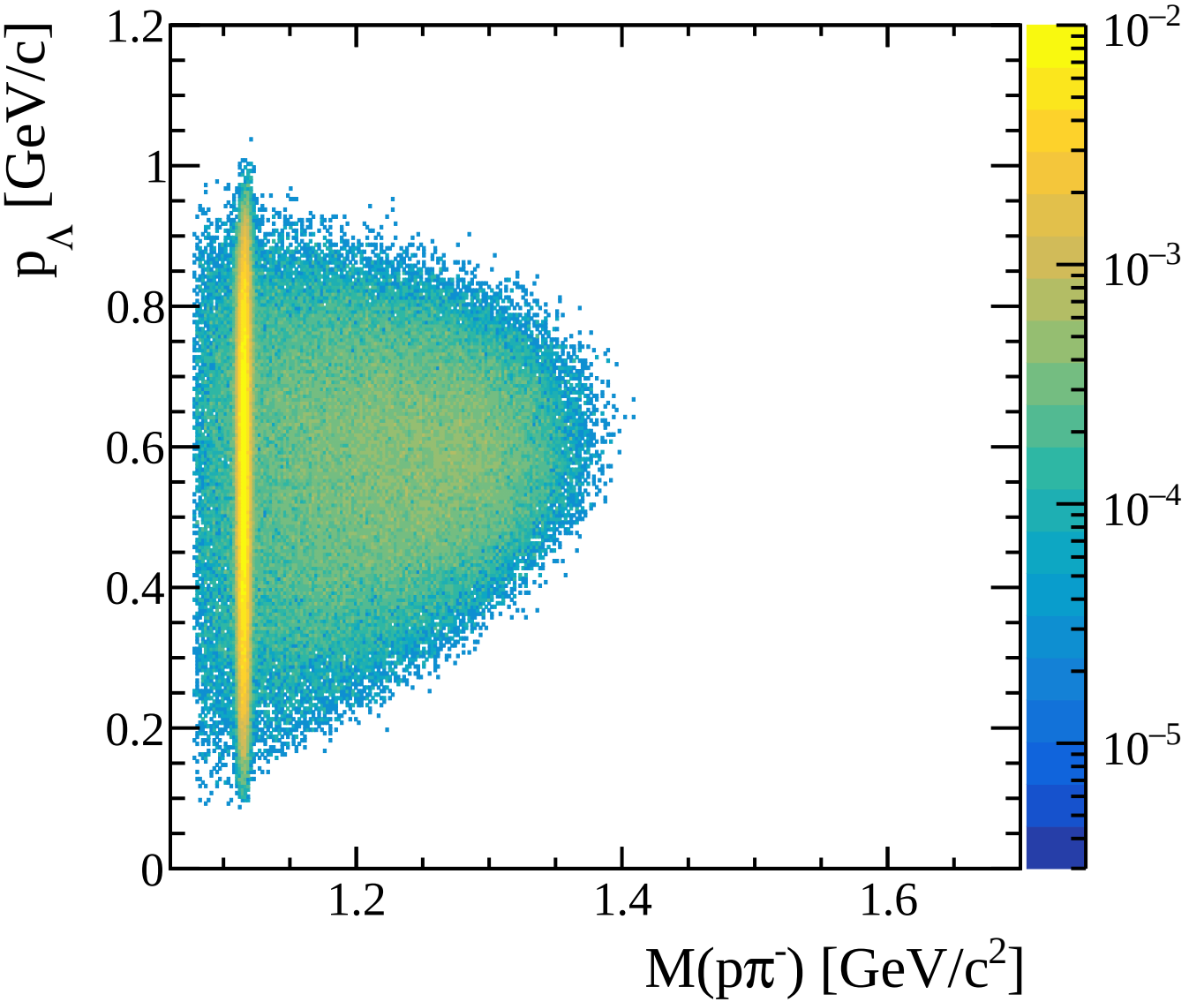

Λη: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
M(pπ⁻) [GeV/c²]
10⁻²
10⁻³
10⁻⁴
10⁻⁵

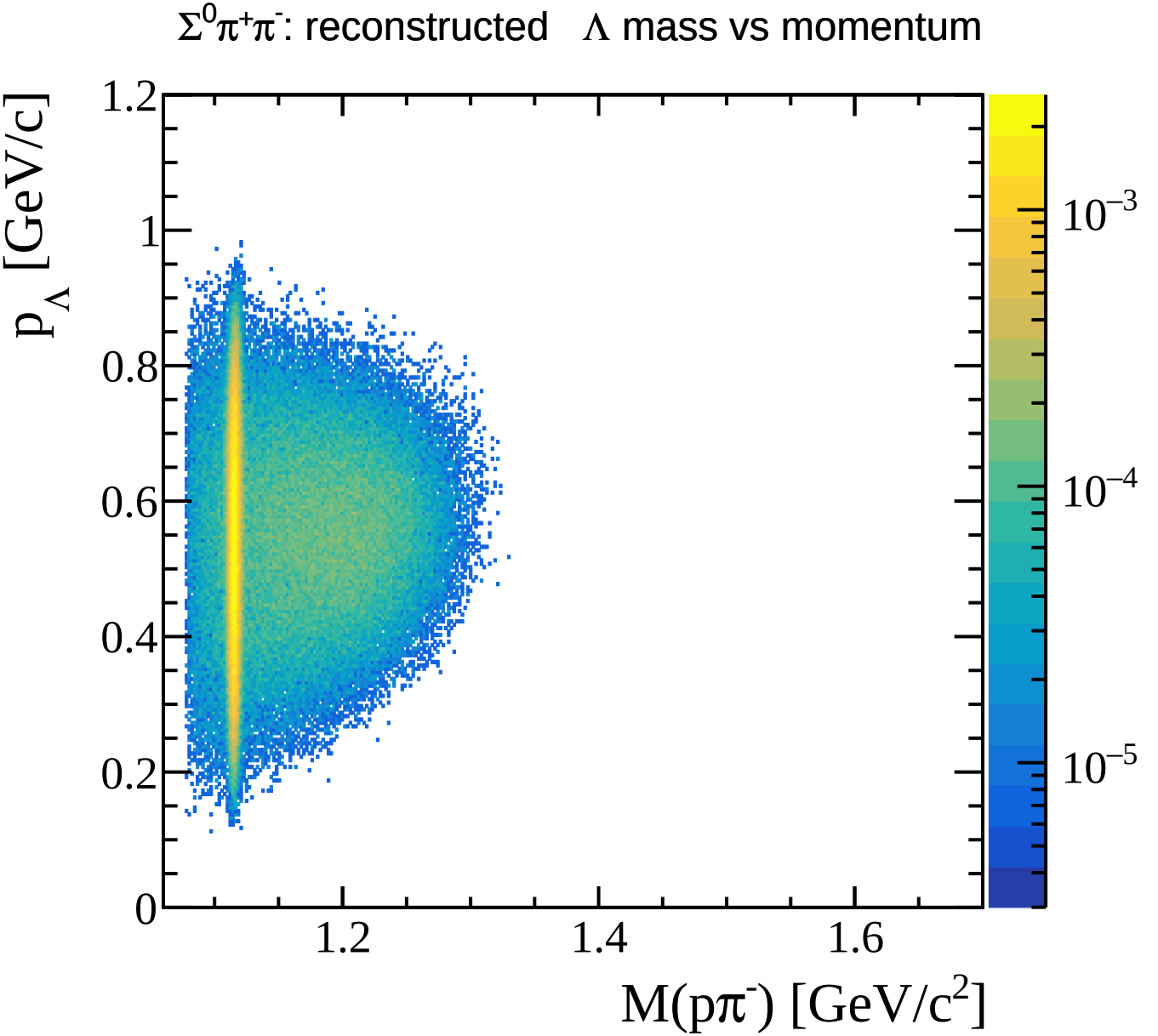

$\Lambda\pi^0$: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
M(pπ^-) [GeV/c^2]
10^{-3}
10^{-4}
10^{-5}

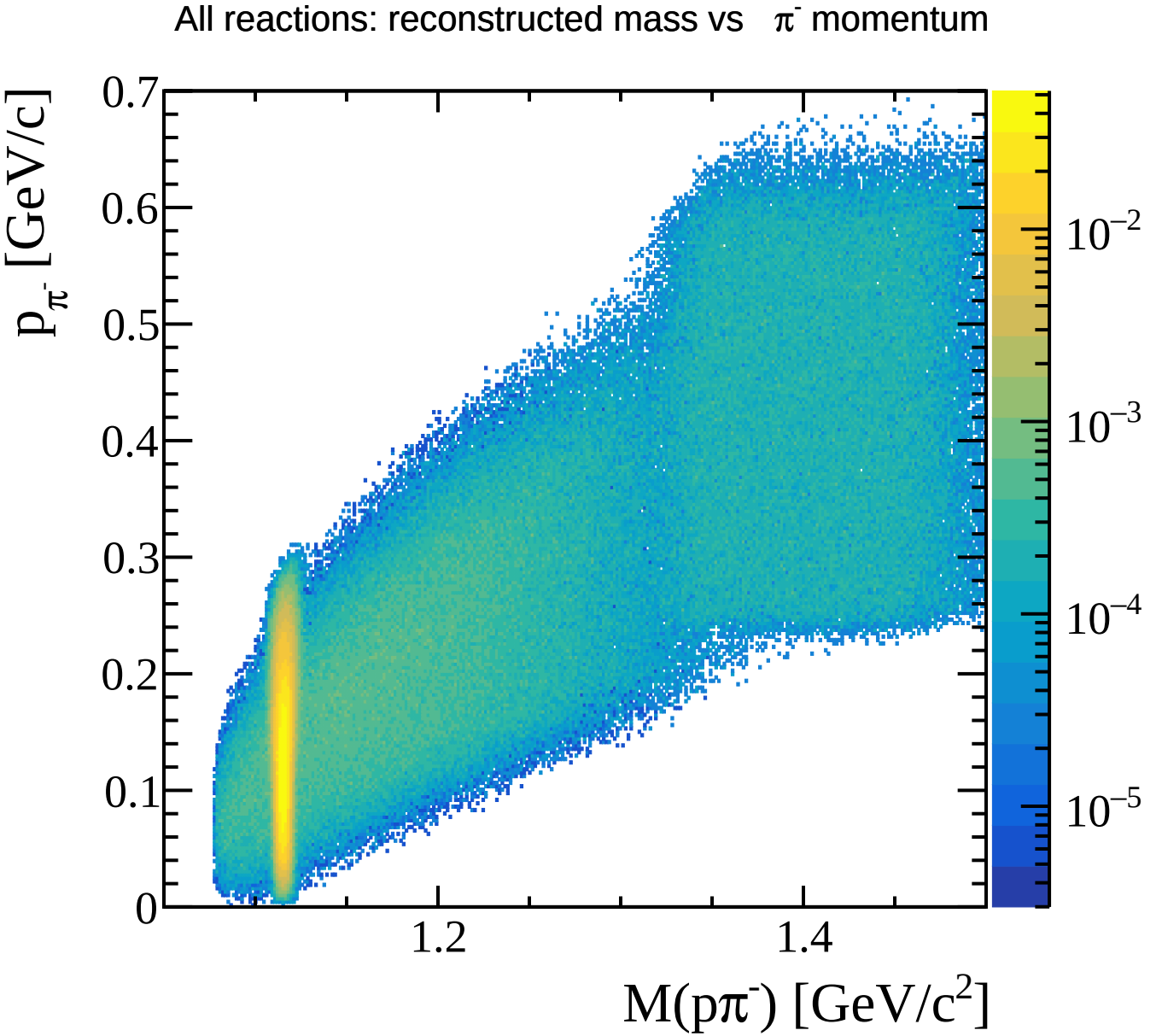

$\Sigma^0\pi^0$: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
M(pπ^-) [GeV/c^2]
10^{-3}
10^{-4}
10^{-5}

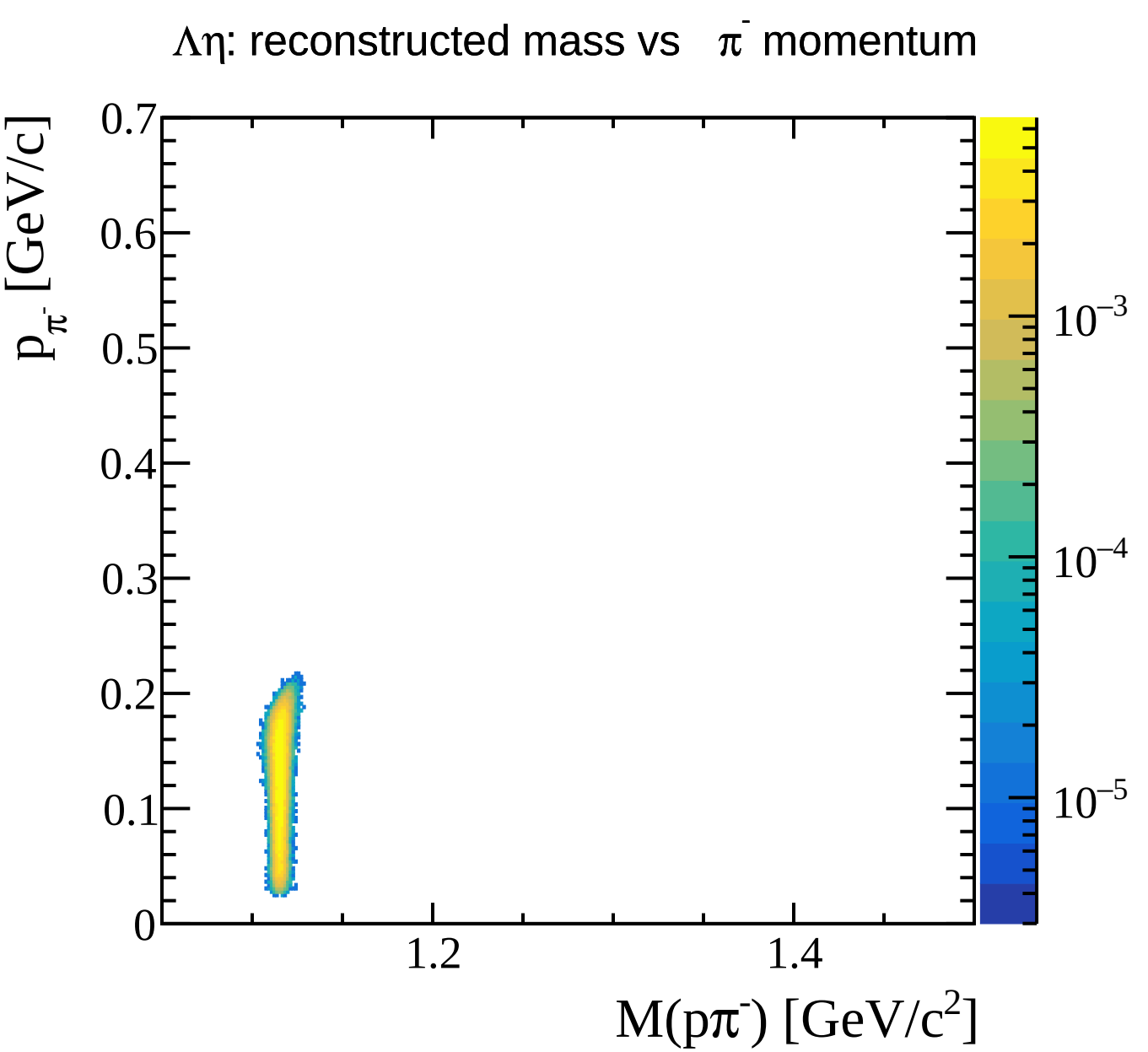

$\Sigma^+\pi^-$: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
M(pπ^-) [GeV/c^2]
10^{-4}
10^{-5}

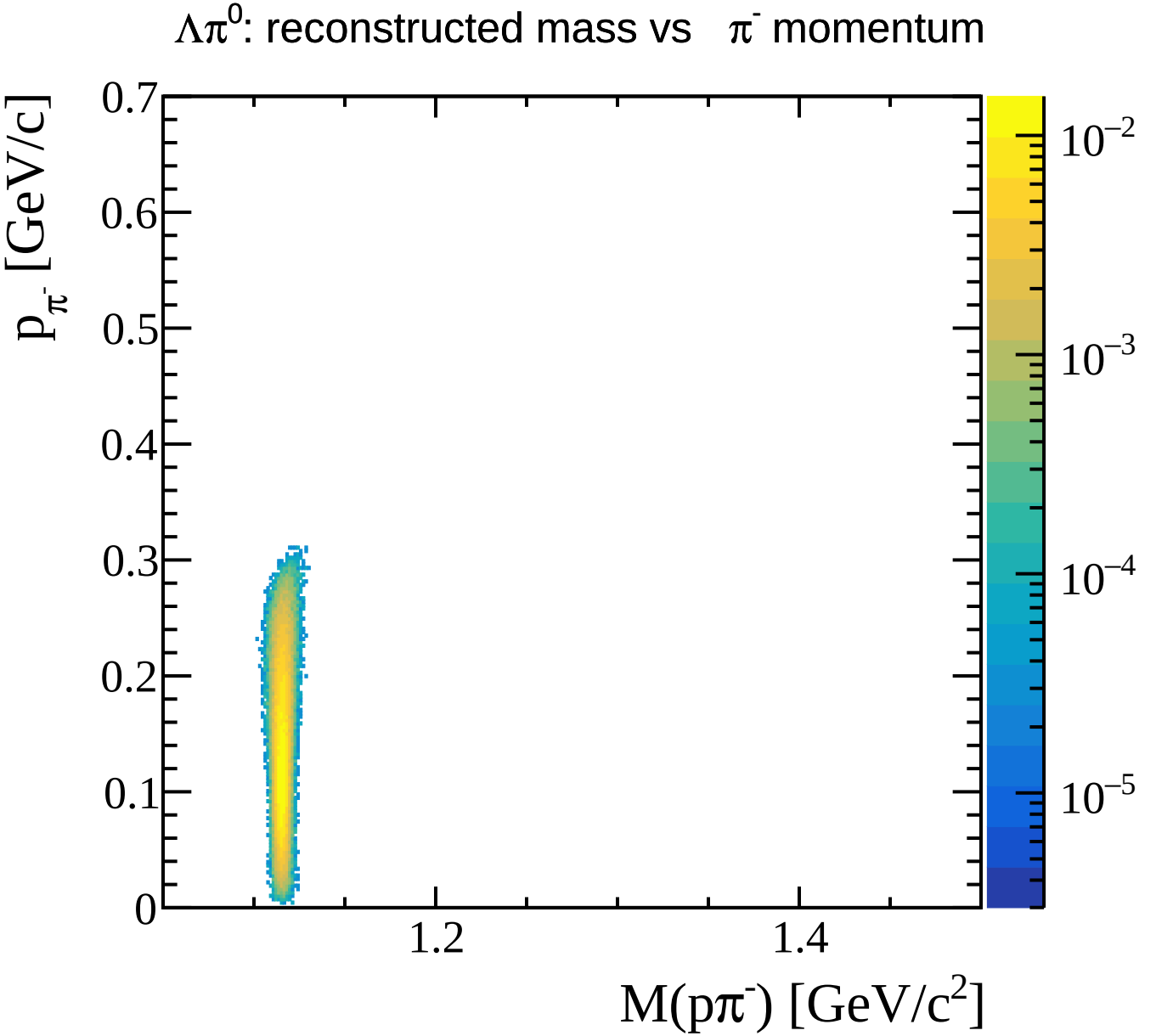

$\Sigma^+\pi^-\pi^0$: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
M(pπ^-) [GeV/c^2]
10^{-4}
10^{-5}

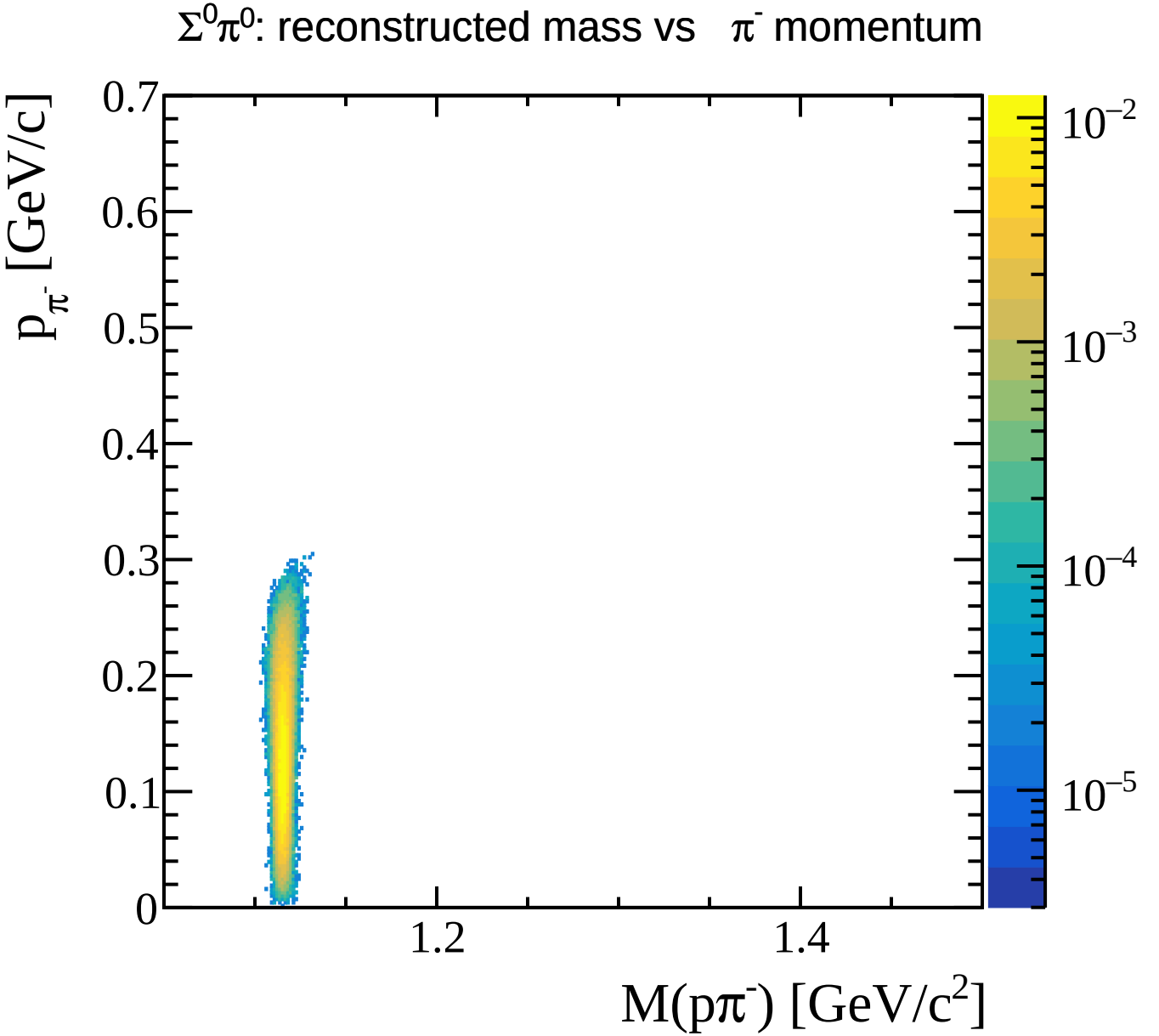

Λπ+π-: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
M(pπ-) [GeV/c2]
10−2
10−3
10−4
10−5

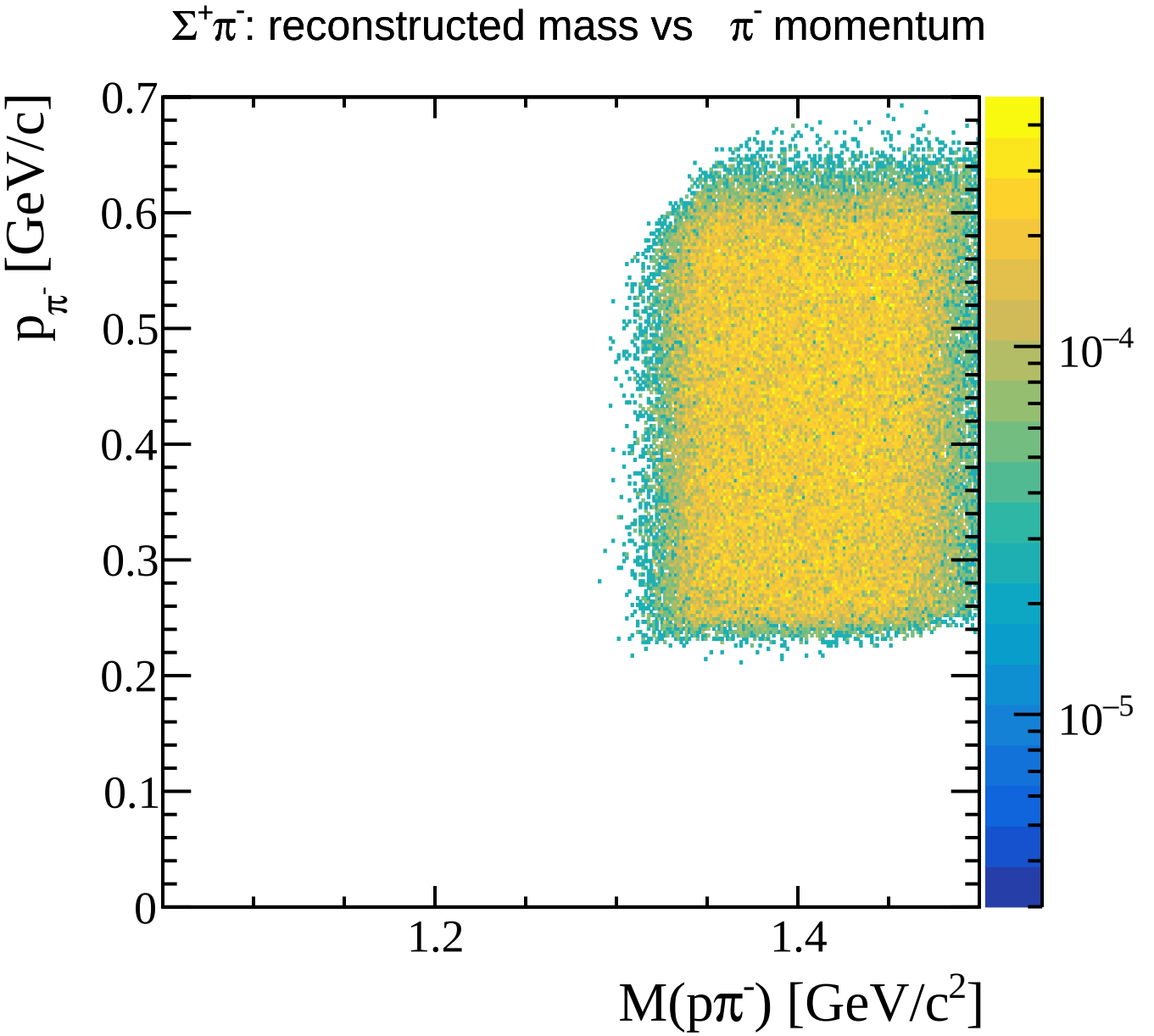

$\Sigma^0\pi^+\pi^-$: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
M(pπ^-) [GeV/c^2]
10^{-3}
10^{-4}
10^{-5}

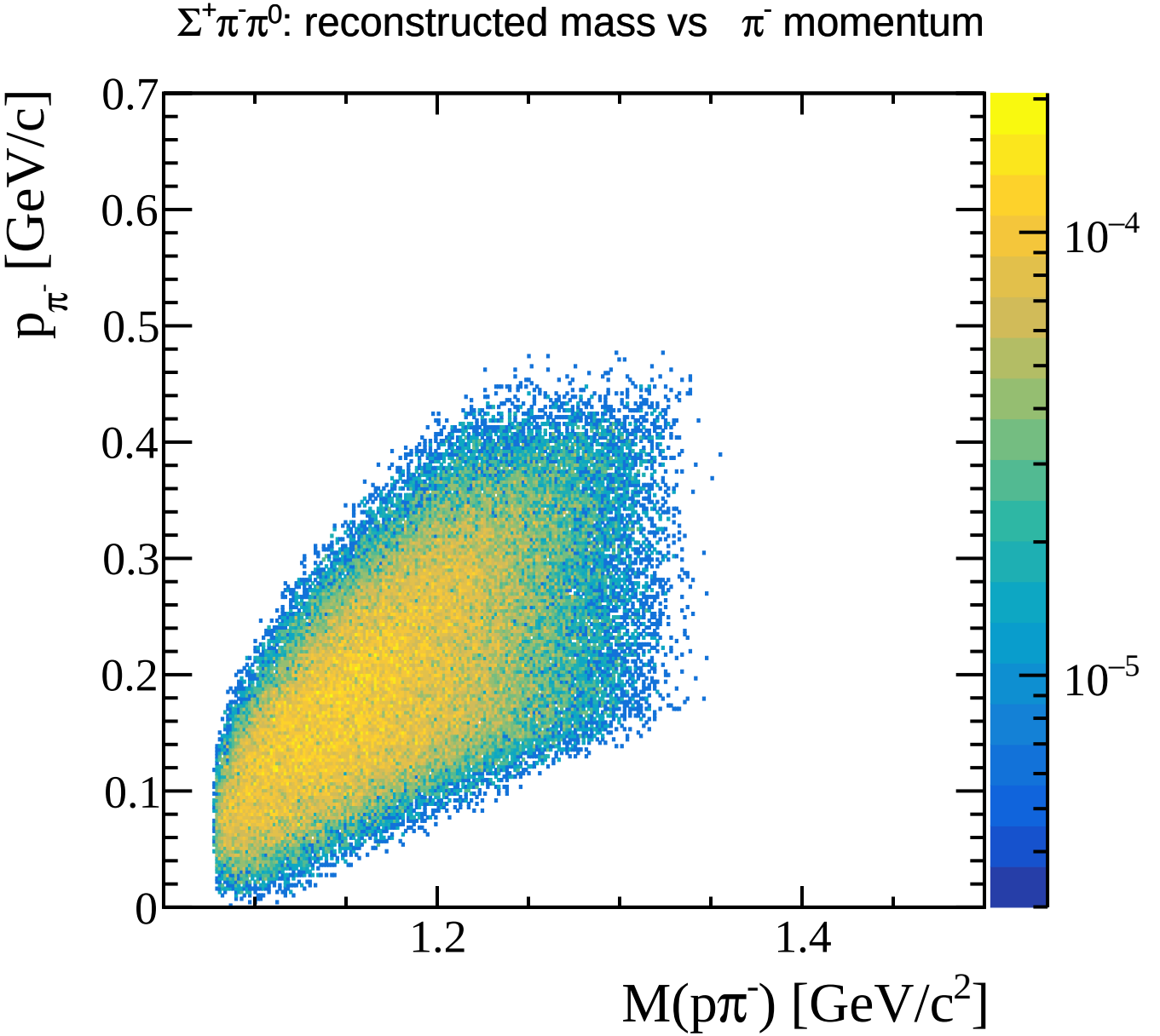

All reactions: reconstructed mass vs π⁻ momentum
p_π⁻ [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
M(pπ⁻) [GeV/c²]
10⁻²
10⁻³
10⁻⁴
10⁻⁵

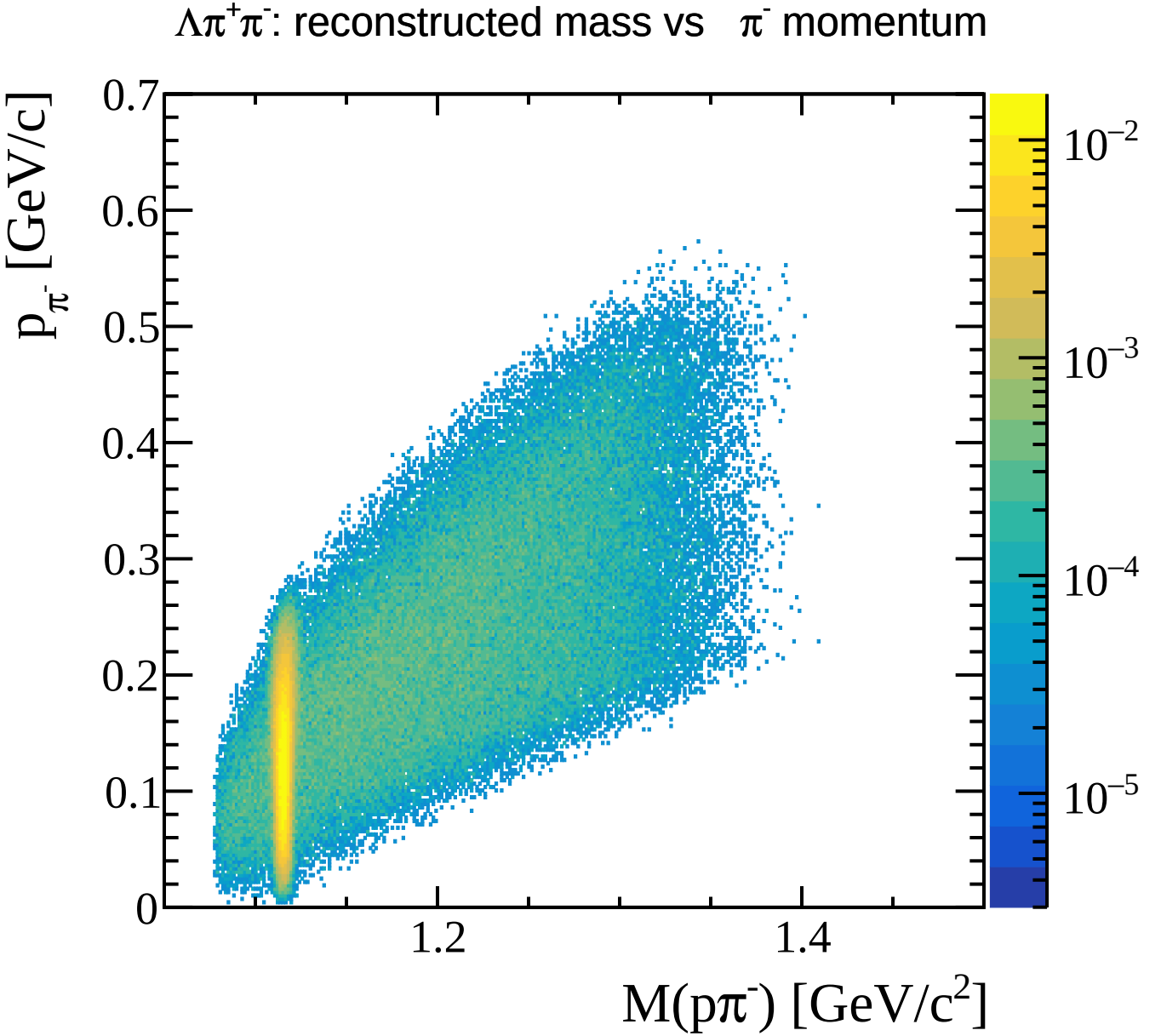

Λη: reconstructed mass vs π- momentum
p_π- [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
M(pπ-) [GeV/c²]
10⁻³
10⁻⁴
10⁻⁵

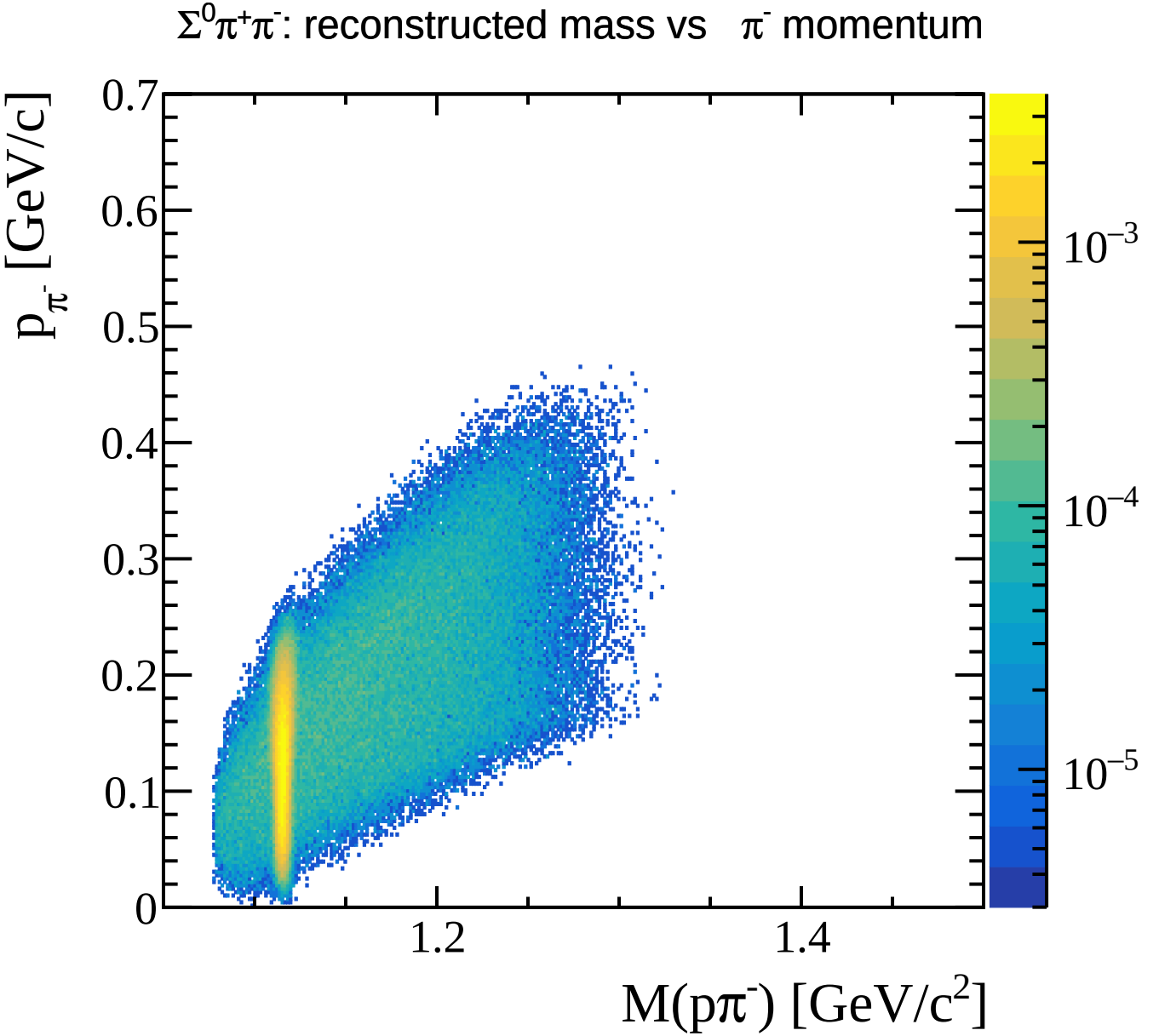

Λπ0: reconstructed mass vs π- momentum
pπ- [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
M(pπ-) [GeV/c2]
10−2
10−3
10−4
10−5

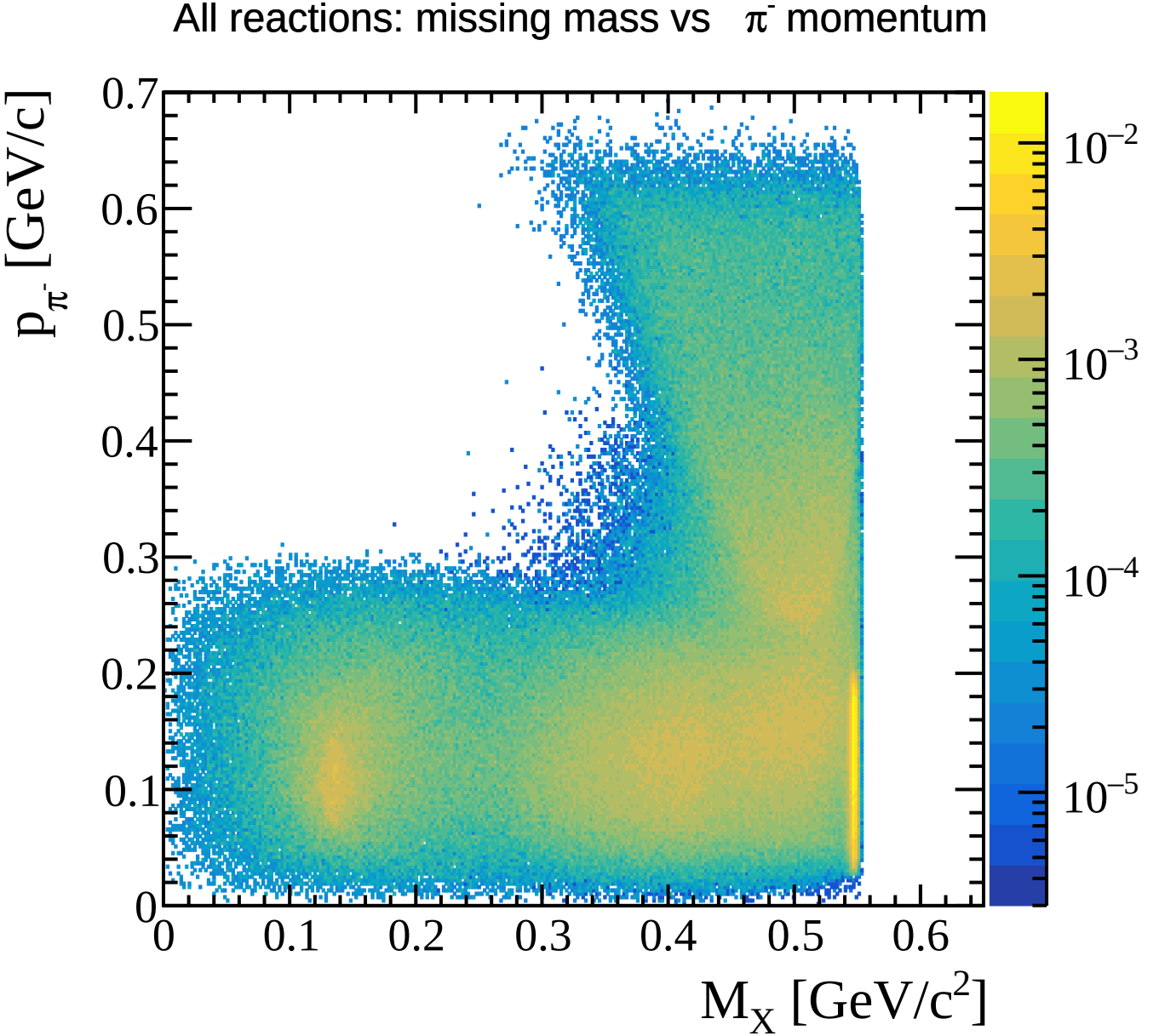

$\Sigma^0\pi^0$: reconstructed mass vs π^- momentum
p_{π^-} [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
M(pπ^-) [GeV/c^2]
10^{-2}
10^{-3}
10^{-4}
10^{-5}

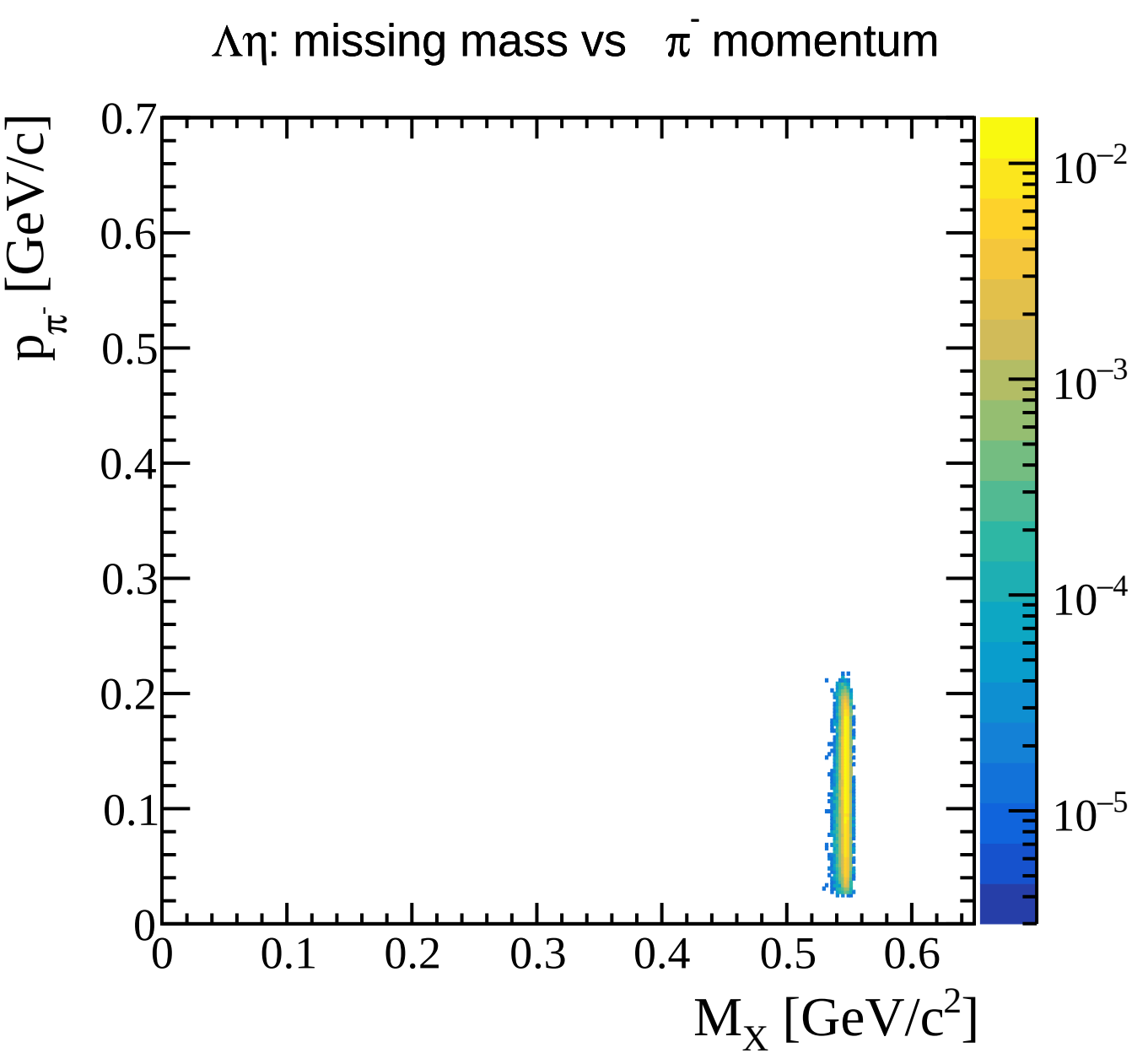

Σ+π-: reconstructed mass vs π- momentum
p_π- [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
M(pπ-) [GeV/c2]
10−4
10−5

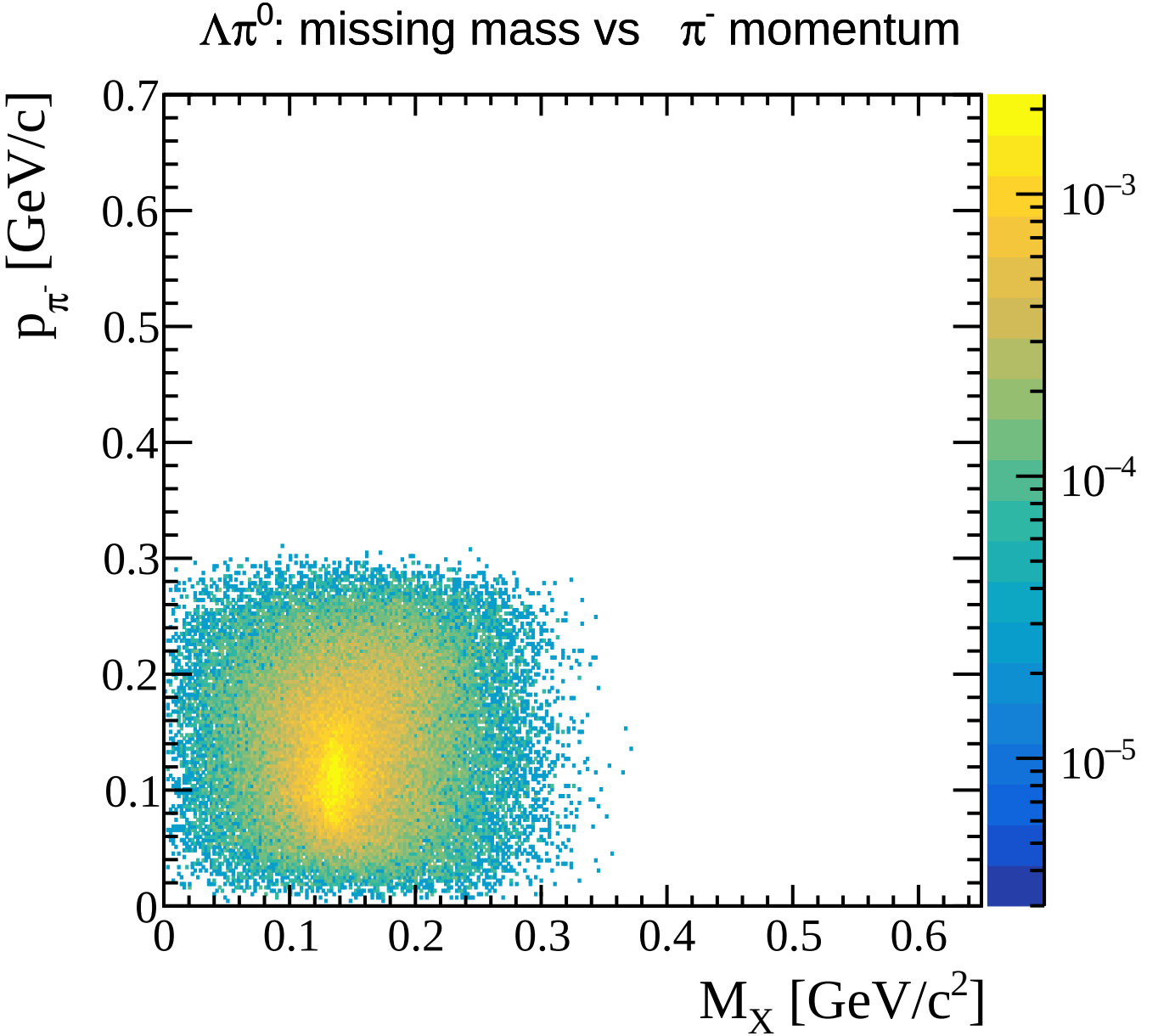

Σ+π-π0: reconstructed mass vs π- momentum
p_π- [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
M(pπ-) [GeV/c2]
10−4
10−5

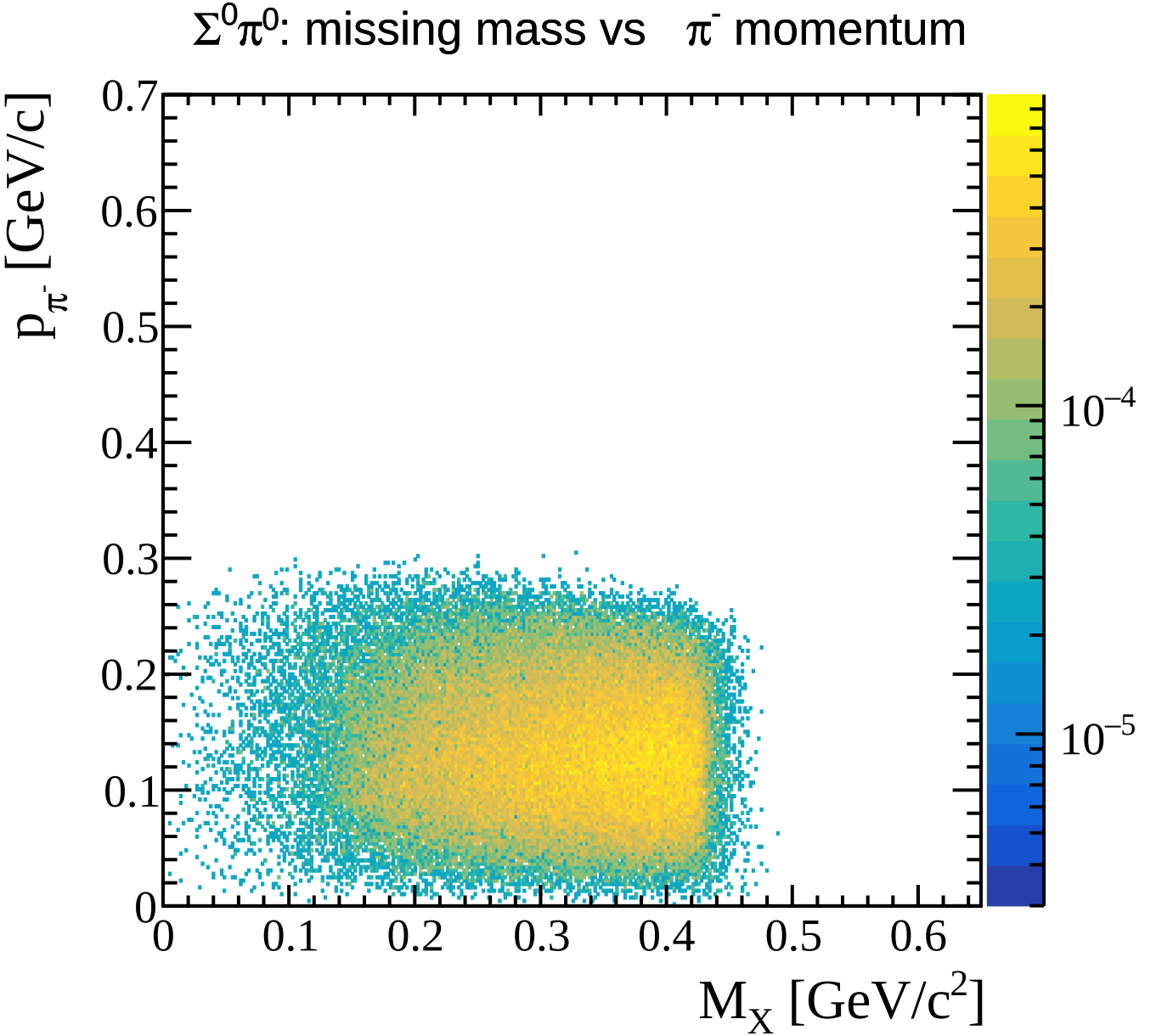

Λπ+π-: reconstructed mass vs π- momentum
p_π- [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
M(pπ-) [GeV/c2]
10−2
10−3
10−4
10−5

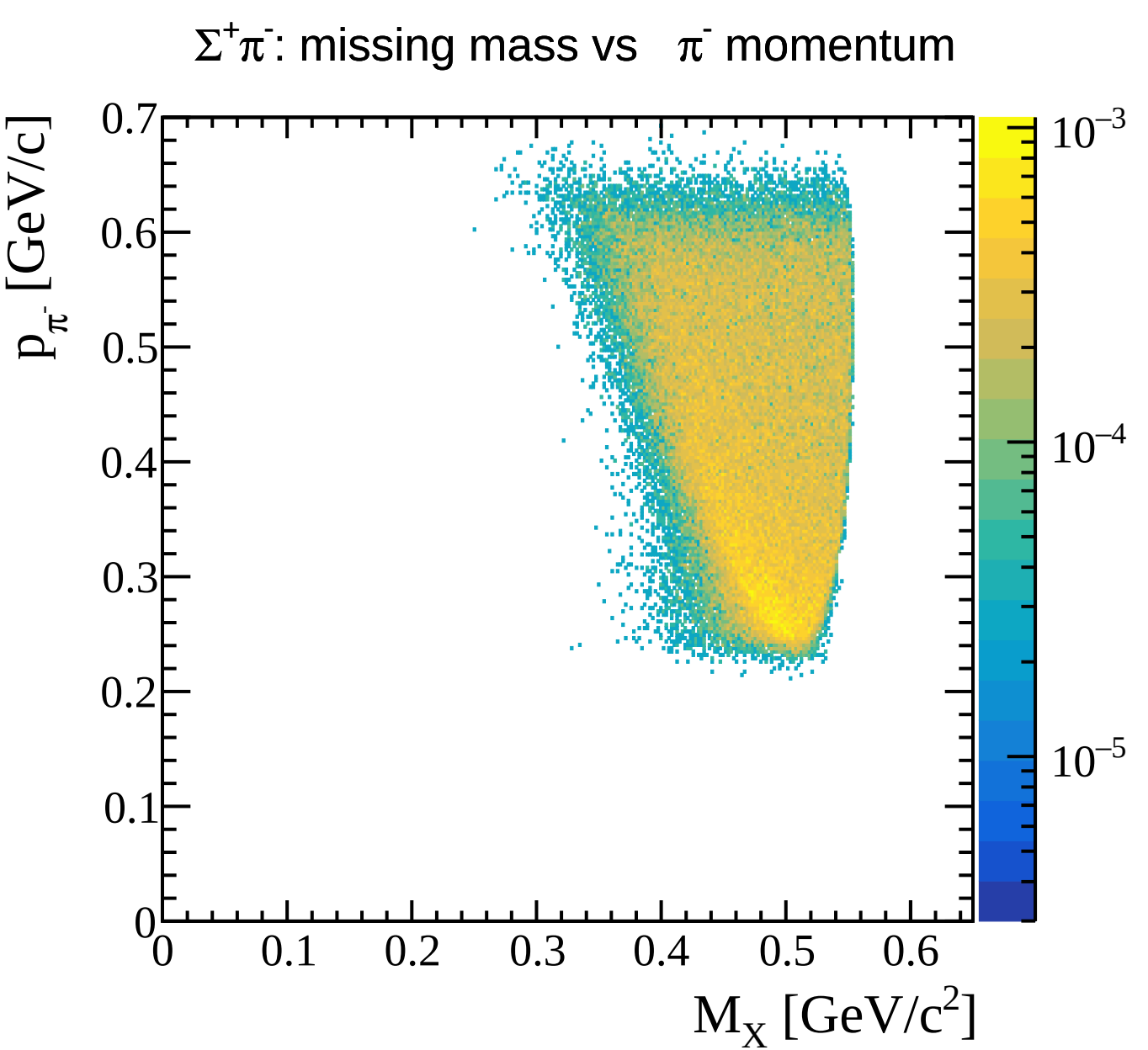

$\Sigma^0\pi^+\pi^-$: reconstructed mass vs π^- momentum
p_{π^-} [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
$M(p\pi^-)$ [GeV/c^2]
10^{-3}
10^{-4}
10^{-5}

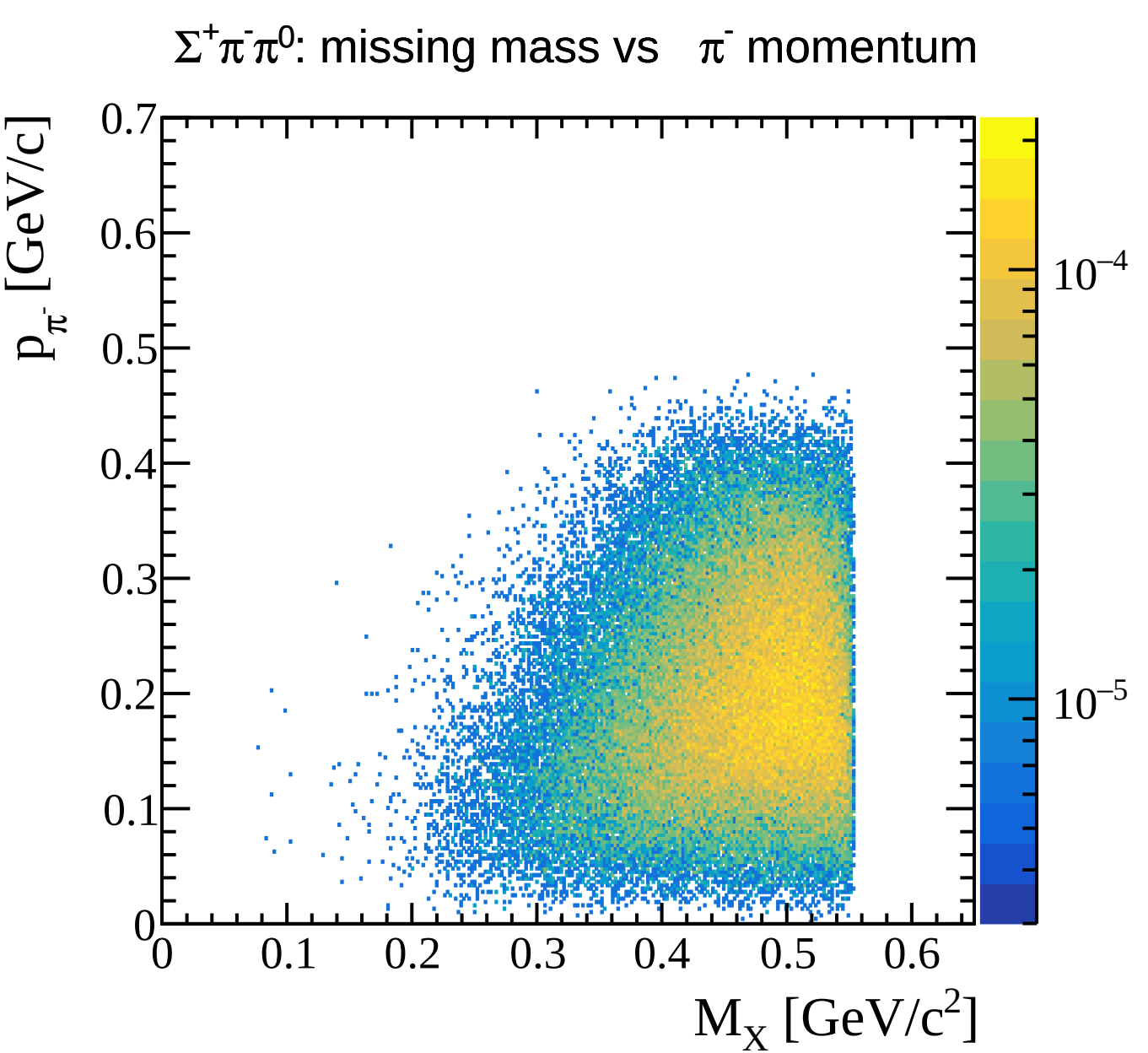

All reactions: missing mass vs π⁻ momentum
p_π⁻ [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c²]
10⁻²
10⁻³
10⁻⁴
10⁻⁵

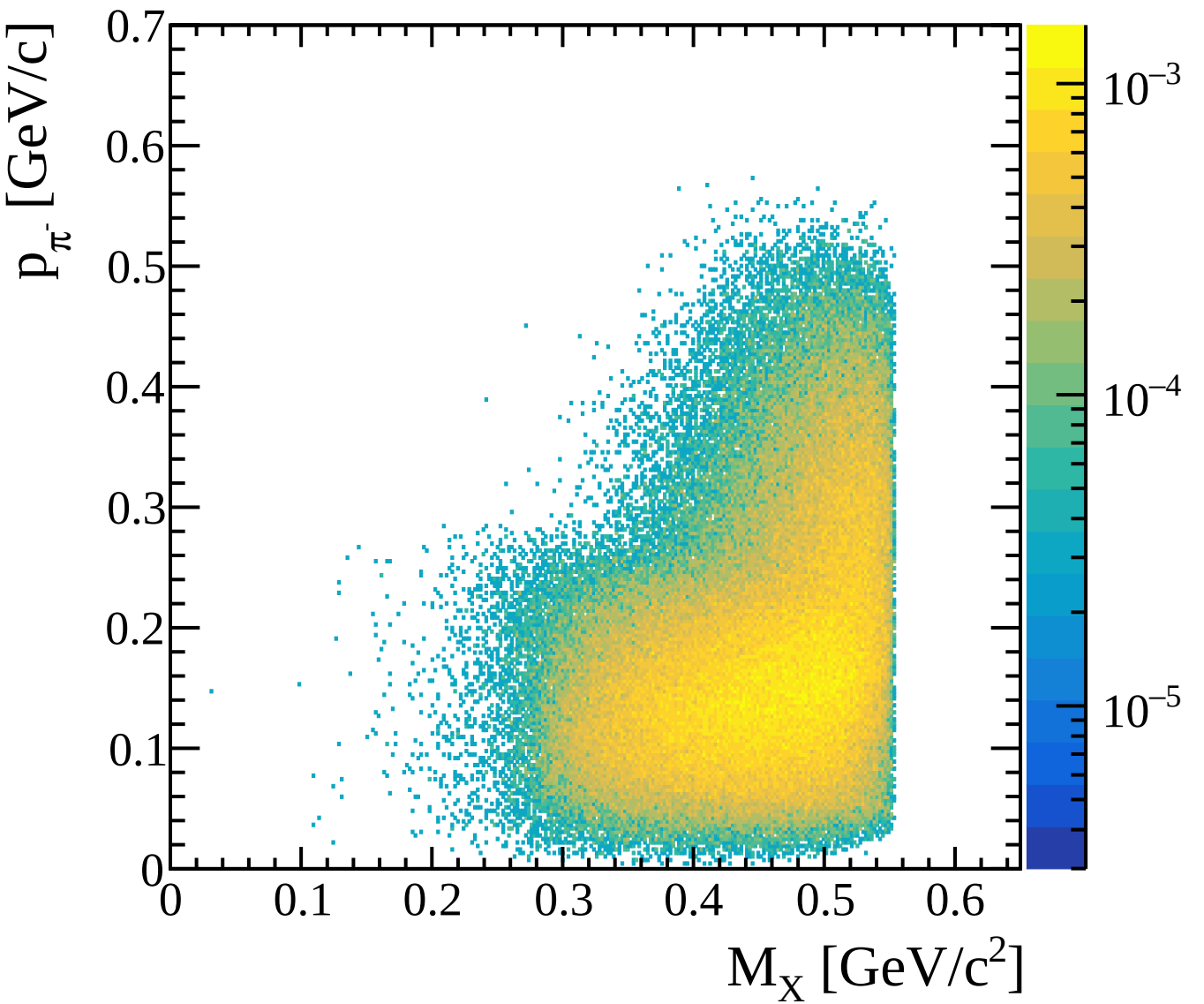

Λη: missing mass vs π⁻ momentum
p_π⁻ [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c²]
10⁻²
10⁻³
10⁻⁴
10⁻⁵

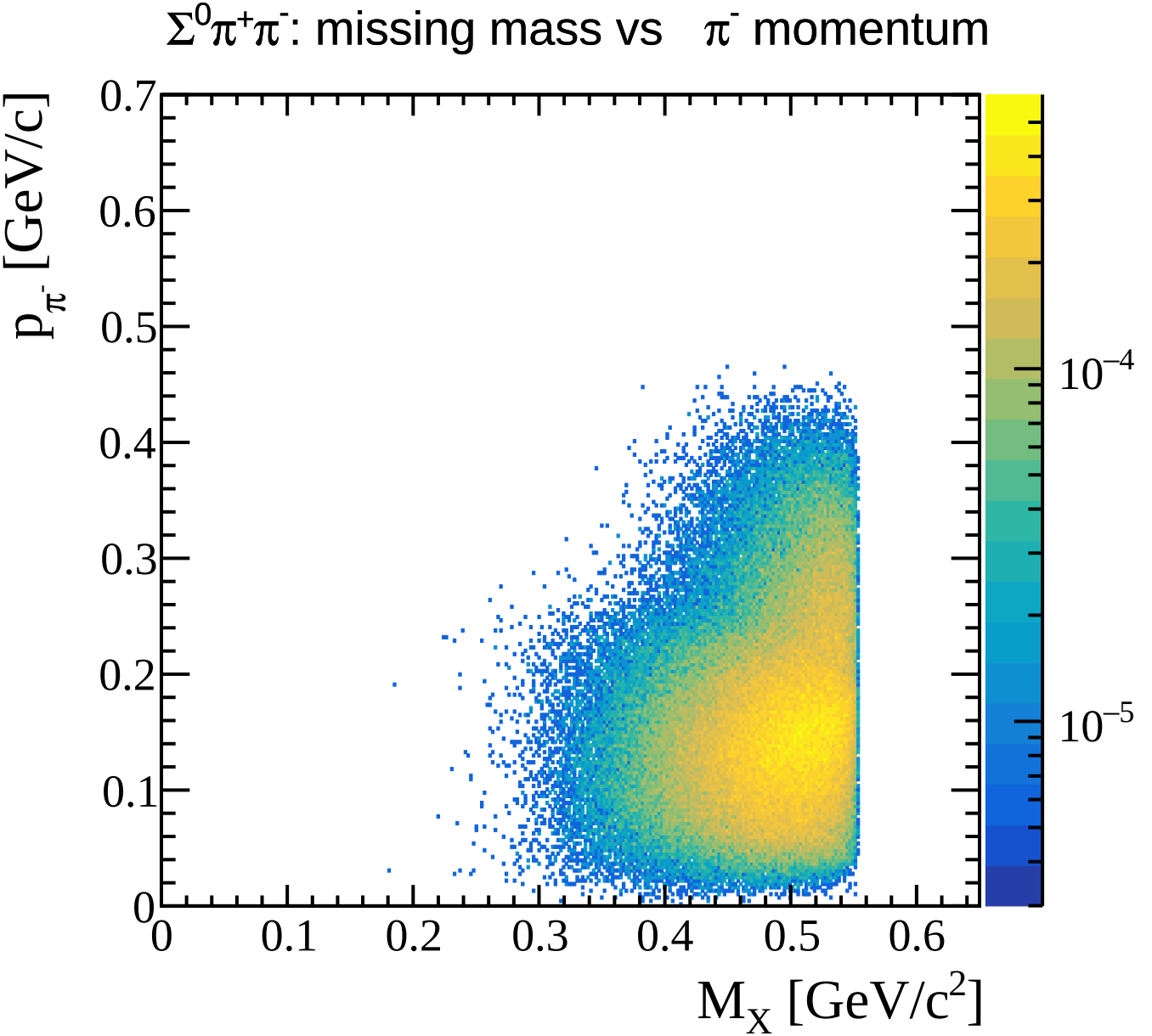

$\Lambda\pi^0$: missing mass vs π^- momentum
p_{π^-} [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]
10^{-3}
10^{-4}
10^{-5}

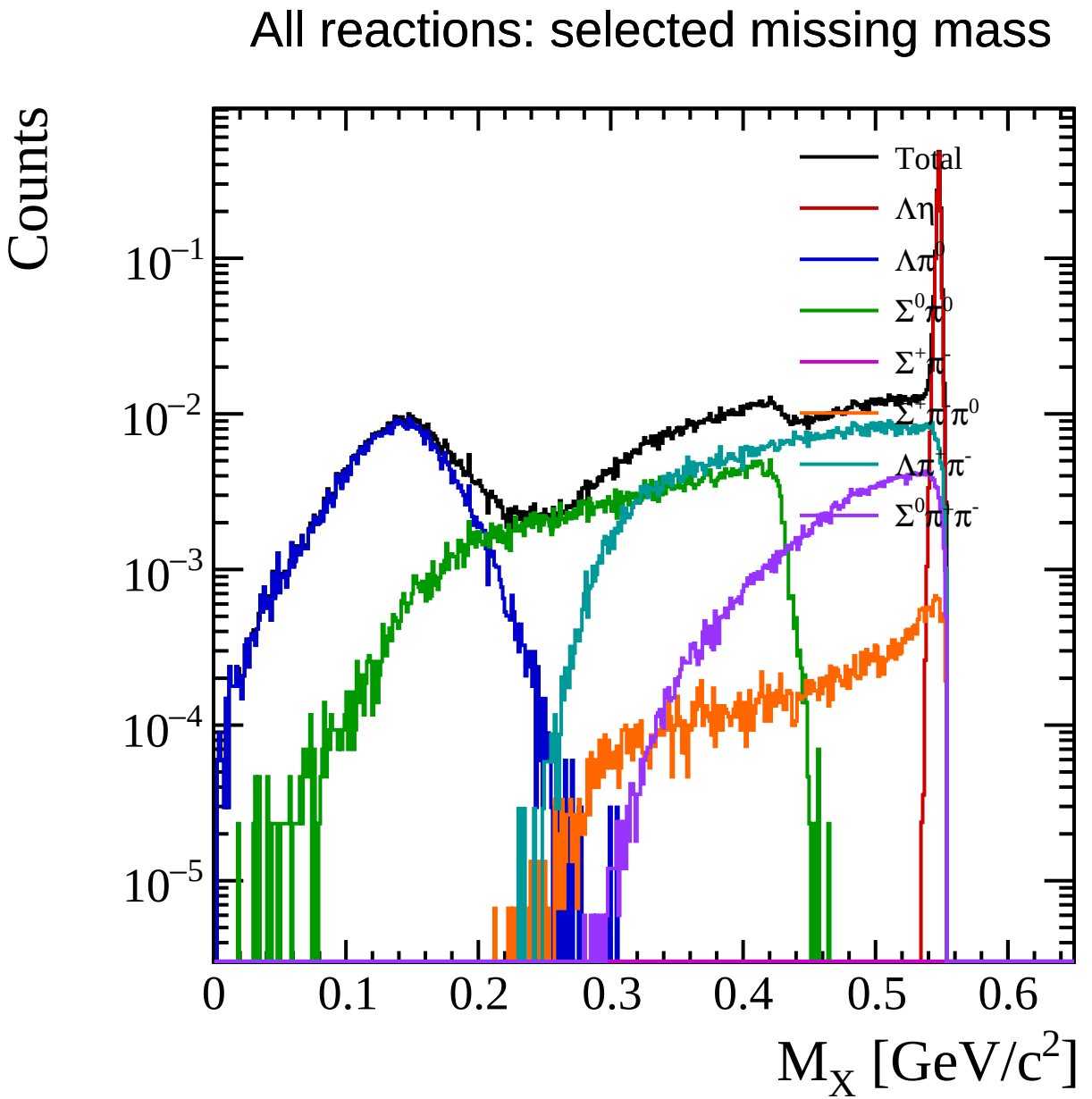

$\Sigma^0\pi^0$: missing mass vs π^- momentum
p_{π^-} [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]
10^{-4}
10^{-5}

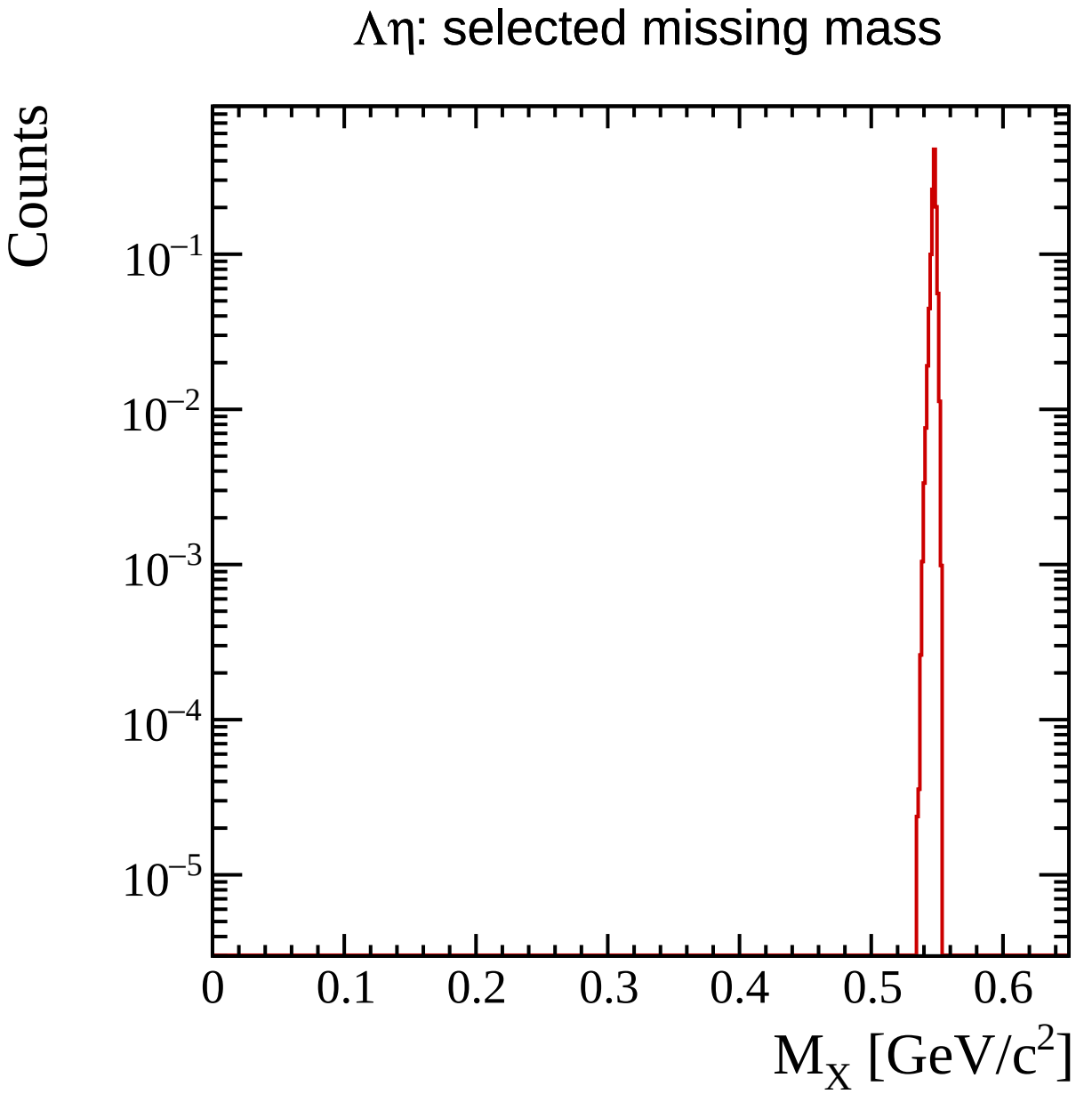

$\Sigma^+\pi^-$: missing mass vs π^- momentum
p_{π^-} [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]
10^{-3}
10^{-4}
10^{-5}

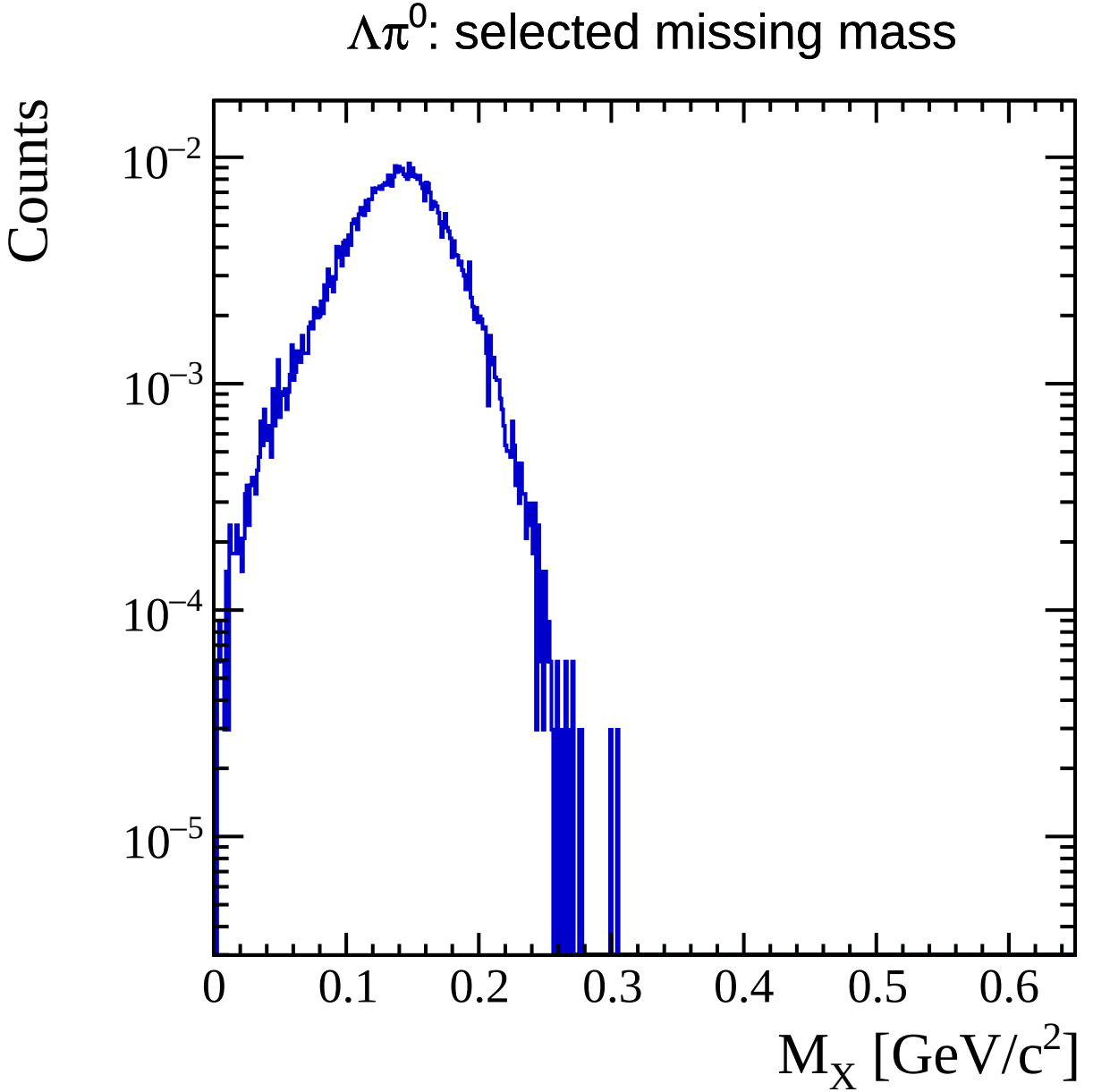

$\Sigma^+\pi^-\pi^0$: missing mass vs π^- momentum
p_{π^-} [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]
10^{-4}
10^{-5}

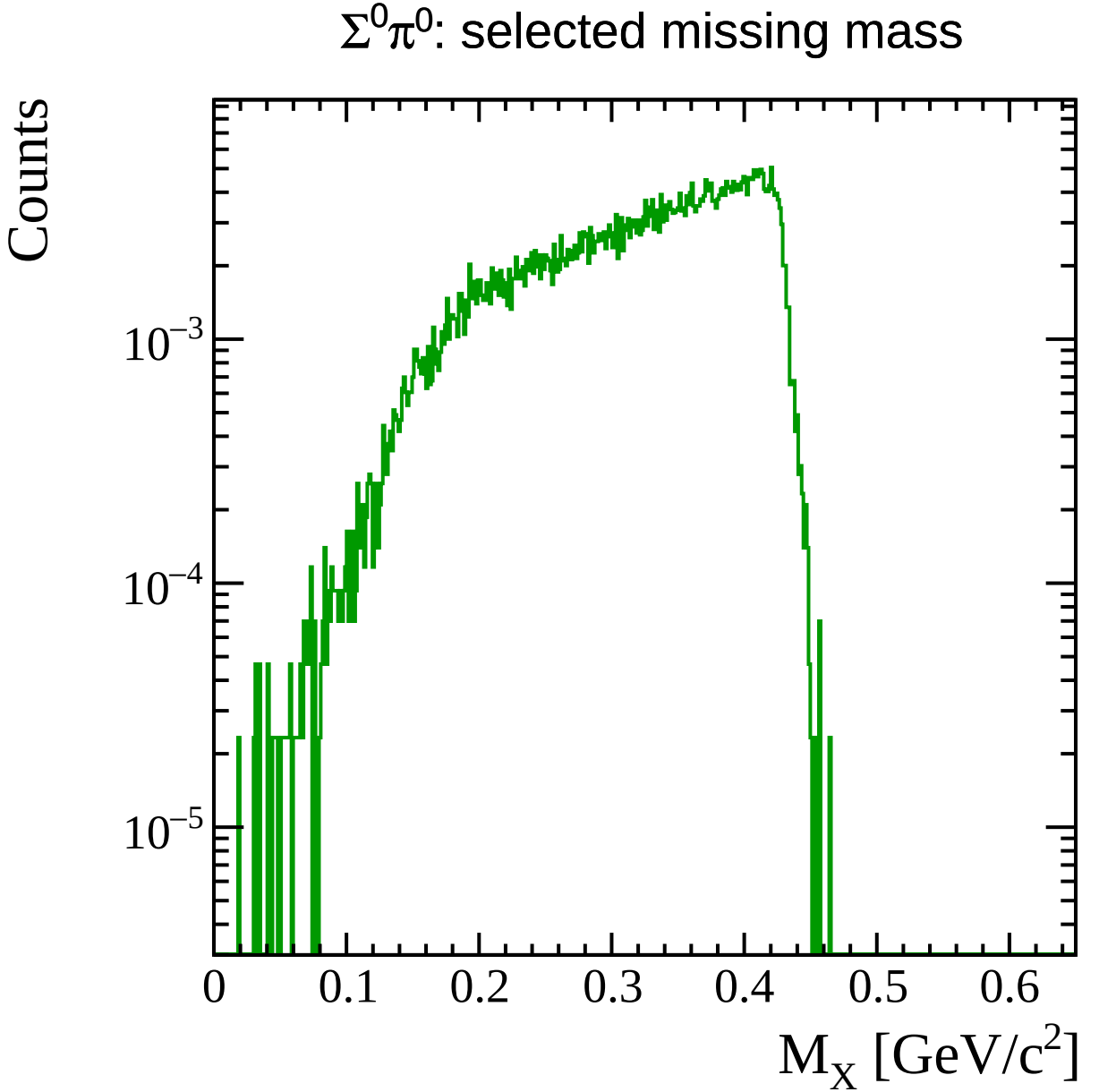

$\Lambda\pi^+\pi^-$: missing mass vs π^- momentum
p_{π^-} [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]
10^{-3}
10^{-4}
10^{-5}

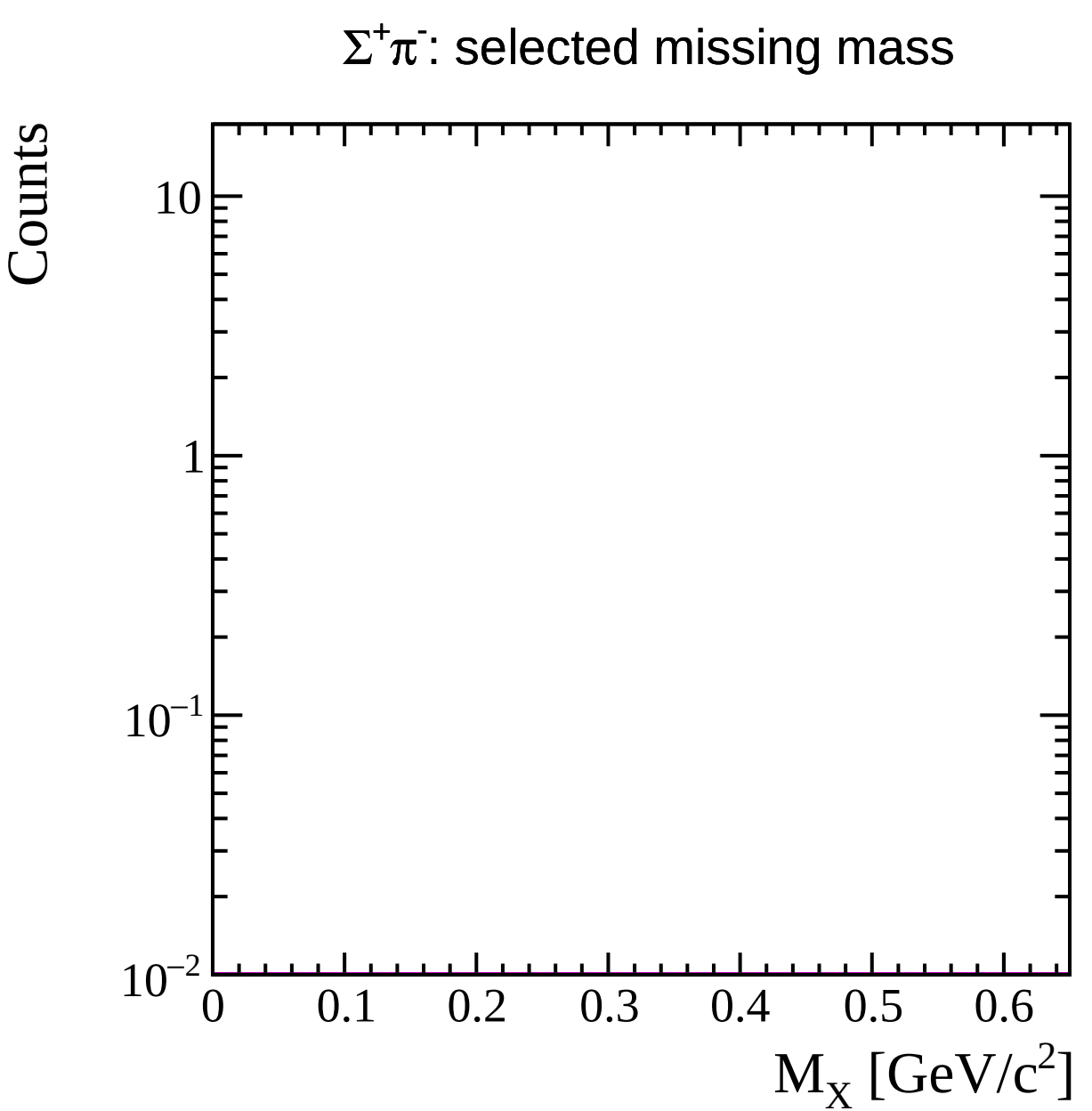

$\Sigma^0\pi^+\pi^-$: missing mass vs π^- momentum
p_{π^-} [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]
10^{-4}
10^{-5}

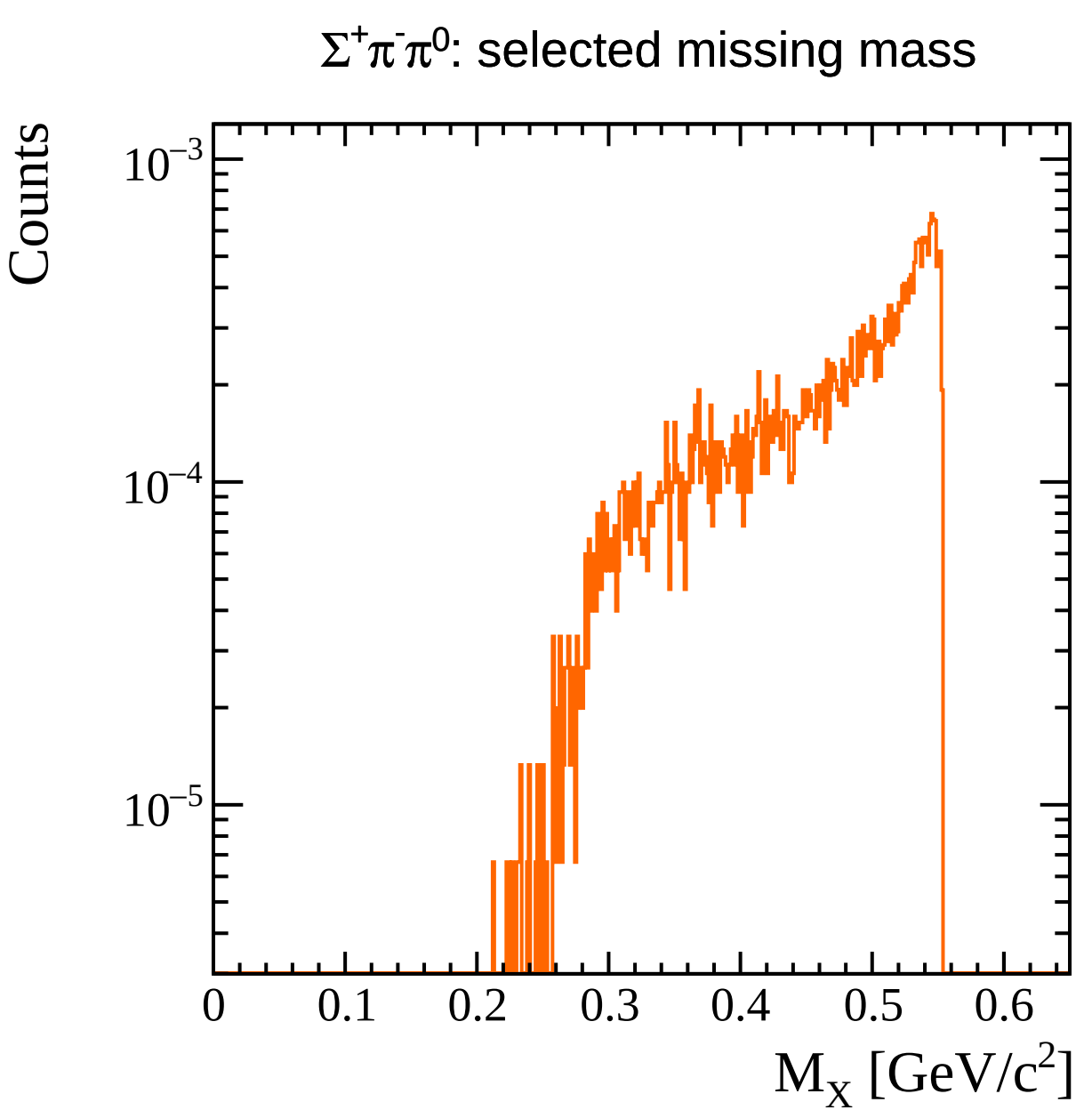

All reactions: selected missing mass
Counts
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]
Total
$\Lambda\eta$
$\Lambda\pi^0$
$\Sigma^0\pi^0$
$\Sigma^+\pi^-$
$\Sigma^+\pi^-\pi^0$
$\Lambda\pi^+\pi^-$
$\Sigma^0\pi^+\pi^-$

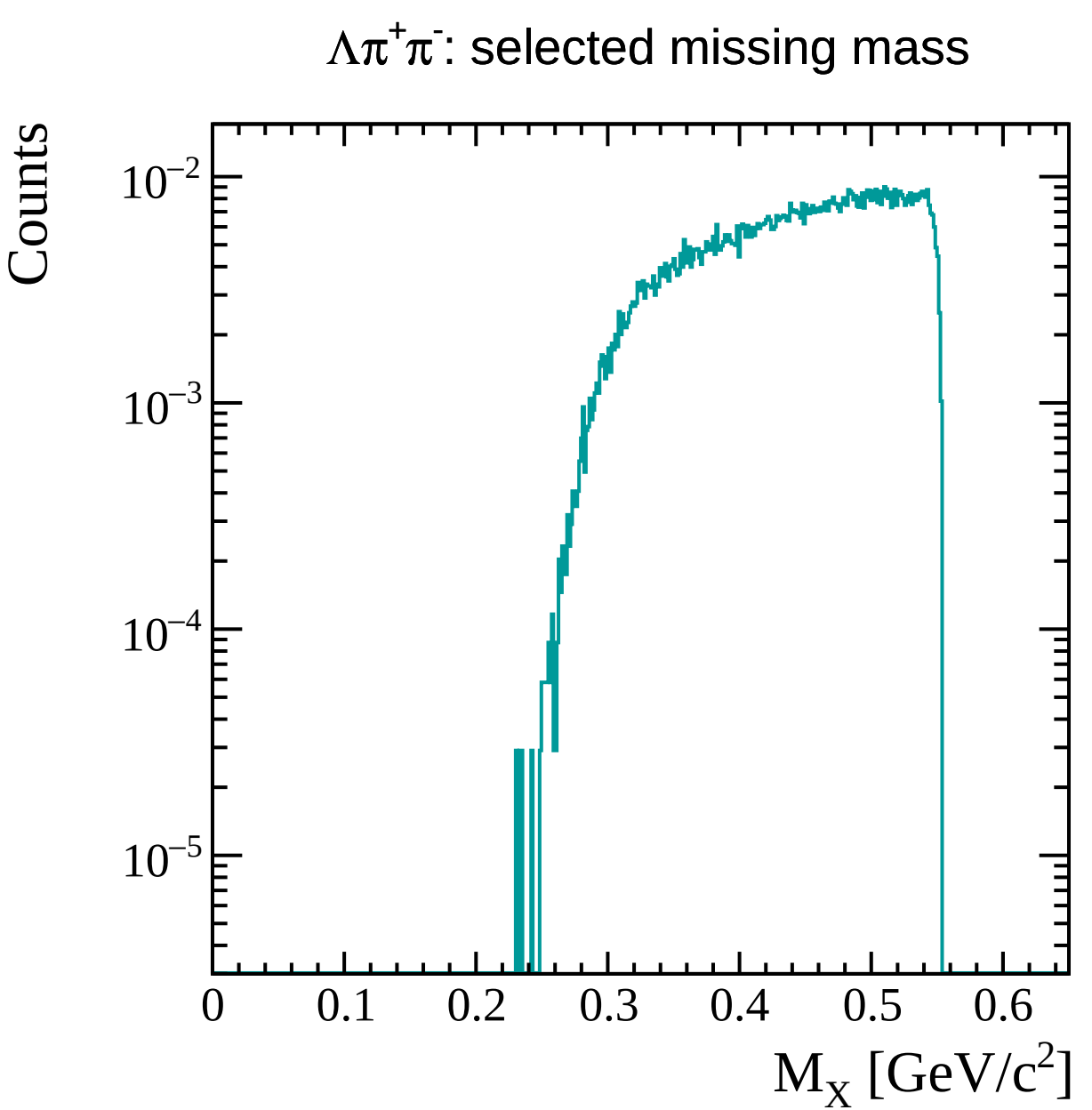

Λη: selected missing mass
Counts
10^−1
10^−2
10^−3
10^−4
10^−5
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]

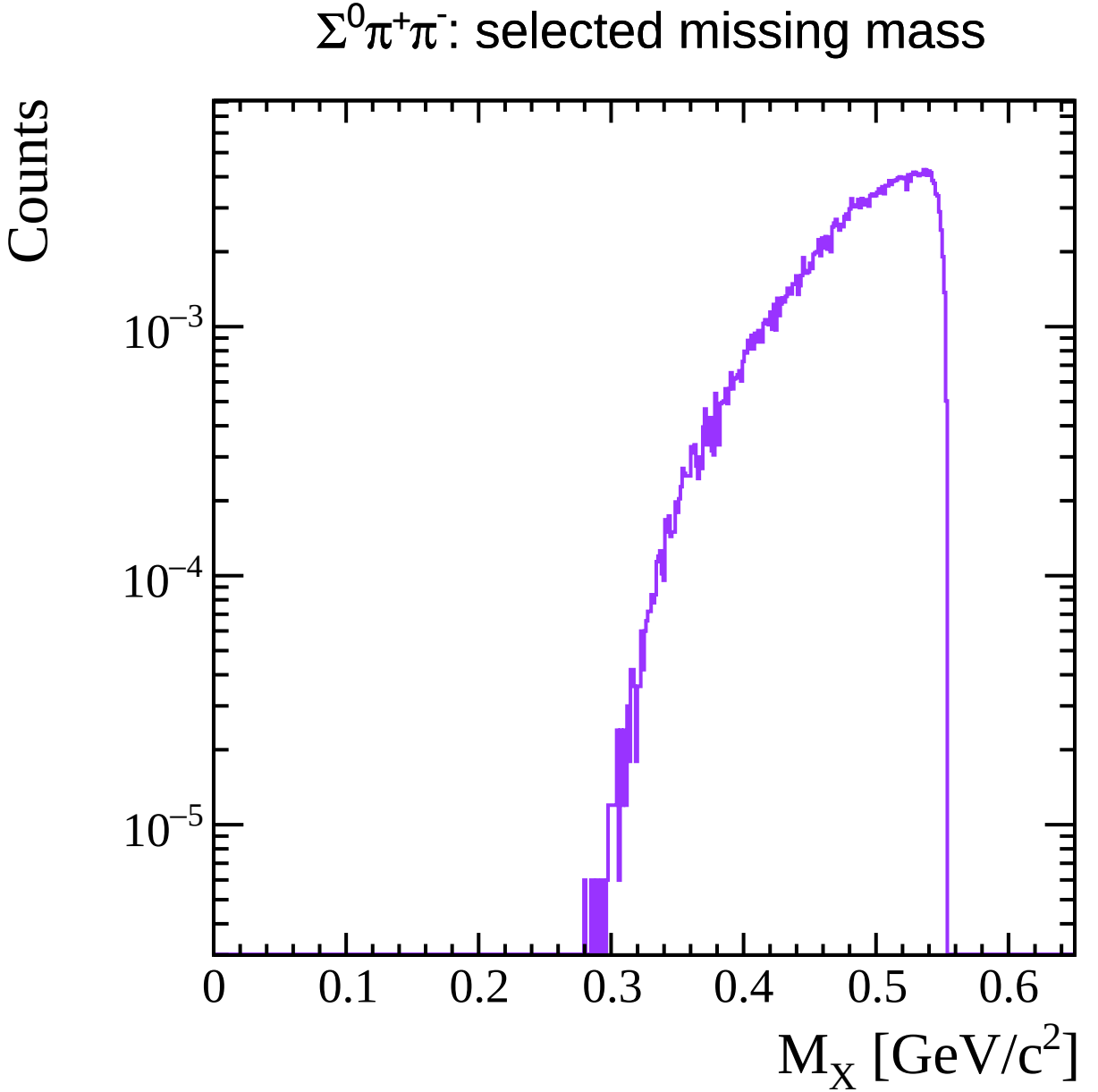

$\Lambda\pi^0$: selected missing mass
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]

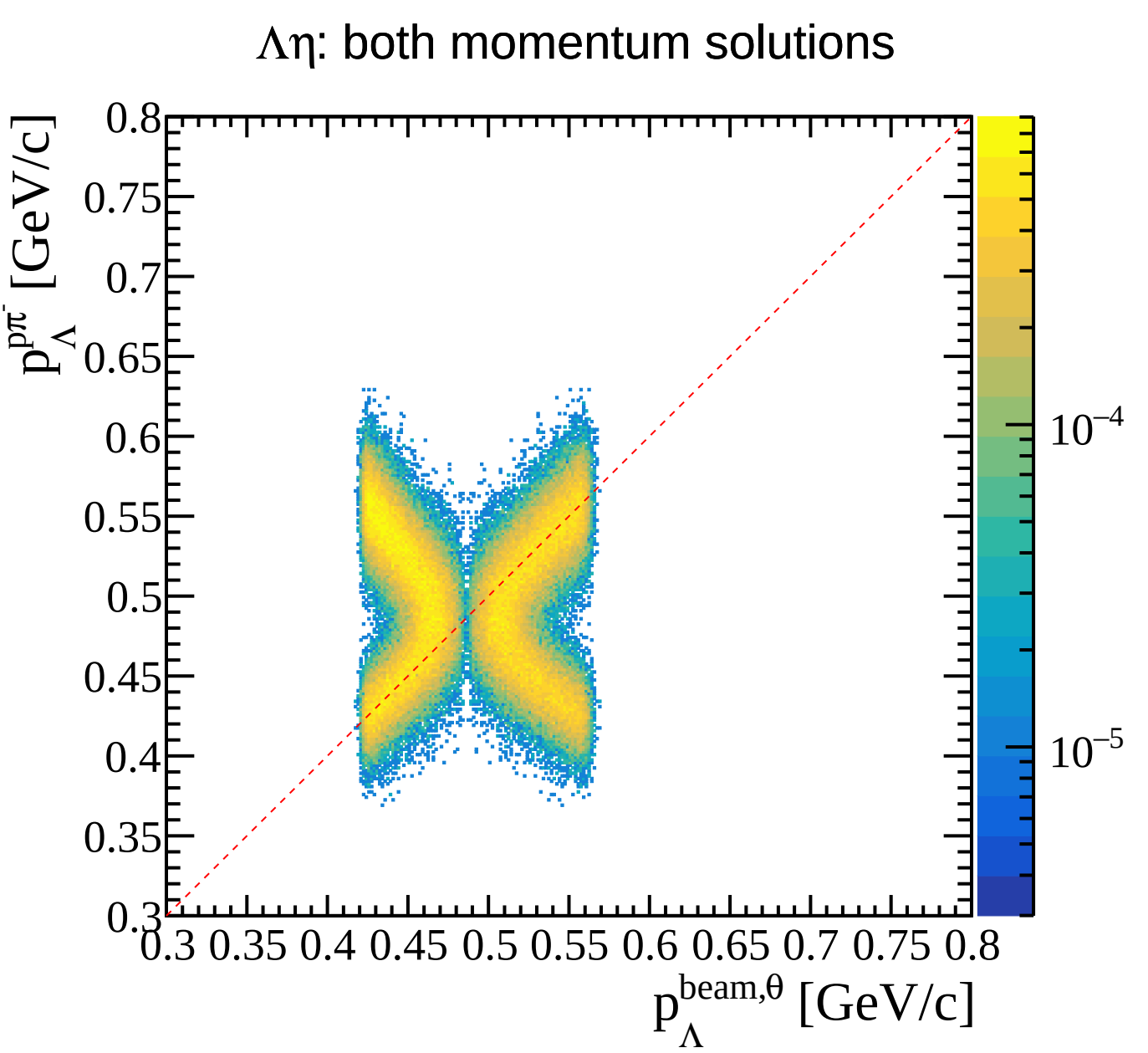

$\Sigma^0\pi^0$: selected missing mass
Counts
10^{-3}
10^{-4}
10^{-5}
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]


$\Sigma^+\pi^-$: selected missing mass
Counts
10
1
10^{-1}
10^{-2}
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]


$\Sigma^+\pi^-\pi^0$: selected missing mass
Counts
10^{-3}
10^{-4}
10^{-5}
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]


$\Lambda\pi^+\pi^-$: selected missing mass
Counts
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]


$\Sigma^0\pi^+\pi^-$: selected missing mass
Counts
10^{-3}
10^{-4}
10^{-5}
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]


Λη: both momentum solutions
$p_\Lambda^{p\pi^-}$ [GeV/c]
$p_\Lambda^{beam,\theta}$ [GeV/c]
0.8
0.75
0.7
0.65
0.6
0.55
0.5
0.45
0.4
0.35
0.3
0.3 0.35 0.4 0.45 0.5 0.55 0.6 0.65 0.7 0.75 0.8
10^{-4}
10^{-5}